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Children's Psychological and Mental Health Detection Application by Drawings Analysis Based on AI.



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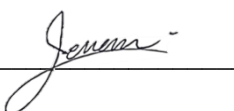
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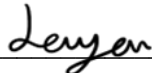
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ABSTRACT

Psychological and Mental health is an important part of human health that affects various aspects of life. Especially nowadays, where even children from the age of 6 to 15 years may face different changes and challenges from an early age that leave a strong impact on them even after they become adults. However, many children cannot express or explain their feelings and thoughts properly. Due to that, Psychological and Mental health specialists found a way to detect mental issues by observing and analyzing different signs from children's drawings. When they start drawing, they can describe their point of view and their mental perception. Furthermore, this process of analyzing and detecting mental disorders by professionals is still considered a complex and time-consuming process. From this point, Anamel comes up with a solution, by employing artificial intelligence to analyze children's drawings and provide diagnosis rates with high accuracy, while reducing the time and cost of paying specialists fees. Prior research in this field mostly focused on detecting Psychological mental issues only by answering a set of questions. Meanwhile, only one research discussed the ability to detect positive and negative feelings from children's drawings which also was developed into an application. However, a notable gap is the limited diagnosis of specific mental issues with high accuracy from both professional questions and children's drawings. Another gap is the unavailability of an application fully supporting the Arabic language. Overall, our project aims to develop an application by using machine-learning algorithms and models that are trained to analyze children's drawings and explain different elements from drawings to detect psychological and mental health disorders.

Keywords: Psychology, Mental health, Drawings, CNN, Machine Learning, Artificial Intelligence, Computer Vision ,Classification, YOLO.

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Chapter 1 – Introduction

1.1 Purpose of the Project

1.1.1 Why Anamel?

1.2 Purpose of this Document

1.3 Overview of this Document

1.4 Existing System

1.4.1 Existing system description

1.4.1.1 Literature Review

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1. INTRODUCTION

1.1 Purpose of the Project

Many children around the world are affected by mental health issues that may have long-lasting effects on their lives academically and personally. Meanwhile, Ignoring or not noticing these issues could negatively impact children's development process and limit their abilities, especially for children who cannot express themselves and explain their feelings clearly. Therefore, psychology specialists found drawing to be a powerful means of communication with children, as it is the secret key to understanding the hidden world within them. Where they express their thoughts and feelings and describe their point of view to the world [1]. This concept was of interest to experts and psychological professionals, who conducted many studies in this field and identified clinical semantics [2] for analyzing children's drawings and detecting their psychological meanings in mental health centers. However, it is a complex, time-consuming, and costly process, for specialists and parents. Moreover, this process contains several steps before and after the drawing analyzing stage, such as asking specific questions by psychologists to parents and analyzing their responses manually [3].

The work on this project started by identifying user's needs by interviewing several psychology specialists such as: Razan Al-Hamdani, Rula Muhanna, Anwar Al-Hadi, and children's drawings analysts such as: Muhammad Al-Tamami, and Maha Abdullah. We asked them about using drawings to express children's feelings and the possibility of analyzing psychological and mental health from these drawings. They praised the importance of drawing as a necessary step to uncover the feelings of children and adolescents that cannot be expressed by words. Since the analysis and discovery is done in a long manual process, they assured us that the application will be a big help in saving time and effort. After we took a complete picture of the field of drawing psychology, we also attended training courses to analyze children's drawings with certified trainers such as Mr. Muhammad Al-Tamami. On the other hand, a questionnaire was developed to ask parents about their need for the application and the results showed their interest and desire for the idea.

Due to that, we observed the need to use technology to develop and simplify this process, hence, we came up with Anamel. In the same context, the ESRA application was the first and only application for analyzing drawings with limited information only by analyzing the drawing positively or negatively without any other details and with an accuracy ranging between 55-79% [4]. Anamel's project is the first application designed to assist psychologists by analyzing children's drawings and collecting initial information about the child. Anamel also aims to assist parents and help them check on their children's mental health by allowing them to upload pictures of their children's drawings to the application and obtain accurate results from home. These results are provided by a trained artificial intelligence model that can detect the meanings of the drawings and their symbols to produce correct results depending on the clinical semantics. Anamel will also provide a way to communicate with trusted psychologists in this field to confirm the condition and provide a suitable treatment method after the analysis stage.

1.1.1 Why “أنامل - Anamel”?

Analysis of children's and teenagers' drawings to diagnose psychological and mental conditions is a critical field as children are unable to express their feelings properly. Although this field is related to psychology, in our current era, by combining computer vision and artificial intelligence in this field, we can diagnose the psychological and mental conditions of children and teenagers by gaining insight into children's psychological and emotional states through modern technology accurately and quickly using deep learning and artificial intelligence algorithms. The name “أنامل - Anamel” is an Arabic word that means fingertips, which we chose to point to the children's drawing process with their small fingers, and to provide early detection and development of personalized treatment plans.

1.2 Purpose of this Document

The purpose of "Anamel" is to develop an auxiliary application for psychiatrists and parents to analyze children's drawings using artificial intelligence to determine their psychological and mental state. Psychology of art is considered one of the branches of psychology because it is linked to the diagnosis and treatment of mental health through drawing through

Chapter 1 Introduction

reading and analyzing drawings. Psychologists use drawings to visualize the hidden feelings of the patient, especially children and adolescents. It is possible that the child does not express what is inside him skillfully with words, but he can express it through his drawings. Therefore, the goal of this project is to help parents understand their children's world and feelings by asking several questions in advance, taking a photo of the child's drawings, and then asking several subsequent questions to reach the analysis result with high accuracy. We provide them with ways to communicate with many specialists in this field. Our goal is also to help psychiatrists analyze drawings and diagnose a child's psychological condition more quickly, saving them time and effort in communicating with their patients. The most important psychological problems that can be identified from drawings, such as the family drawing test or free drawing, will be analyzed, where negative emotions such as sadness, anxiety, anger, and aggressiveness can be discovered. Thus, positive emotions such as joy and happiness can also be discovered.

1.3 Overview of this Document

In this document, we follow the Waterfall methodology, which is 5 steps. It begins with drawing the application requirements and depends on the needs of the users, as they were identified through several personal interviews that we conducted with psychologists and drawings analysts, and a course in analyzing children's drawings. Then we planned and designed the system architecture to create diagrams and design documents that define how the system works. Then we went through the process of collecting and counting data. Finally, the implementation, testing, and maintenance phase. As for the organization of the document, the first chapter contains an introduction and description of the project, defining its purpose. It also describes the current system and its problems. The second chapter includes data analysis through Data flow diagrams, system requirements including clients and users, functional requirements, and non-functional requirements such as (requirements for look and feel, ease of use, security, performance, and portability), and proposed and alternative solutions. In the third chapter design considerations are design constraints including hardware, software, and end-user characteristics. It also presents architecture strategies, algorithms used, reuse of software components, project management strategies,

development methods, and finally Future enhancements and plans. For the fourth chapter, system design, we focus on system architecture and program flow, the major and sub-modules, and finally a detailed description of system components.

1.4 Existing System

1.4.1 Existing system description

Despite the importance of the field of drawing analysis and its use in psychological diagnosis, it has not witnessed significant technical development, as there is only one application in the world for analyzing children's drawings with limited capabilities [1]. This literature review brings together fifteen research papers that explore studies and applications in analyzing children's drawings to diagnose their psychological condition. And mental health, in addition to research papers related to the general diagnosis of the psychological and mental condition.

1.4.1.1 Literature Review

Ali et al. [1] proposed creating an application based on artificial intelligence to recognize children's emotions through their drawings. This is due to children experiencing many different emotions such as happiness, sadness, fear, and feelings that are difficult to express. In the ESRA application that analyzes graphics and classifies them into only two categories: positive and negative emotions, a convolutional neural network model was applied to classify graphics, where three pre-trained models were used to train the model: ResNet18, ResNet34, and ResNet50. The study conducted four experiments using different combinations of datasets consisting of 102 drawings from a local school and 521 drawings from online sources. The deep learning model trained using the Fastai library in Python, classified graphics into positive or negative emotions with an accuracy of between 55% and 79%, and the study stressed the need for further development to improve the accuracy of the artificial intelligence model. With the addition of more feelings and emotions.

Furthermore, Yuan Y et al. [2] referred to analyzing children's drawings to identify potential psychological problems by using a shallow convolutional neural network (CNN) to extract features from

drawings, specifically those depicting trees. The model uses a multi-label classification method using the sigmoid function in the fully connected layer. The study included a data set for the school's fifth grade with more than 300 drawings of males and females. The drawings were analyzed based on psychological problems such as anxiety, stress, aggression, division, insecurity, and inability to learn. These problems are identified through a preliminary psychological analysis conducted by psychoanalysts of children's drawings. The classification accuracy using the CNN model in the study reached 70.4%. The study emphasized the importance of early identification of potential psychological problems in children and the potential benefit of the artificial intelligence model in providing an auxiliary diagnosis to psychological counselors.

Moreover, Zeeshan et al [3]. presented a two-step fine-tuning-based method to train a multi-label convolutional neural network (CNN) architecture via a person drawing (DAP) based test. The results confirmed that the two-step CNN architecture outperformed traditional pre-trained CNNs as it was 81.1% accurate in registering gross details, 99.2% accurate in registering attachments, and 79.3% in registering head detail categories for DAP samples.

Also, Moetesum et al. [4] presented the use of CNNs to analyze neuropsychological graphics, with a focus on brain graphics testing (BGT) as a means of early detection of neuropsychiatric disorders. By using pre-trained CNNs, it reduces the computational load and addresses data scarcity issues. The results showed high accuracy in shape recognition via CNN models with an accuracy of up to (98.33%). They also highlight the effectiveness of the ResNet101-LDA group with an accuracy ranging from 79.16% to 97.61% in identifying deformities. The study showed that this approach has proven effective and accurate in examining neurological and psychological conditions.

Additionally, Kim et al [5]. proposed to create a drawing interface using drawing recognition techniques and a fine motor skills workbook to evaluate children's drawing skills. The interface was used with 89 children, and the study also noted that children aged 5 years

and older showed better fine motor skills. Several algorithms have been used in the interface for drawing recognition such as Tanimoto Parameter and Random Forest + Bagging machine learning technology to determine the label of the drawing based on its features. The interface also includes a pedagogical feedback system to provide instructions for the correctness of children's drawing. Based on the results, the interface explained that it determines children's fine motor skills more accurately than the traditional “drawing test” with a pencil, while being useful to children in preparing for school.

In the same context of our project, Lee et al. [6] study explores the potential of using artificial intelligence (AI) to analyze children's drawings for psychological assessment, specifically focusing on the House-Tree-Person Test (HTP). By employing object detection technology, it achieves a mean Average Precision (mAP) of about 92.6% to 94.1%. The resulting psychological analysis table exhibits an average accuracy of 94.4%. This approach aims to overcome limitations in previous classification-based studies, offering a more comprehensive extraction of information from images. If successful, this AI-based method could revolutionize psychological testing by enabling online or smartphone-based testing services.

Next, Chen S et al [7]. Developed a digital art system based on artificial intelligence and virtual reality to improve primary school students' ability to draw and recognize colors. The study sample consisted of 30 students in the experimental group that received learning with the help of artificial intelligence, and 30 students in the control group, who are fourth-grade students in a primary school in the city of Taitung, Taiwan. The study showed the role of artificial intelligence in recognizing outlines, matching colors, and calculating proportions, in addition to... To use smart glasses in augmented reality panels. The system recognizes fonts using OpenCV and generates recommended hue color matching using the pix2pix method based on Conditional Generative Adversarial Networks (cGAN). The study also used the Torrance Tests of Creative Thinking to evaluate students' imagination and creativity. The study showed that the system

improved students' performance in drawing, imagination, and creativity.

In addition, Likforman-Sulem et al. [8] introduced a developed database named EMOTHAW that connects emotional states to handwriting and drawing. The database included 129 samples from the participants. The emotional states of the participant, anxiety, depression, and stress are evaluated using the Depression-Anxiety-Stress Scale (DASS) questionnaire. The data is collected using a digitizing tablet and includes pen positions, time stamps, pressure, pen azimuth, and altitude. The research shows an analysis of the database EMOTHAW using random forests, a machine learning algorithm, to identify features that are indicative of emotional states. The top-ranked features for each emotional state are identified and used to build random forest classifiers associated with each emotional state. The results show that stress and anxiety recognition perform better than depression recognition. The research also provides accuracy, sensitivity, and specificity evaluation measures obtained from cross-validation experiments.

However, Rakhmanov et al. [9] focus on testing several methods of image classification, using machine learning and computer vision algorithms to classify DAPT (Draw-A-Person Test) images. A collection of 1000 sketches was collected from primary school students in Nigeria, aged 4 to 11. The sketches are manually classified according to the age. The research investigates the possibility of combining sketch recognition with the analysis of cognitive tests. The main reason why the research conducted several image classification tests using machine learning and computer vision algorithms is to see if the researchers can develop a model to classify images automatically. The research first conducts a classification test only using the pixel value of the pictures and SVM and ANN to classify them, the accuracy result is 18% and 17%. Then they used HOG (Histogram of oriented) to extract features from pictures in all 8 dataset using also SVM and ANN to classify them and the accuracy ranged from 17% to 48% with SVM trained on Double_inv dataset obtaining the highest result, showing that the double data did performed better.

Then they used BoVW and faced a challenge with the dimension of input vectors and used two different ways to classify the dataset: manual and formal methods, the manual method result reached 54% of accuracy and 62% accuracy with the double dataset. The research also shows that a test was conducted using the several types of CNN structures and algorithms, but highest accuracy achieved in this experiment is 62% with BoVW manual method.

As well, Binti [10] delved into the creation of a sketch recognition tool employing deep learning to observe the psychological growth of children. she collected a dataset of 1,051 children's sketches. the researcher asserts rapid design and prototyping cycles. and employed the Rapid Application Development (RAD) approach. The programming language used in this research is Python with the convolutional neural network (CNN) model for image recognition. The highest accuracy achieved is 89%. The research aimed to overcome the limitations and constraints created by manual identification methods and the lack of psychological identification tools for children's mental and emotional development. The research showed some traditional methods such as the Drawing Person Test (DAPT) and some other psychological drawing tests were explored as standard assessment techniques. One of the future recommendations of this research is to develop a user interface for the tool and expand its application in different fields.

Apart from this, Lawrie [11] delved in the analyze of children's drawings as indicators of mental well-being, and the main goal of this research is to assess the validity of the emotional indicators themselves as well as the accuracy of LASSO regressions, the dataset used in this research is a set of 453 drawings from the children in India where 219 household have at least on child with CLP (cleft lip/palate) present and they were consisted in the survey sample, each child was asked to draw a picture of their family and indicate where each family member are. The method used in this research include manual evaluating for each drawing and assigning a score of 1 to each emotional indicator variable if its present but this is a time-consuming process. So, the researcher shows an interest for further research that

could be using machine learning for analysis and showing signs of emotional distress in the children. Although HFDs are still an important tool, fast and cost-effective to analyzing children's emotions.

Moreover, Long et al. [12] investigated the parallel developmental changes in children's ability to both produce and recognize line drawings of visual concepts. The study included children aged from 2 to 10 years to participate in two tasks, drawing and recognition. The research shows the analysis of the drawings using evaluation both deep convolutional neural network model of object recognition and human annotations of the object parts. The interface was navigable by children and the first page showed a short consent form for parents to enter their child's age, this was the only demographic information that was collected. To guide the child through the experiment a video prompts of an experimenter was provided. The game stated to be for only one person at a time, and children were able to stop the game any time but each trial will end after 30 seconds or after the child press the next button. For future works the researchers propose that a full understanding of how children produce and recognize drawings of visual concepts will allow a unique and novel perspective on the both the development and the nature of visual concepts.

In addition, Tutin et al. [13] proposed mental health disorder prediction by a decision support system (DSS) based on AI. The study aimed to find solutions for problems of mental health assessment tools, that require a long time and get low participation rates. And the results must be analyzed manually by experts. The paper found improving the DSS and the psychological tests by artificial intelligence and advanced analytics an efficient solution to discover mental problems. Then it used a Network Pattern Recognition (NEPAR) algorithm to build assessment tools, identify the questions for the participants, and reduce the number of questions in the test. Furthermore, they predicted mental issues by training machine learning models with participant's responses to the test, and historical data. In conclusion, the results showed that the DSS can automatically diagnose mental disorders

using only 28 questions without any human input, with an accuracy level of 89%. That means that this method was successful in developing the DSS that can replace traditional paper-based examinations, decreasing the possibility of missing data for predicting mental disorders. It also has fewer questions than its counterparts which leads to higher participation and completion rates. Future work for this paper is using deep learning models and the Psikometrist framework datasets for more improvement.

Furthermore, Srividya et al. [14] presented machine learning algorithms for mental health behavioral modeling. The paper addressed the challenge of predicting mental disorders of different groups, such as high school students, college students, and working professionals. The common notion between these categories is the stress and depression daily life they go through. The study aimed to identify people with mental illness by using machine learning algorithms. At first, they prepared a questionnaire supported by experts. Then based on the person's score, which depends on his response, his mental health status was predicted. In addition, it used clustering techniques to identify groups and potential classes for further processing, as well as classification algorithms such as SVM, KNN, and Random Forest for finding mental health illnesses. The results were on two target groups in the ages of 18 to 21 years and 22 to 26 years that showed that SVM, KNN, ensemble (bagging), and tree ensemble (random forest) performed best with 90% accuracy. In conclusion, they successfully developed a framework for discovering the mental health status of people by building prediction models using clustering and classification algorithms. Future work involved more sections of society and more types of mental illnesses. Also, it implied physiological parameters and deep learning methods, such as recursive neural networks for higher classification accuracy.

Finally, Veltman & Browne.[15] This paper was conducted to find out whether psychologists and teachers can determine if a child is being subjected to abuse through his drawings. No matter how bad this child is receiving. Two studies were conducted. In the first, they divided 33 psychologists and 10 teachers and showed them 12 sets of

children's drawings. The other study was conducted on the same teachers and psychologists, but they showed two series of 28 drawings to ten-year-old children. Psychologists were able to identify all the children who had been abused, while the teachers were somewhat confused in the interpretation. But we can reach the point where teachers and psychologists can tell that a child is being abused through his drawings. Note that two techniques were used in drawing, the first being Favorite Day Drawings (FKDs), where each child is asked to draw his favorite day, and the other technique is Kinetic Family Drawings (KFDs), where each child is asked to draw his entire family and himself with them

1.4.2 Problems in the existing system

Through these fifteen studies, a comprehensive understanding emerges regarding the analysis of children's drawings to diagnose their psychological and mental conditions. While some research papers emphasize the accuracy and helpfulness of using technology in diagnosing psychological and mental illnesses. By compiling the findings from these various studies, this literature review confirms the possibility of using artificial intelligence and technology to analyze children's drawings. Based on the only application that analyzes drawings, ESRA [1], which emphasized the importance of improving the accuracy of the model and developing to detect more emotions, the Anamel application will seek to create A unique application for analyzing the drawings of children and teenagers to discover their psychological and mental state with many additional precise features for correct diagnosis.

Chapter 2 – System Analysis

2. Data Analysis

2.1 System requirements

2.1.1 Functional and data requirements

2.1.2 Non-functional requirements

2.1.2.1 Look and Feel Requirements

2.1.2.2 Usability requirements

2.1.2.3 Security requirements

2.1.2.4 Performance requirement

2.1.2.5 Portability requirements

2.2 Data flow diagrams

2.2.1 DFD Level 0

2.2.2 DFD Level 1

2.2.3 DFD Level 2

2.3 Use Case Diagram

2.4 Scenarios

2.5 Activity Diagram

2.6 Sequence Diagram

2.7 Component Diagram

2.8 Class Diagram

2.9 ERD

2.10 Proposed Solutions

2.11 Alternative Solutions

Note : [Click on the Diagram Name for a Closer View](#)

2. Data Analysis

2.1 System Requirements

The System Analysis chapter delves into the details of the proposed system according to the functional and non-functional requirements of the main system and subsystem and explains them through various diagrams. Finally, future plans.

2.1.1 Functional and Data Requirements

Person

1. Sign-up

- 1.1 The user shall be able to sign up.
 - 1.1.1 The system will show a sign-up page for the user to create an account.
 - 1.1.2 The system will ask the user to fill in the following fields (Name, Birth date, E-mail, Phone number, and password) the password field is repeated for confirmation.
 - 1.1.3 The system will make sure the fields are filled properly.
 - 1.1.4 The system will show a pop-up message if an error occurs.

Specialist

2. Sign-up

- 2.1 The Specialist shall be able to sign up.
 - 2.1.1 The system will show a sign-up page for the Specialist to create an account.
 - 2.1.2 The system will ask the Specialist to fill the following fields (Name, Professional license number, Birthdate, E-mail, Phone number, password).
 - 2.1.3 The system will make sure the Professional license number and other fields are correct.
 - 2.1.4 The system will show a pop-up message if an error occurs.

Person/Specialist

3. Verification

- 3.1 The user shall verify his/her account.
 - 3.1.1 The system will send a verification code to confirm the user's identity.
 - 3.1.2 The system will ask the user to try again if the code is incorrect.
 - 3.1.3 The system will save the account in the database and transfer the user to the home page if the code is correct.

Person/Specialist/Admin

4. Sign-in

- 4.1 The user shall be able to Sign-in.
 - 4.1.1 The system will ask the user to enter his/her e-mail and password.
 - 4.1.2 The system will make sure the e-mail and password are written correctly.
 - 4.1.3 The system will compare the user's e-mail and password with the ones registered in the database.
 - 4.1.4 The system will transfer the user to the home page if the e-mail and password are correct.

5. Forgot Password

- 5.1 The user shall be able to change his/her password when forgotten.
 - 5.1.1 The system will ask the user to enter his/her email address.
 - 5.1.2 The system will send a verification code to the user's email address.

5.1.3 The system will ask the user to enter a new password.

5.1.4 The system will save the new password in the database.

6. Browse Home Page

- 6.1 The user shall be able to browse the home page.

6.1.1 The system will display the home page for the user after verifying the account or after logging in.

6.1.2 The system will display the main functions of the application to the user.

Person/Specialist

7. Show Profile

- 7.1 The user shall be able to show his/her profile.

7.1.1 The system will display the user profile page with the information in the database.

8. Edit Profile

- 8.1 The user shall be able to edit his/her profile.

8.1.1 The system will allow the user to modify and update personal information.

9. Add Child

- 9.1 The user shall be able to add children to the account.

9.1.1 The system will allow the user to enter the child's information and save it for future analysis.

10. View Child

- 10.1 The user shall be able to view added children.

10.1.1 The system will display the children's interface for the user which contains all the children's information when the user clicks on the “view children” button that exists both on the profile and the homepage.

11. Edit Child

- 11.1 The user shall be able to Edit the child's information

11.1.1 The system will allow the user to update the child's information whenever the user needs to.

12. Remove Child

- 12.1 The user shall be able to remove a child from the profile.

12.1.1 The system will remove the child from the user account when the user clicks the remove child button.

13. Answer Questionnaire

- 13.1 The user shall be able to answer the questionnaire.
 - 13.1.1 The system will ask the user to answer the pre-questions of the child's family environment and other questions before analyzing the drawing.
 - 13.1.2 The system will display the pre-questions as closed questionnaires for specific and clear results.

14. Take Photo

- 14.1 The user shall be able to take a photo of the child drawing for analysis.
 - 14.1.1 The system will allow the user to take a photo of the child's drawing and re-take it if the photo is unclear.
 - 14.1.2 The system will analyze the photo and inform the user when the analysis is done.

15. Upload Photo

- 15.1 The user shall be able to upload an image of the child's drawing.
 - 15.1.1 The system will allow the user to upload the child's drawing.
 - 15.1.2 The system will analyze the photo and inform the user when the analysis is done.

16. Confirm State

- 16.1 The user shall be able to confirm the state of the analysis.
 - 16.1.1 The system will ask the user to answer the post-question about the child's mental health and some disorders after the analysis of the child drawing to confirm the state of the analysis.
 - 16.1.2 The system will display the post-questions as a closed questionnaire for specific and clear results.

17. Show Analyzing State

- 17.1 The user shall be able to show the analyzing state.
 - 17.1.1 The system will display the analyzing state of the child drawing with the disorders and their percentage after the post-questions.
 - 17.1.2 The system will allow the user to download the analyzing state as a PDF.
 - 17.1.3 The system will provide a button to contact the specialist to make it easy for the parent or the child to take a consultation.

18. Download Analyzing State

- 18.1 The user shall be able to download the analyzing state of the drawing.
 - 18.1.1 The system will provide a download button after the analysis ends to enable the user to have a PDF format file of the analysis.

19. Browse Analyzed States

- 19.1 The user shall be able to browse all the state analyses.
 - 19.1.1 The system will display a page for the user to see all the previous analyses.
 - 19.1.2 The system will show a comparison of the disorders for all the previous analyses.

20. Report Problem

- 20.1 The user shall be able to report a problem or add a suggestion.
 - 20.1.1 The system will display a form for the user to fill in the fields, the subject of the message, if it was a report for a problem or adding a suggestion, and then describe it in the text area.

Person

21. Search for a specialist

- 21.1 The user shall be able to search for a specialist.
 - 21.1.1 The system will have a search bar to help the user reach a specific specialist.

22. Browse specialist

- 22.1 The user shall be able to browse and view the specialist.
 - 22.1.1 The system will allow the user to view the specialist section.
 - 22.1.2 The system will display the specialist page with the specialist's information and their contact accounts.

23. Contact specialist

- 23.1 The user shall be able to contact the specialist.
 - 23.1.1 The system will allow the user to contact the specialist when clicking their contact accounts buttons such as Snapchat and WhatsApp.

Specialist

24. Register as a consulting specialist

- 24.1 The user shall be able to register as a consulting specialist.
 - 24.1.1 The system will provide a form for the specialist to register as a consulting specialist for our application users.
 - 24.1.2 The system will check the registered specialist information and send a message of approval or disapproval.
 - 24.1.3 If approved the system will display the specialist information and contacts after the analysis when the user browses or when a user searches for the specialist name.

Admin

25. Check Specialist Request

- 25.1 The admin shall be able to view and accept requests from specialized users.
 - 25.1.1 The system will display a list of newly added specialists to be accepted into the application.
 - 25.1.2 The system will allow the administrator to accept or reject the request.

26. Check Reported Problem

- 26.1 The admin shall be able to view reported problems from users.

26.1.1 The system will display a list of submitted problems.

Person/ Specialist/Admin

27. Sign-out

- 27.1 The user shall be able to sign out.

27.1.1 The system will enable the user to sign out after clicking the sign-out button.

2.1.2 Non-functional requirements

2.1.2.1 Look and feel requirements.

2.1.2.1.1 The system layout will be attractive for both adults and children.

2.1.2.1.2 The system application colors were wisely chosen to support the idea of the project.

2.1.2.2 Usability requirements

2.1.2.2.1 The system will be easy to learn and use.

2.1.2.2.2 The system will support Arabic and English languages for a better understanding.

2.1.2.2.3 The system will display the error messages correctly and clearly.

2.1.2.3 Security requirements

2.1.2.3.1 The system will send a verification code to the user's email address.

2.1.2.3.2 The system must ensure the users are who they claim to be when log-in

2.1.2.4 Performance requirement

2.1.2.4.1 The system will be fast and efficient in analyzing children's drawings.

2.1.2.4.2 The system will be available 24/7.

2.1.2.4.3 The system will handle any number of users.

2.1.2.4.4 The system will be accurate and clear when displaying the analysis results.

2.1.2.5 Portability requirements

2.1.2.5.1 The system will be able to run on multiple platforms.

2.2 Data Flow Diagrams

A data flow diagram is a graphical representation of the flow of data within a system.

- 2.2.1 DFD Level 0

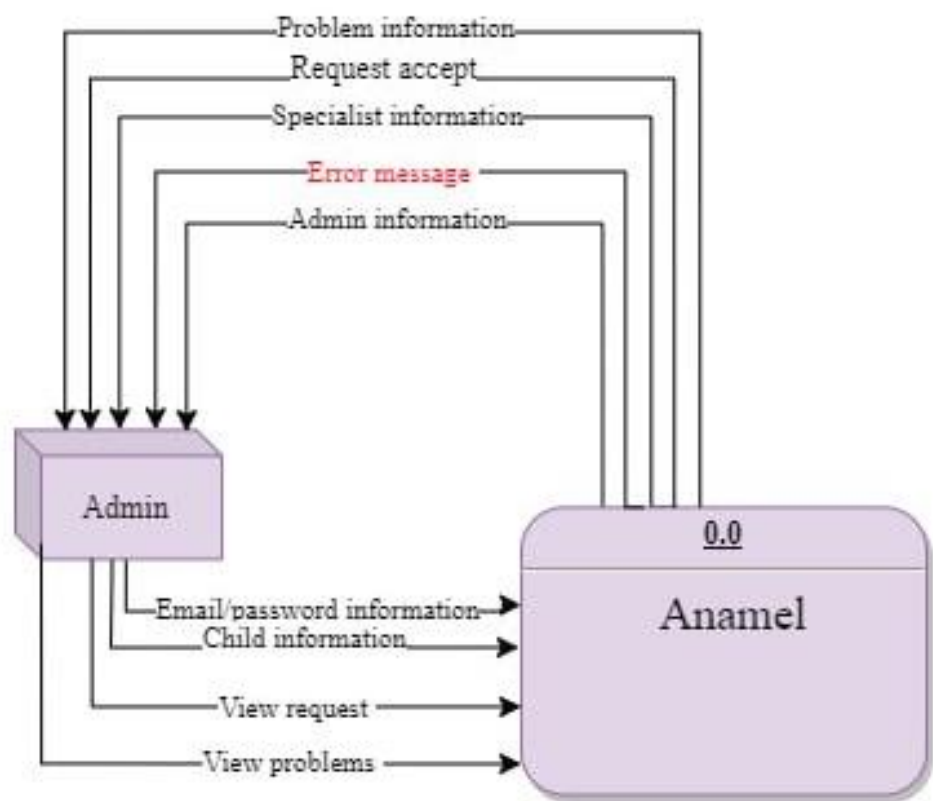


Figure 1: Context Diagram (Level 0)-(Admin)

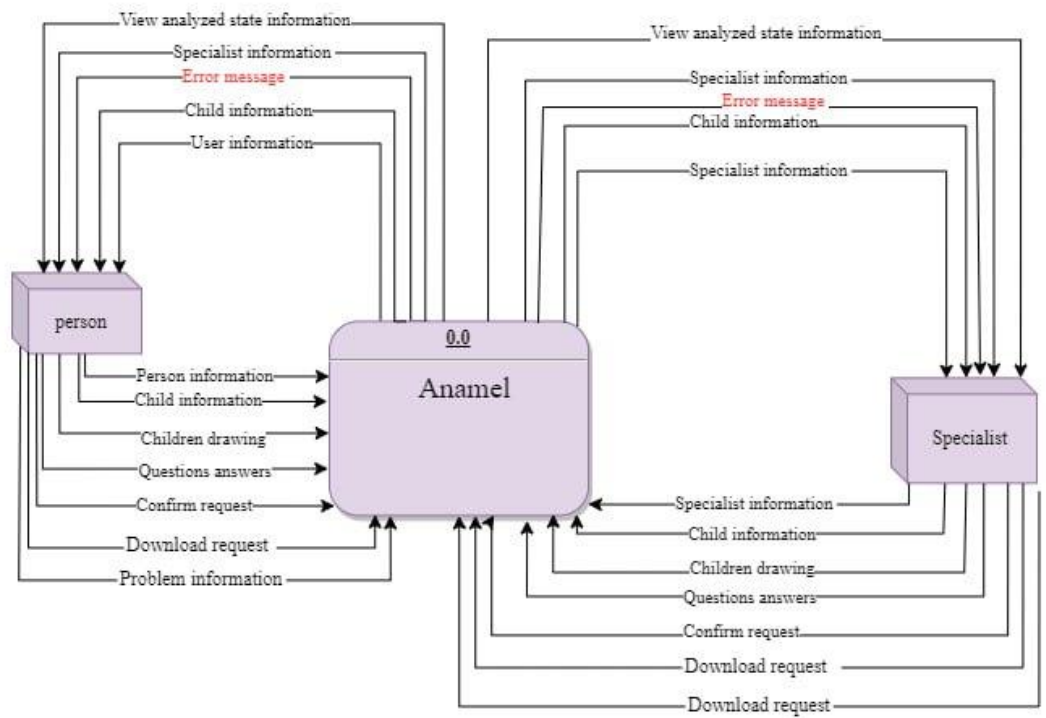


Figure 2: Context Diagram (Level 0)-(Person-Specialist)

- 2.2.2 DFD Level 1

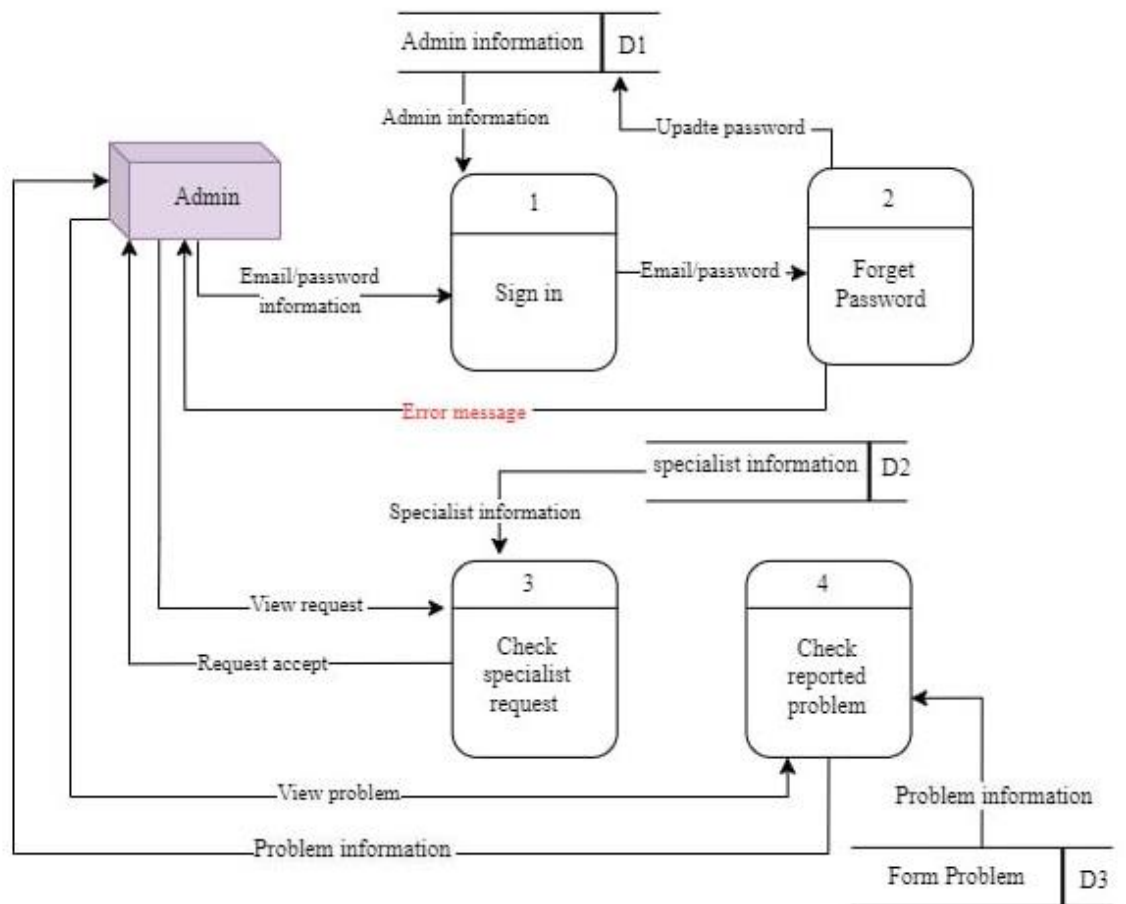


Figure 3: Data Flow Diagram (Level 1)-(Admin)

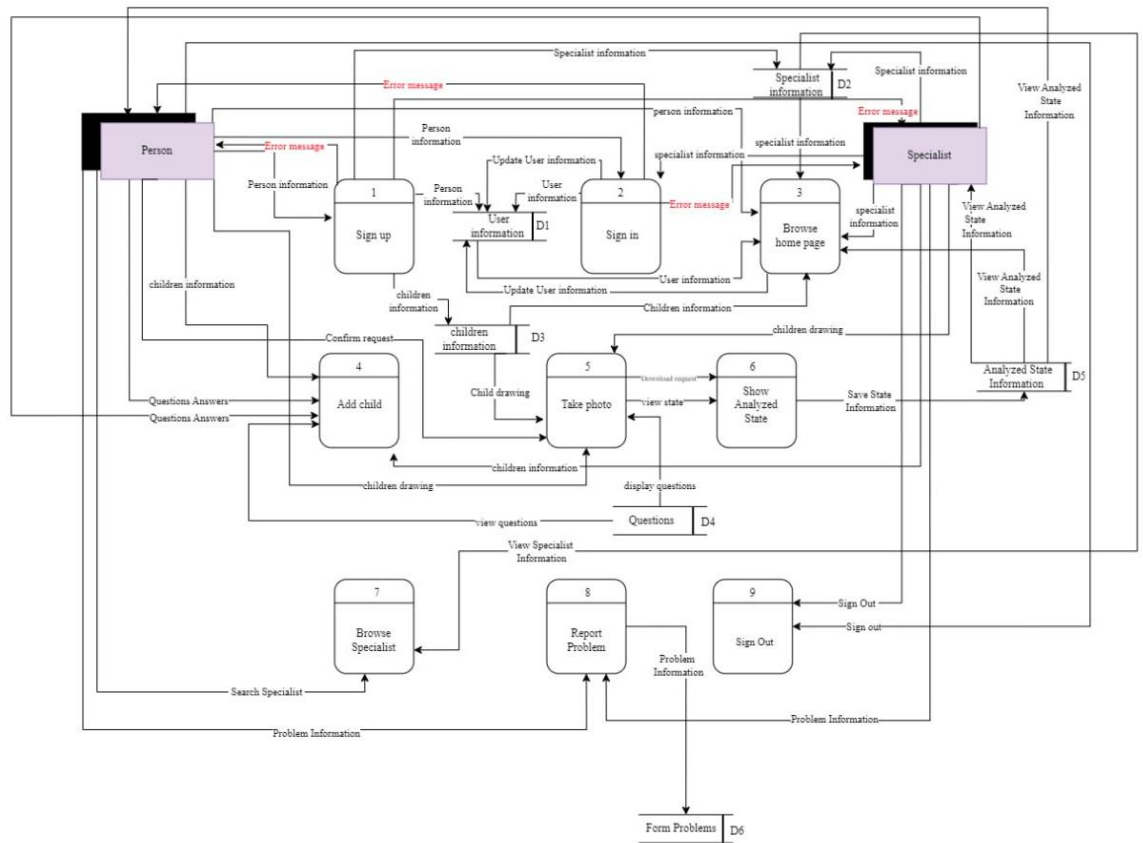


Figure 4: Data Flow Diagram (Level 1)-(Person-Specialist)

- 2.2.3 DFD Level 2

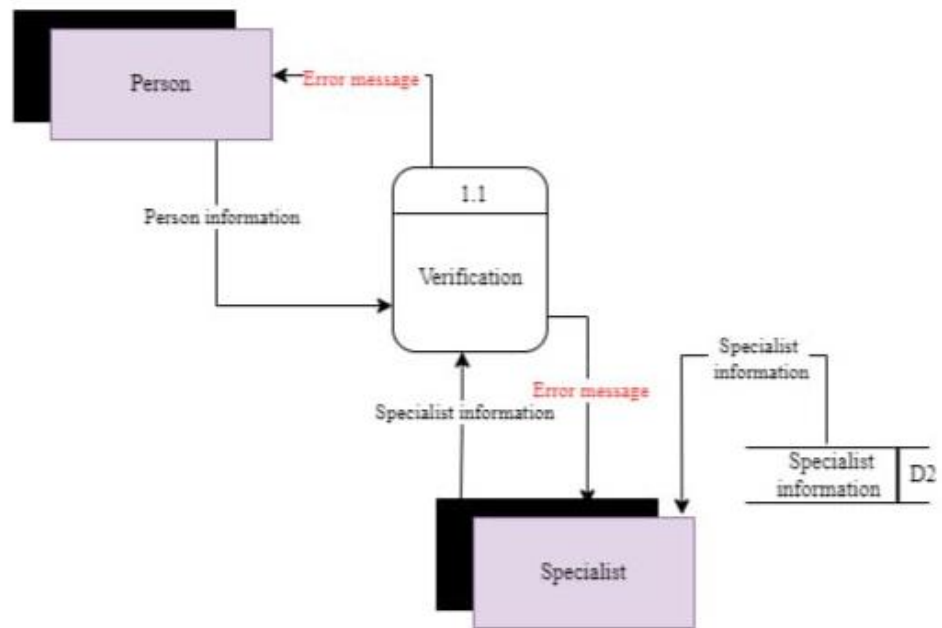


Figure 5: Data Flow Diagram (Level 2) -Verification

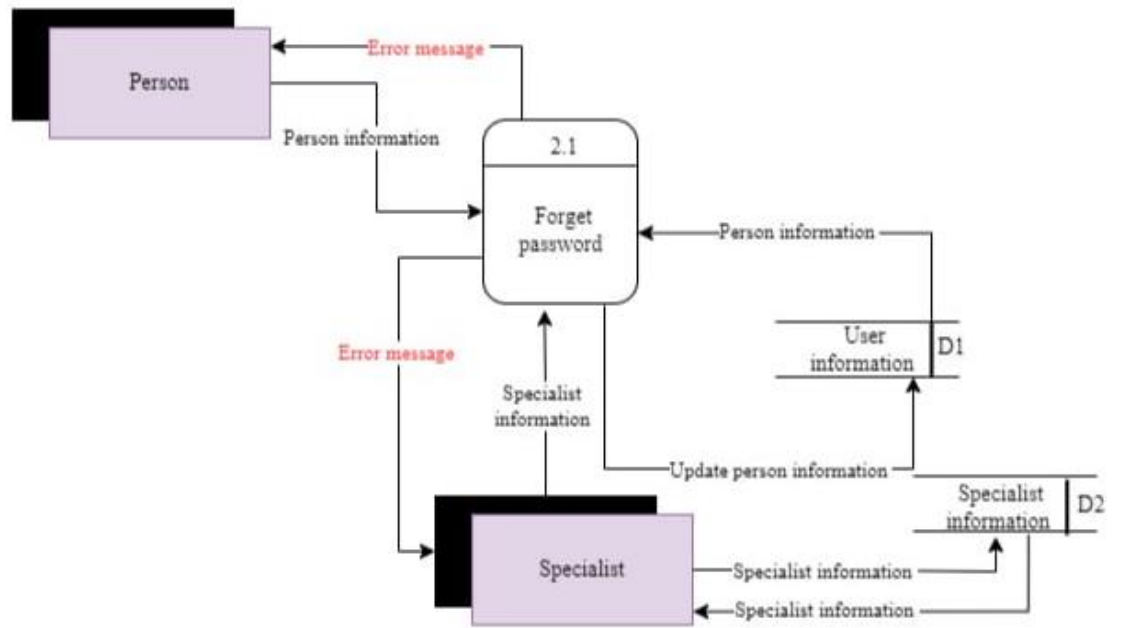


Figure 6: Data Flow Diagram (Level 2) -Forget password

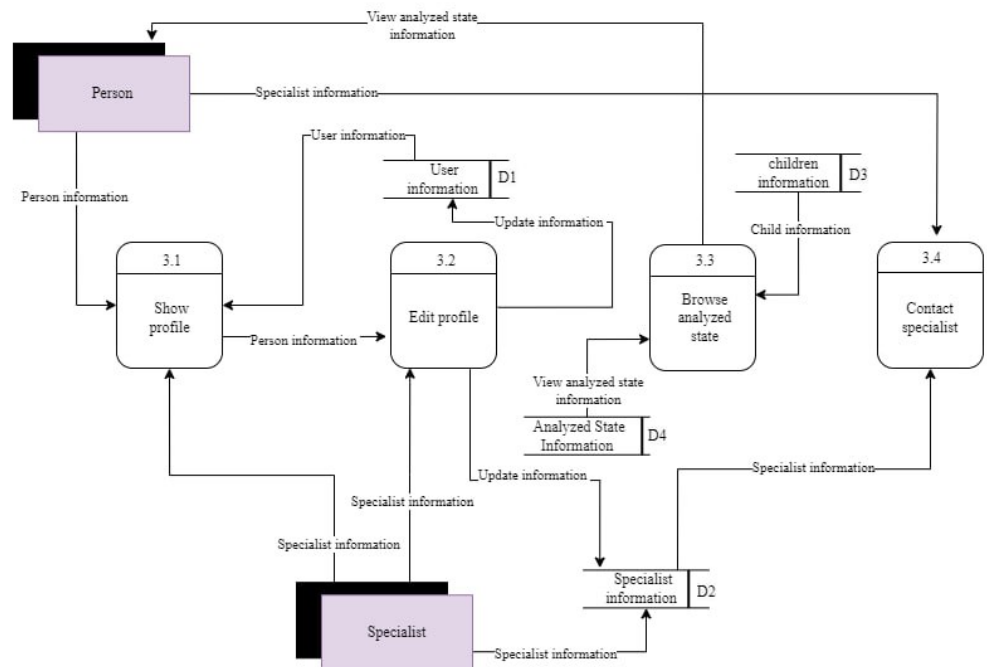


Figure 7: Data Flow Diagram (Level 2) -Profile-Brows analyzed state-contact

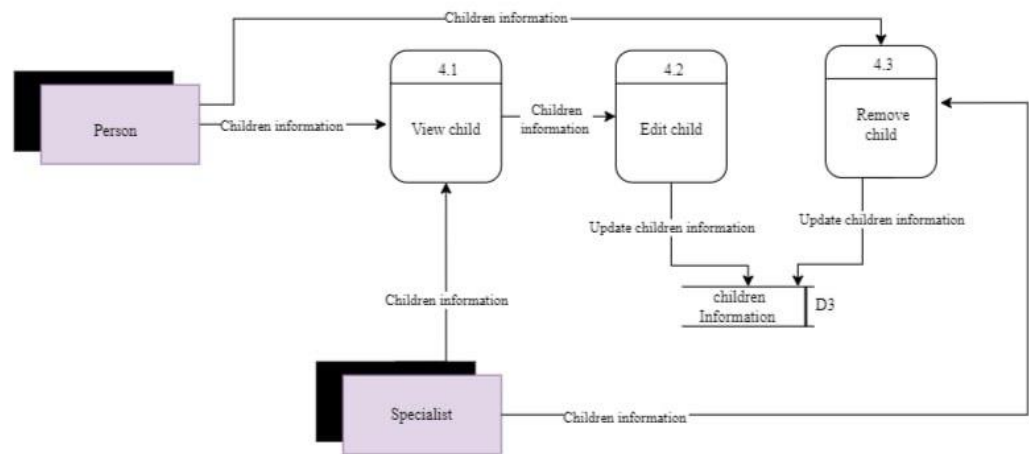


Figure 8: Data Flow Diagram (Level 2) -child

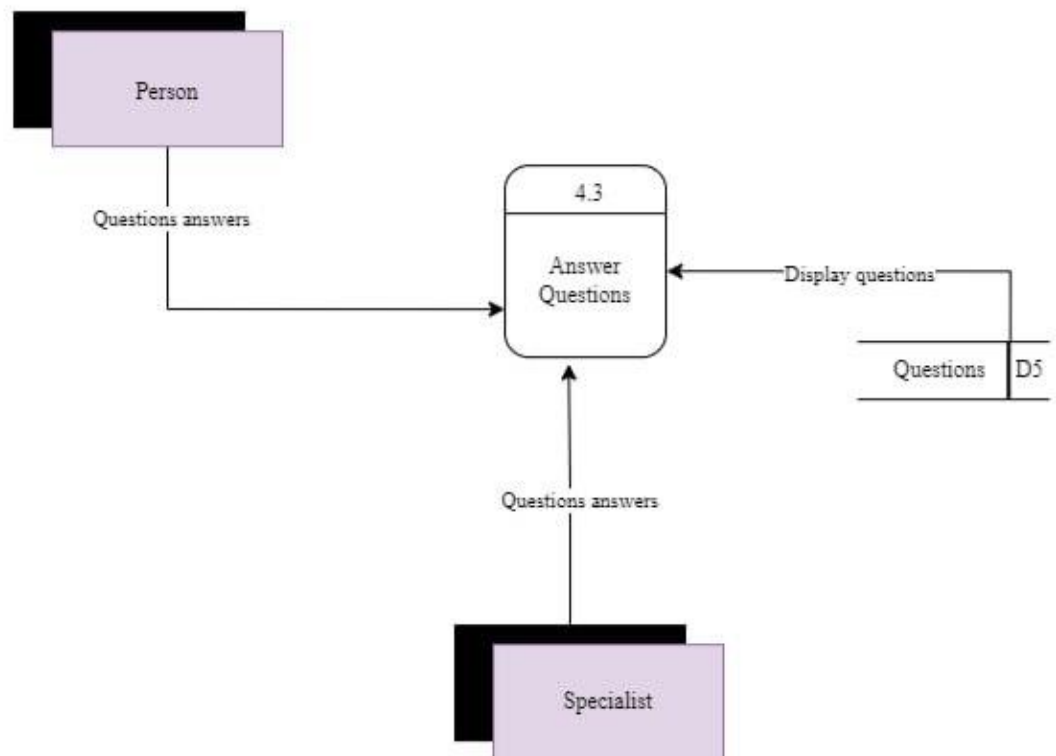


Figure 9: Data Flow Diagram (Level 2) -Answer questions

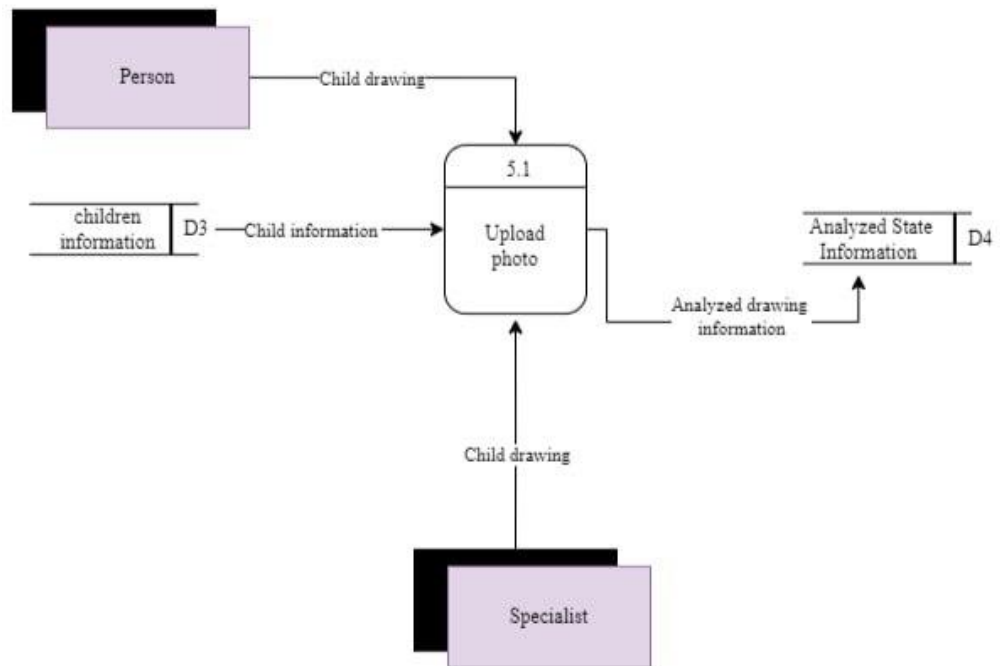


Figure 10: Data Flow Diagram (Level 2) -upload photo

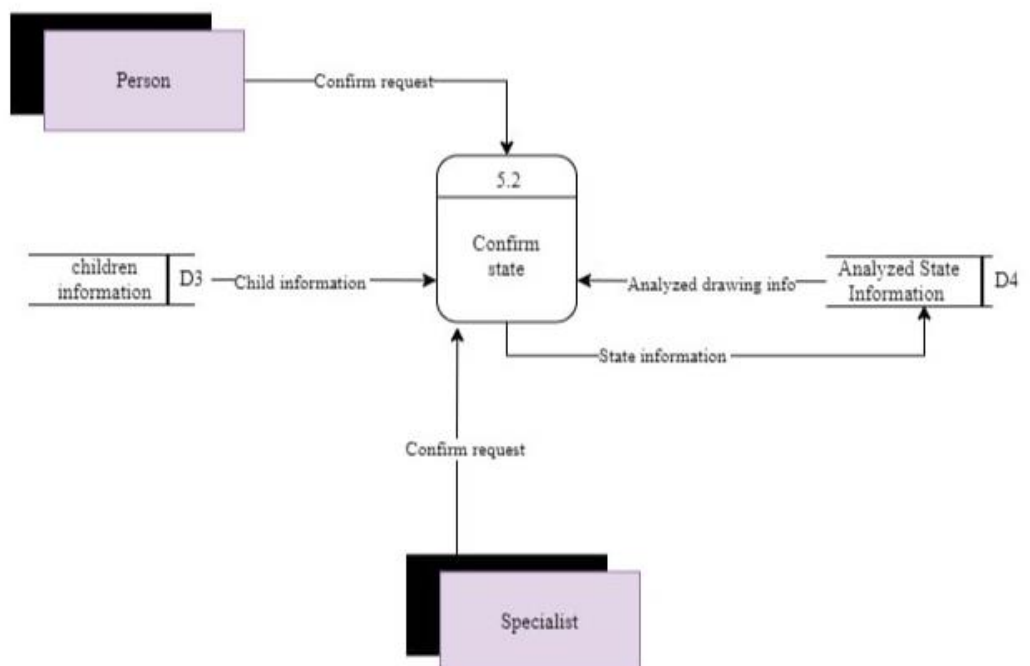


Figure 11: Data Flow Diagram (Level 2) -confirm state

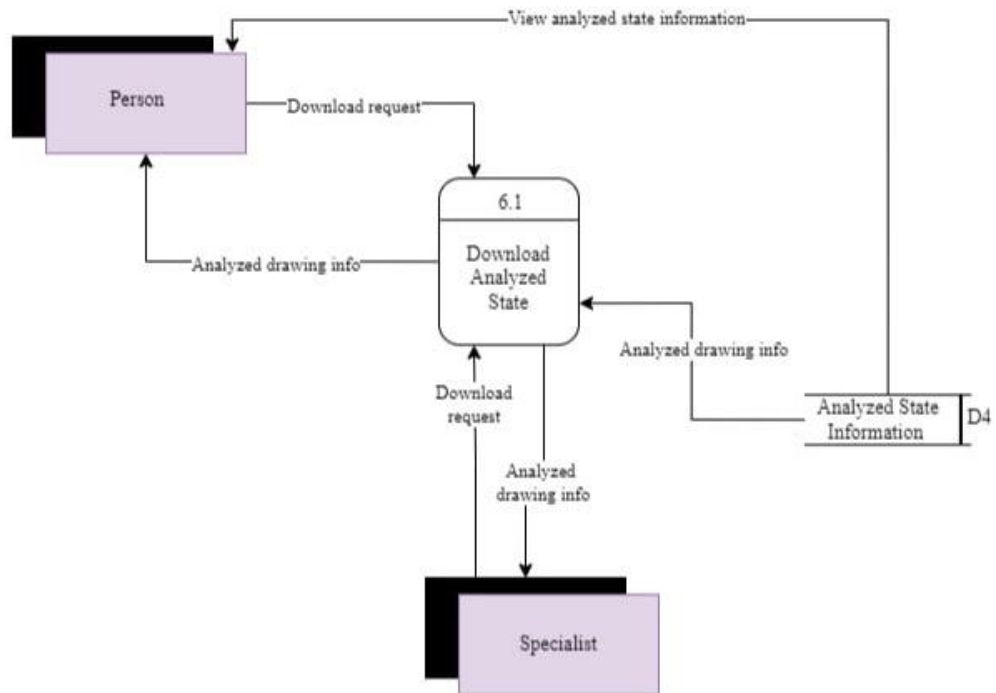


Figure 12: Data Flow Diagram (Level 2) -download analyzed state

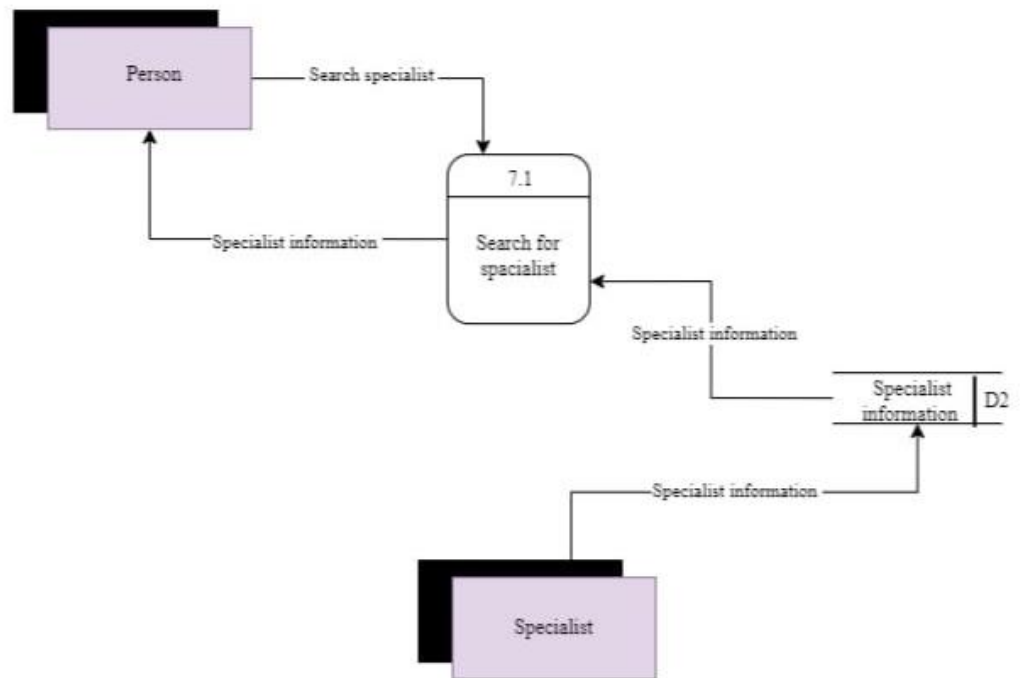


Figure 13: Data Flow Diagram (Level 2) -Search for specialist

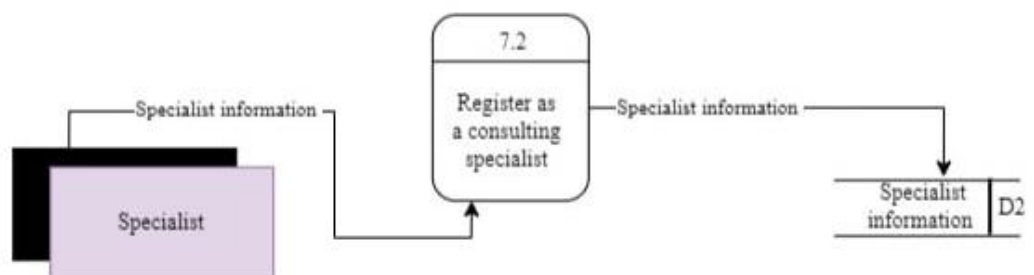


Figure 14: Data Flow Diagram (Level 2) -Register as a consulting specialist

2.3 Use Case Diagram

The use case diagram for the Anamel system explains each task with its user.

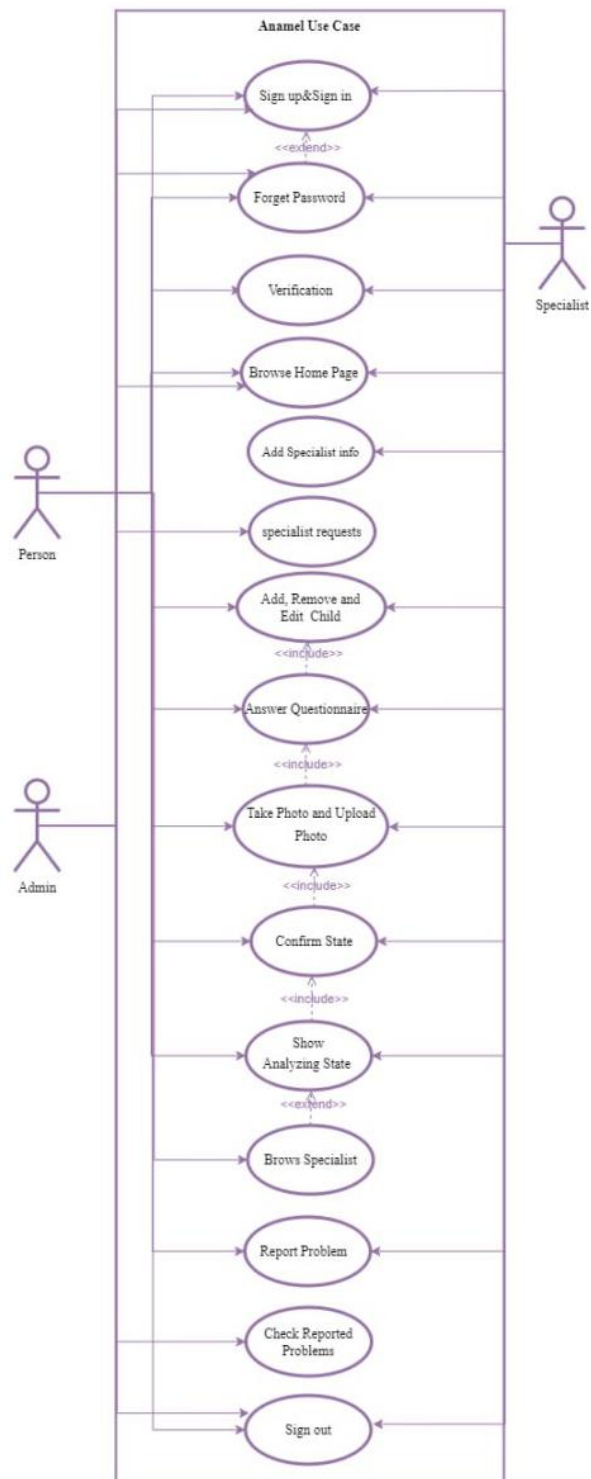


Figure 15: Use Case Diagram for Admin, Specialist, and Person

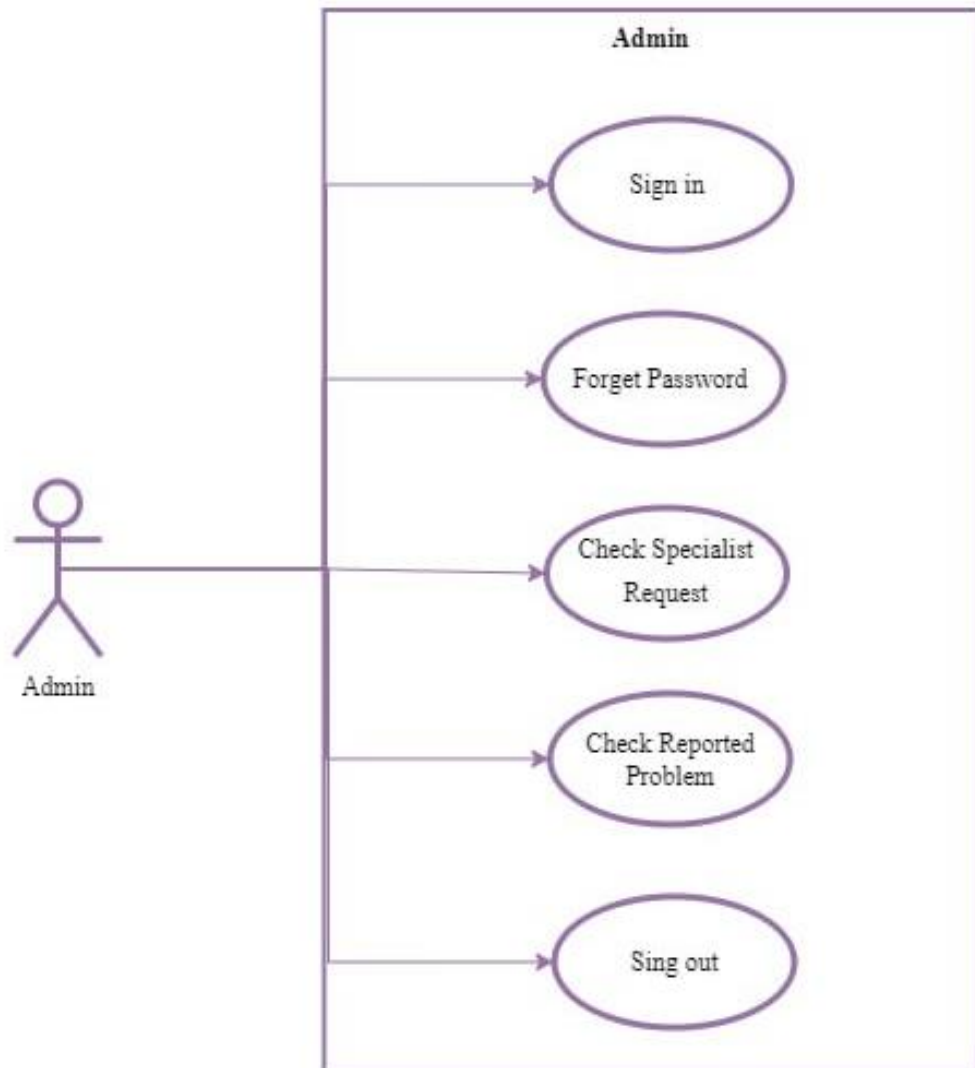


Figure 16: Use Case Diagram for Admin

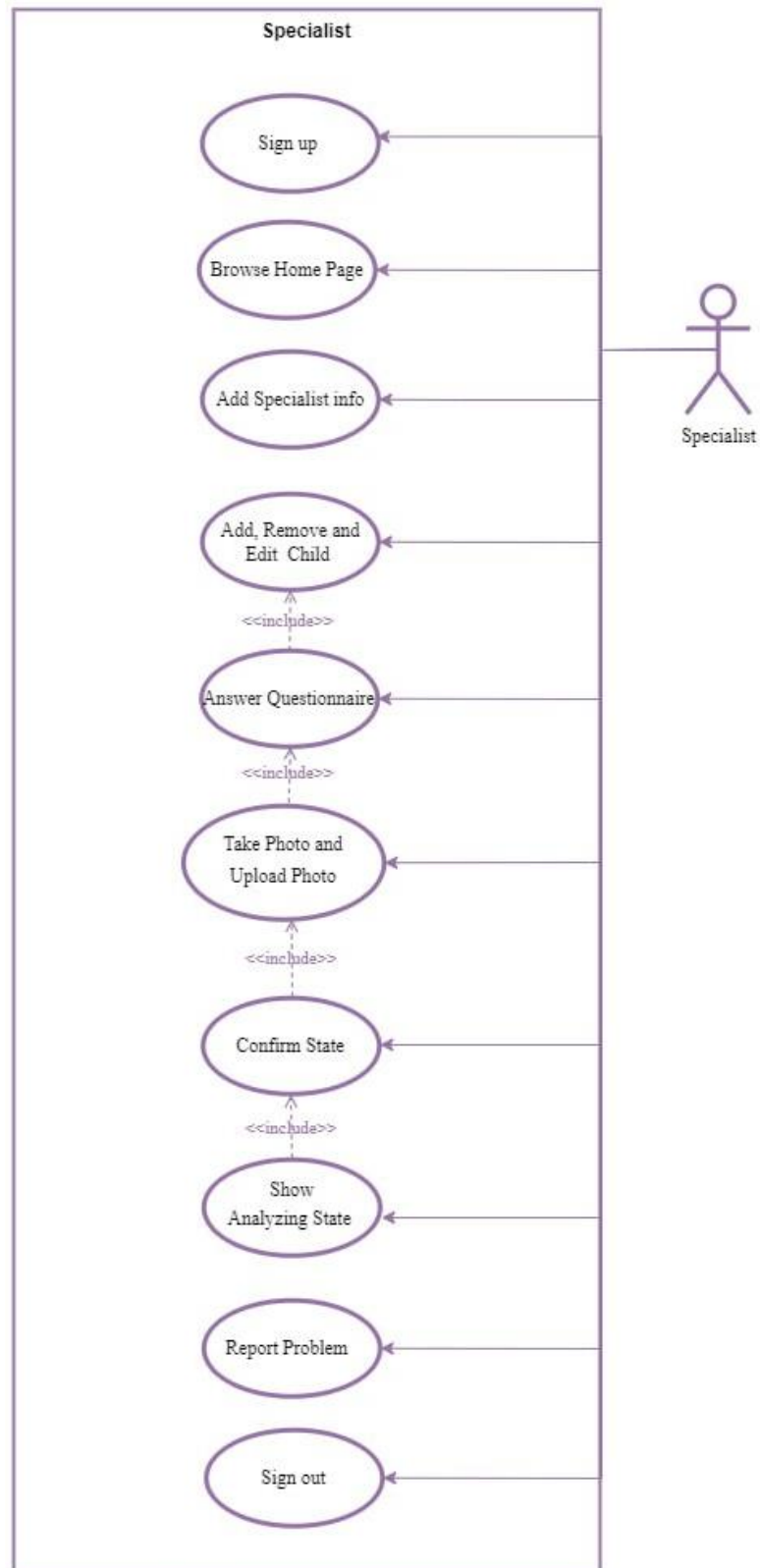


Figure 17: Use Case Diagram for Specialist

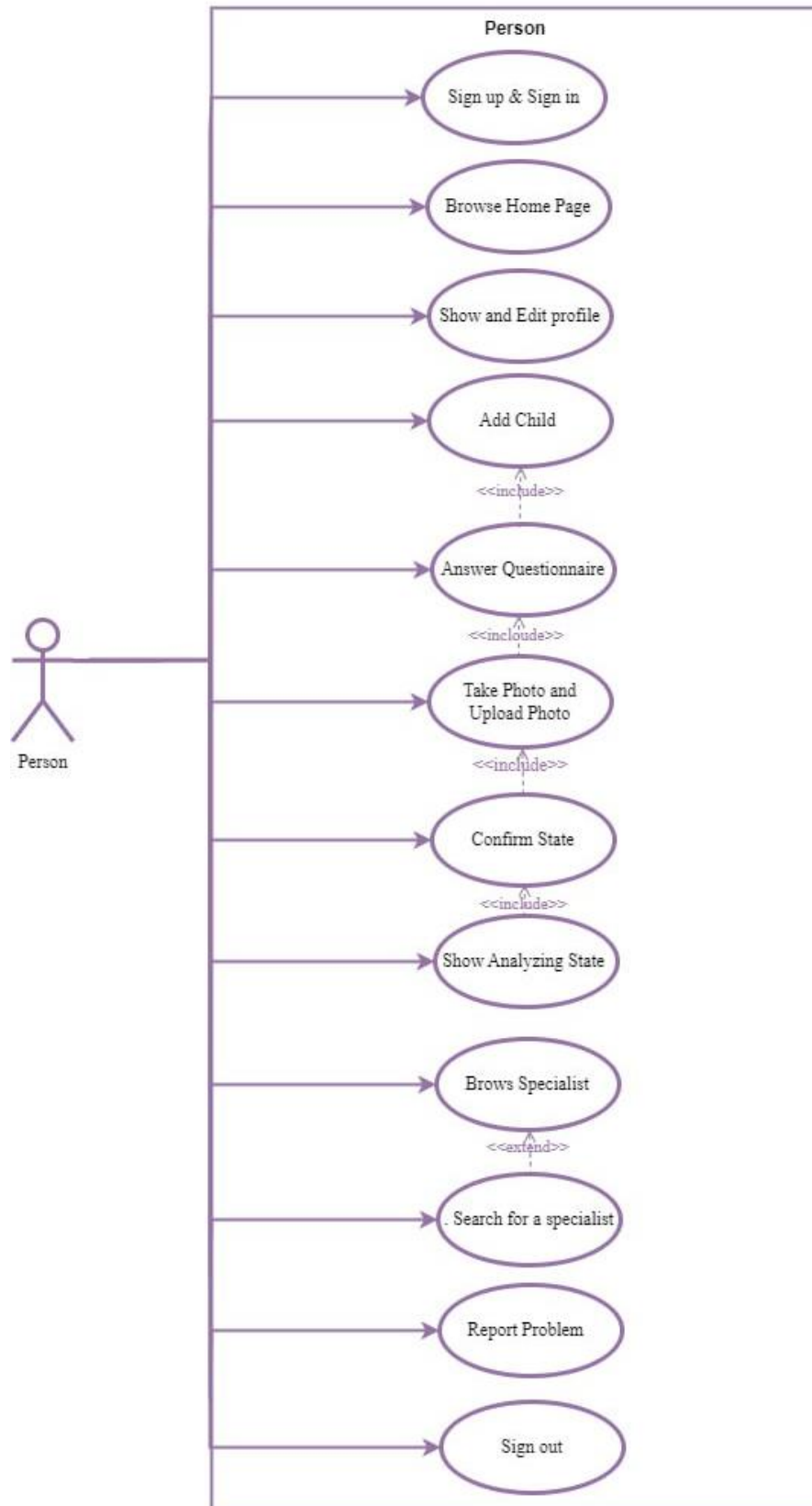


Figure 18: Use Case Diagram for Person

2.4 Scenarios

Sign Up	
Actors	Person, Specialist
Description	1. All users must create an account to access the system. 2. The user must enter all required personal information to sign up. 3. If the user is a specialist: 3.1 he/she must provide their license number to be approved. 4. The system will make sure the fields are filled properly. 4.1 The system will show a pop-up message if an error occurs. 5. The account will be successfully created and all the users information is stored in the data base.
Data	Person info: (Full name, Birth date, E-mail, Phone number, password) Specialist info: (Full name, Professional license number, Birth date, E-mail, Phone number, password).
Pre-condition(s)	To create a new person account the user's age must be 12 +.
Output	User's account is successfully created and the user can access the system.
Comments	None

Table 1: Sign Up Scenario

Verification	
Actors	Person, Specialist
Description	1. The user shall verify their account. 1.1 The system will send a verification code to confirm the user identity. 1.2 The system will ask the user to try again if the code is incorrect. 1.3 The system will save the account in the database and transfer the user to the home page if the code is correct. 1.3.1The system will show a pop-up message if an error occurs.
Data	Verification Code
Pre-condition(s)	The user has signed up.
Output	Account verified.
Comments	None

Table 2: Verification Scenario

Sign In	
Actors	Person, Specialist, Admin
Description	1.The user shall be able to Sign in. 2. The system will ask the user to enter their e-mail and password to sign-in 2.1 The system will make sure the e-mail and password are written correctly by comparing them with the ones registered in the database. 3. The system will transfer the user to the home page if the e-mail and password are correct.
Data	E-mail & Password.
Pre-condition(s)	The user has a signed up and verified account.
Output	System Access.
Comments	None

Table 3: Sign In Scenario

Forget Password	
Actors	Person, Specialist, Admin
Description	1. The user shall be able to change their password when they forget. 2. The system will ask the user to enter their Email. 3. The system will send a verification code to the user's Email. 4. The system will ask the user to enter a new password. 5. The system will save the new password in the database.
Data	Phone number & New Password.
Pre-condition(s)	The user has a signed up and verified account.
Output	New password.
Comments	None

Table 4:Forget Password Scenario

Browse Home Page	
Actors	Person, Specialist, Admin
Description	1.The system will display the user’s home page to access the possible permissions for each user. 2.The system will display the main function of the application to the user. 2.1 If the user was a person, then: the system will display Browse Analyzed State, add a child, communicate with specialists, search for a specialist, view the profile. 2.2. If the user is a specialist, then: the system will display Browse Analyzed State, add a child, view the profile, Register as a consulting specialist. 2.3 If the user is an admin, then: the system will display Check reported problem, Check specialist request.
Data	Person information / Specialist information.
Pre-condition(s)	The user has signed in a verified account.
Output	The ability to go to the page to add a child, view the profile, communicate with a specialist, or browse analyzed states. Register the specialist to appear to the users.
Comments	None.

Table 5: Browse Home Page Scenario

Show Profile	
Actors	Person, Specialist
Description	1.The user shall be able to view his/her profile from the application bar that will appear in the home page by clicking on the “user” icon. 2.The system will display the user’s profile page with the information in the database that contains (full name, date of birth, Email, children)
Data	Parent information /child information / specialist information.
Pre-condition(s)	The user has signed in a verified account.
Output	Retrieve and display personal data saved in the database.
Comments	None.

Table 6: Show Profile Scenario

Edit Profile	
Actors	Person, Specialist
Description	1.The user will be able to edit the profile after opening the profile view and click on the edit icon. 2.The system will display an alert to confirm saving the modification, 2.1 when the user presses OK then: 2.1.1The changes will be saved and modified directly in the database 2.2 When the user presses Cancel: 2.2.1 Modifications will not be saved.
Data	Parent information /child information / specialist information.
Pre-condition(s)	The user has signed in a verified account.
Output	Save modifications and update them in the database.
Comments	None

Table 7: Edit Profile Scenario

Add Child	
Actors	Person, Specialist
Description	1.The user must be able to add children to his/her account by clicking on “Add a child” in the home page. 2.The system will allow the user to enter the child's information and save it for future analysis. 3.The system will display a set of pre-diagnostic questions about the child.
Data	Child information
Pre-condition(s)	The user has signed in a verified account.
Output	Save the child's data in the database.
Comments	None

Table 8: Add Child Scenario

Edit Child	
Actors	Person, Specialist
Description	1. The user shall be able to Edit the child's information. 2. The system will allow the user to update the child's information whenever the user needs to.
Data	Child information
Pre-condition(s)	The user has signed in a verified account.
Output	Save the child's data in the database.
Comments	None

Table 9: Edit Child Scenario

View Child	
Actors	Person, Specialist
Description	1. The user shall be able to view their children. 2. The system will display the children's interface for the user which contains all the children's information when the user clicks on the “view children” button that exists both in the profile and the homepage.
Data	List of children.
Pre-condition(s)	The user has signed in a verified account.
Output	Retrieve and display Children data saved in the database.
Comments	None

Table 10: View Child Scenario

Remove Child	
Actors	Person, Specialist
Description	1.The user removes the child from his profile 2.The system will remove the child from the user's account when the user clicks on the "Remove Child" button.
Data	List of children.
Pre-condition(s)	The user has signed in a verified account.
Output	The child will be removed from the application.
Comments	None

Table 11: Remove Child Scenario

Answer Questionnaire	
Actors	Person, Specialist
Description	1. The system offers the user a questionnaire that must be answered. 2. The system will ask the user to answer the pre-questions of the child's family environment and other questions before the drawing analysis. 3. The system will present the prior questions as closed questions for specific and clear results.
Data	Questions and answers.
Pre-condition(s)	The user has registered a child.
Output	The user is transferred to the interface where he/she can take/upload the child drawing
Comments	None.

Table 12: Answer Questionnaire Scenario

Take Photo	
Actors	Person, Specialist
Description	1. The user takes a picture of the child's drawing to be analyzed 2. If the image is unclear, the user can re-capture it 3. The system will analyze the photo
Data	A picture of the child's drawing.
Pre-condition(s)	The user has registered a child.
Output	A photo the user took by clicking on the button "take a photo".
Comments	None.

Table 13: Take Photo Scenario

Upload Photo	
Actors	Person, Specialist
Description	1.The user upload a picture from the camera album by clicking on the "upload photo" button. 2.The system will analyze the photo.
Data	photo of the child's drawing
Pre-condition(s)	The user has registered a child.
Output	A photo the user uploaded by clicking on the button "upload photo".
Comments	None.

Table 14: Upload Photo Scenario

Confirm State	
Actors	Parent, Specialist
Description	1.The user makes sure of the result of the child's photo analysis by answering the post questionnaire 2.The system asks him/her to answer the post-child health questions about some disorders after analyzing the children's drawing. 3.The system presents the following questions as closed questions to obtain specific and clear results
Data	User Answer.
Pre-condition(s)	A photo of the child's drawing must be uploaded.
Output	The analyzation result of the child's drawing.
Comments	None

Table 15: Confirm State Scenario

Show Analyzing State	
Actors	Person, Specialist.
Description	1. The user will see the analyzation result of the child drawing after clicking on "Show analyzing result". 2. The system will display the analyzing state of the child drawing with the disorders after the post-questions
Data	None
Pre-condition(s)	A photo of the child's drawing must be uploaded, and pre-questions and post questions must be answered.
Output	The analyzed mental health and disorders information from the child picture will be shown
Comments	None.

Table 16: Show Analyzing State Scenario

Browse Analyzed State	
Actors	Person, specialist.
Description	1. The user will browse the pervious analyzed states after clicking on the "Browse analyzed state" button 2. The system will display a page for the user to see all the pervious analysis 3. The system will show a comparison of the disorders for all the pervious analysis.
Data	None
Pre-condition(s)	The user has a previous analyzed state.
Output	The pervious analyzed states of the child mental health and disorders will be shown in a comparison and with percentage
Comments	None

Table 17: Browse Analyzed State Scenario

Report Problem	
Actors	Person, Specialist
Description	1. The user wants to report a problem. 2. The system will display a form for the user to fill the fields.
Data	Subject of the message and a description.
Pre-condition(s)	None.
Output	The system displays a message says " Report has been sent".
Comments	None.

Table 18: Report Problem Scenario

Search for a Specialist	
Actors	Person
Description	1. The user will be able to search for a specialist. 2. The system will have search bar to help the user reach to a specific specialist.
Data	Specialist name.
Pre-condition(s)	None.
Output	The system displays specialist information to the user.
Comments	None.

Table 19: Search for a Specialist Scenario

Register as a consulting specialist	
Actors	Specialist
Description	1.The user shall be able to register as a consulting specialist. 2.The system will provide a form for the specialist to register as a consulting specialist for our application users. 3.If approved the system will display the specialist information and contacts after the analysis when user brows or when a user search for the specialist's name.
Data	Specialist name - Social media accounts - license number
Pre-condition(s)	The Specialist has signed in a verified account.
Output	The system records the specialist's information to be verified and accepted by the admin.
Comments	None.

Table 20: Register as a consulting specialist Scenario

Check specialist request	
Actors	Admin.
Description	1. The admin will be able to view and accept requests from specialists. 2. The system will display a list of newly added specialist's requests to be accepted into the application. 3. The system will allow the admin to accept or reject the request.
Data	specialist's requests.
Pre-condition(s)	The admin has signed in a verified account.
Output	The system will send a message of approval or disapproval to the specialists.
Comments	None.

Table 21: Check Specialist Request Scenario

Check Reported Problem	
Actors	Admin
Description	1. The admin will be able to view reported problems from users after clicking on the "check reported problem" button. 2. The system will display a list of submitted problems.
Data	Reported problems.
Pre-condition(s)	The admin has signed in a verified account.
Output	None
Comments	None

Table 22: Check Reported Problem Scenario

Sign-out	
Actors	Person, specialist.
Description	1. The user wants to sign-out. 2. The system will enable the user to sign-out after clicking the sign-out button.
Data	None.
Pre-condition(s)	The user has signed in a verified account.
Output	The user will sign-out of his profile
Comments	None.

Table 23: Sign out Scenario

2.5 Activity Diagram

The activity diagram illustrates the dynamic behavior of the Anamel system, showcasing how the system responds to events. It delineates the sequence of actions divided between the user and the Anamel system, commencing from input and proceeding towards the corresponding output.

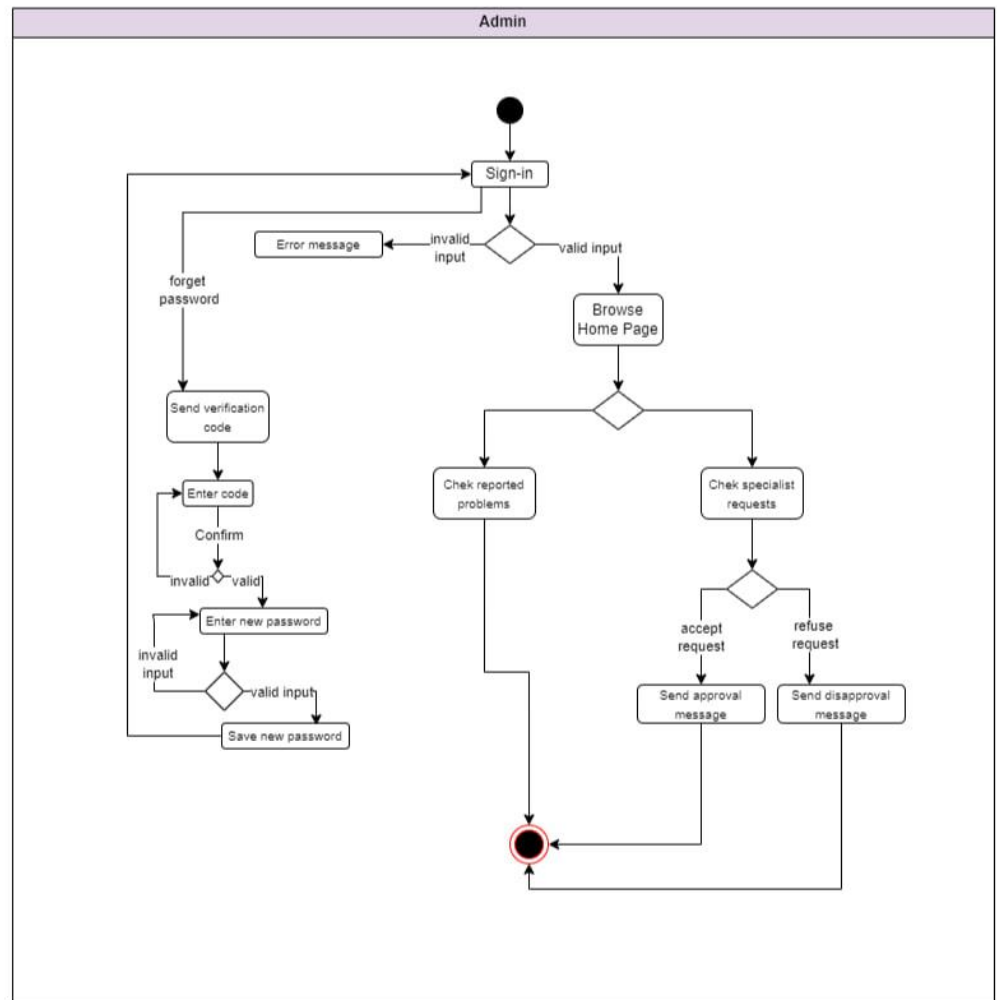


Figure 19: Anamel Activity Diagram

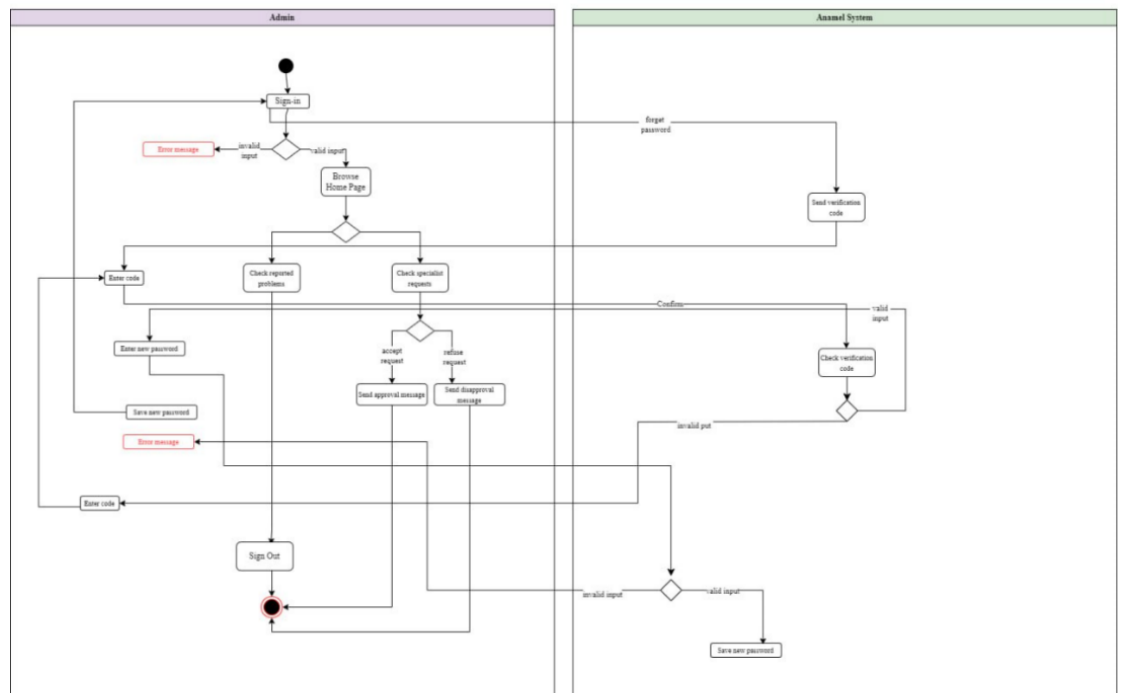


Figure 20: Admin Activity Diagram.

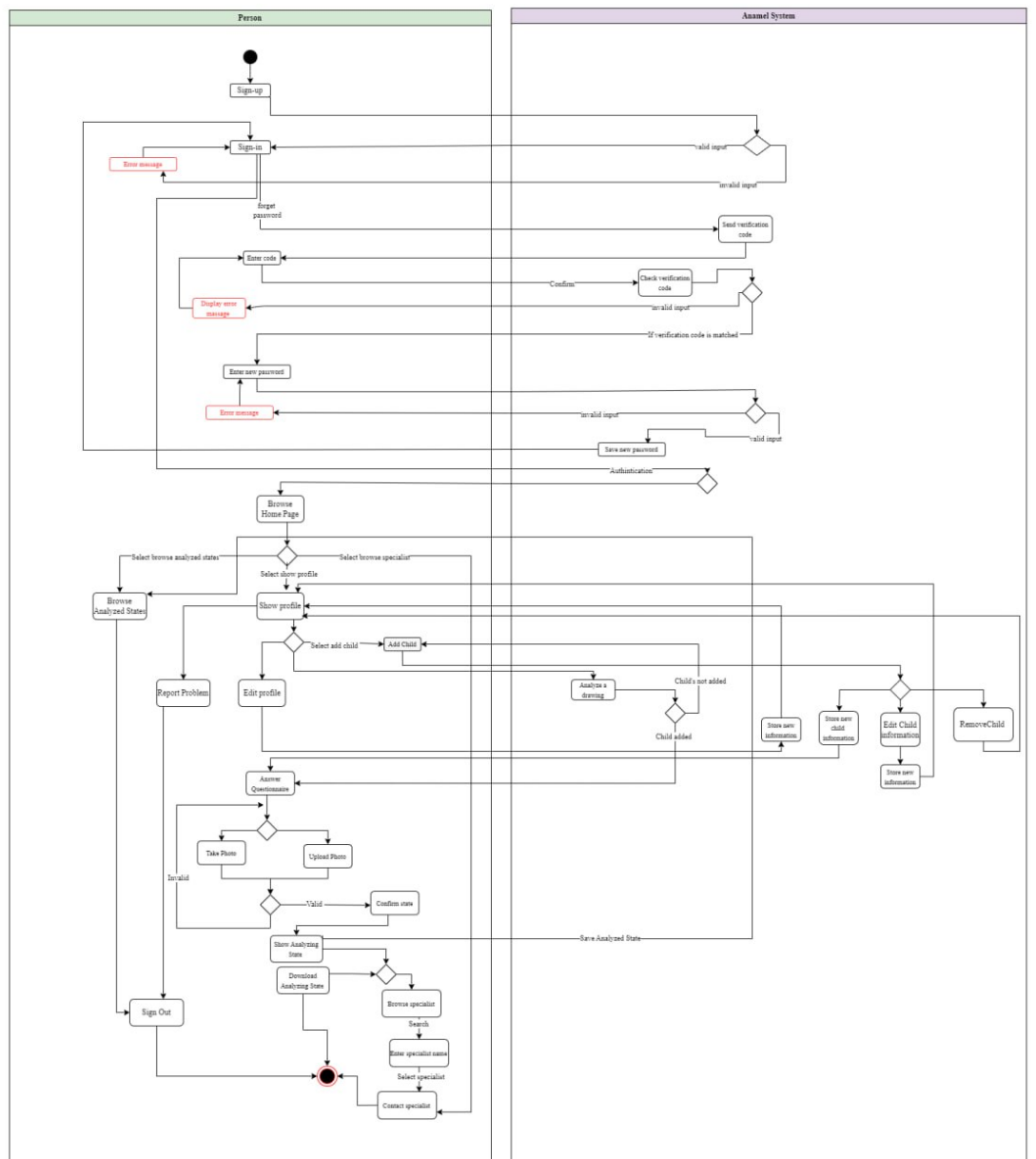


Figure 21: Person Activity Diagram

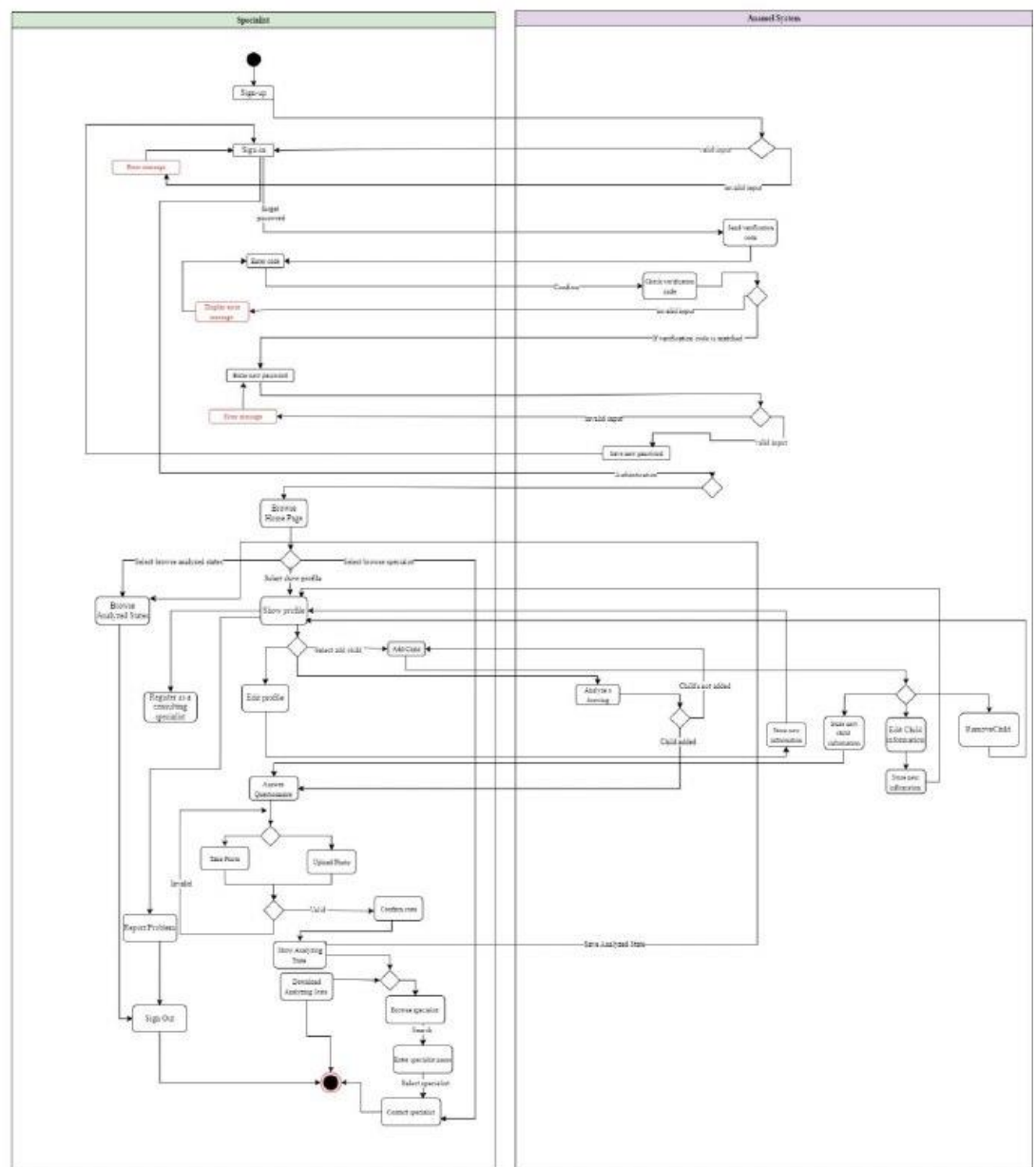


Figure 22: Specialist Activity Diagram

2.6 Sequence Diagram

The sequence diagram demonstrates interactions that take place throughout a use case instance between Anamel system components and actors.

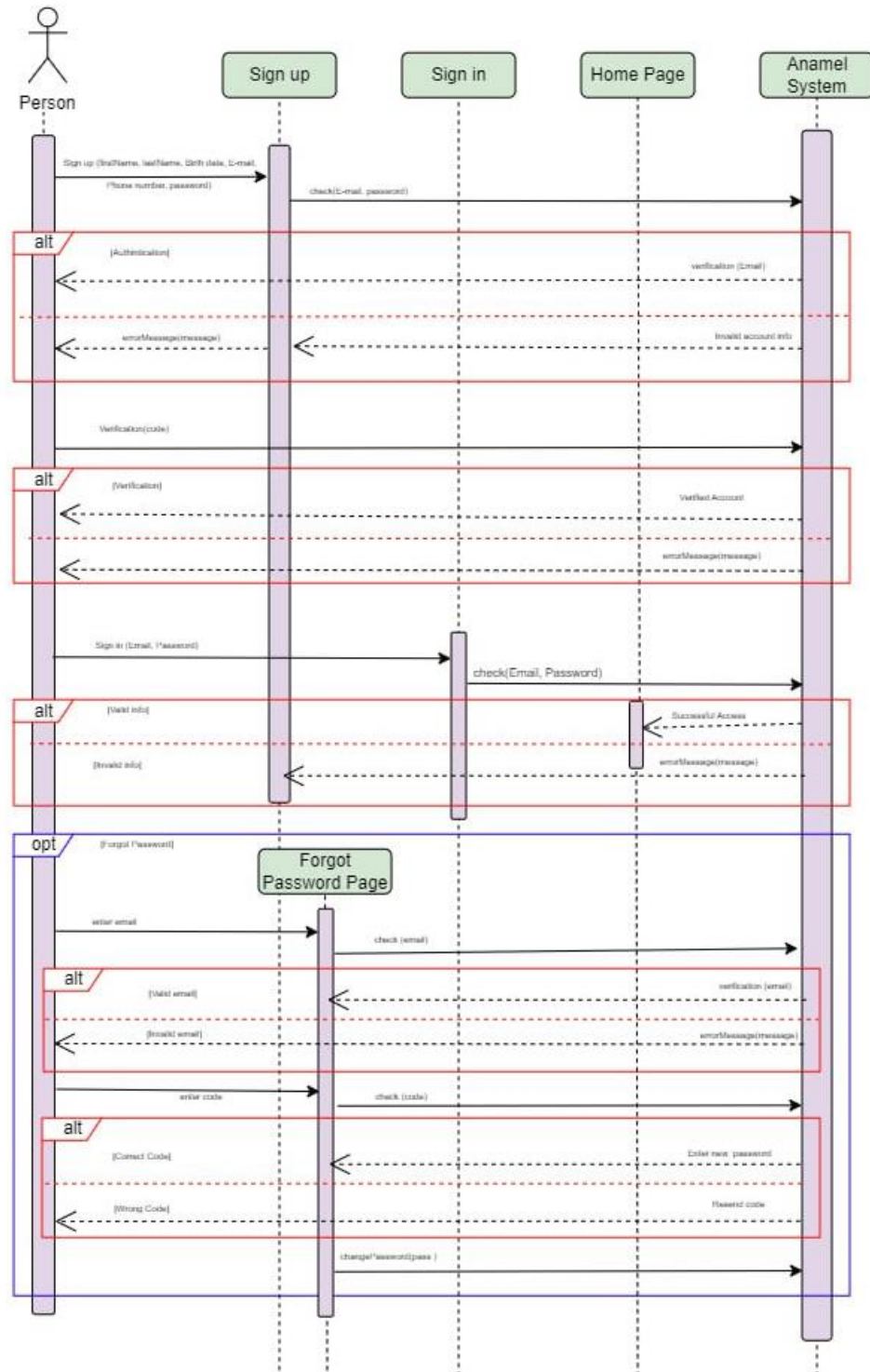


Figure 23: Person Sign Up and Sign in Sequence Diagram

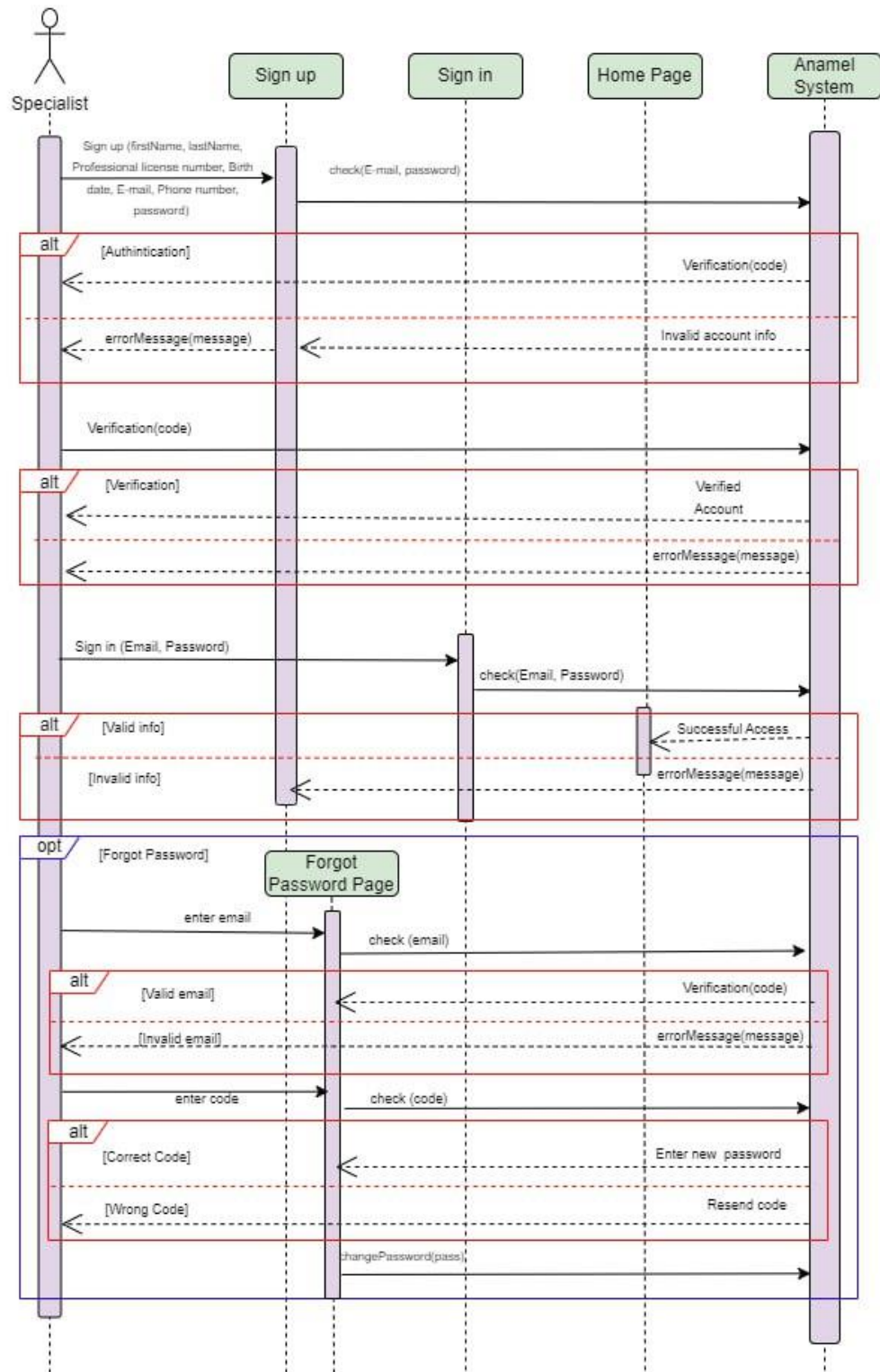


Figure 24: Specialists Sign Up and Sign in Sequence Diagram

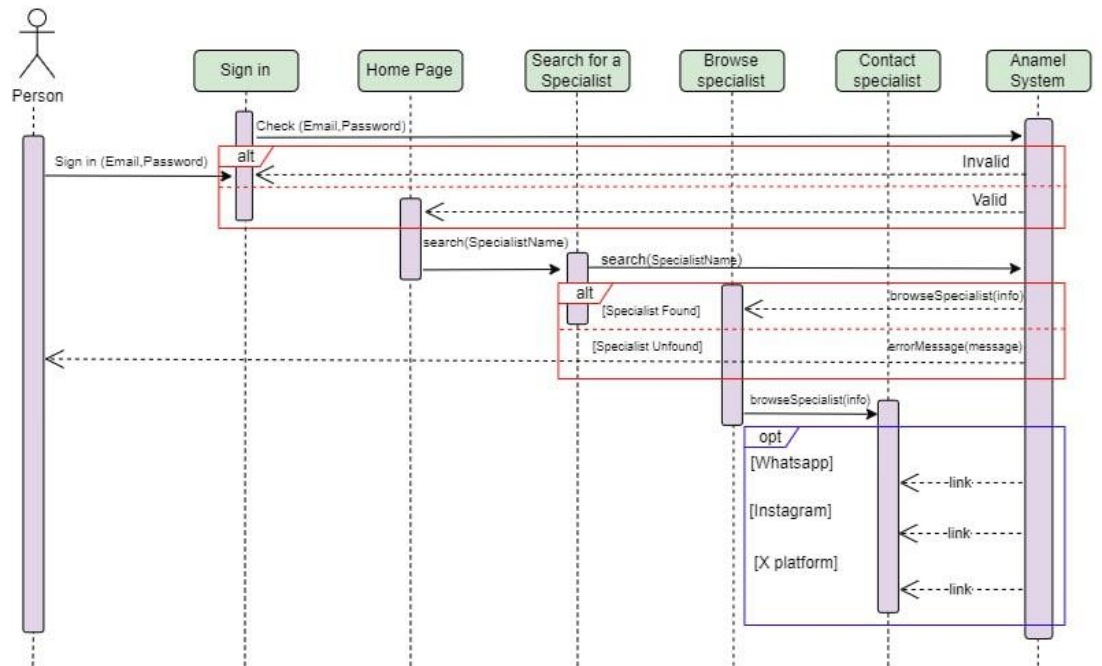


Figure 25: Person Search and Contact Specialist Sequence Diagram

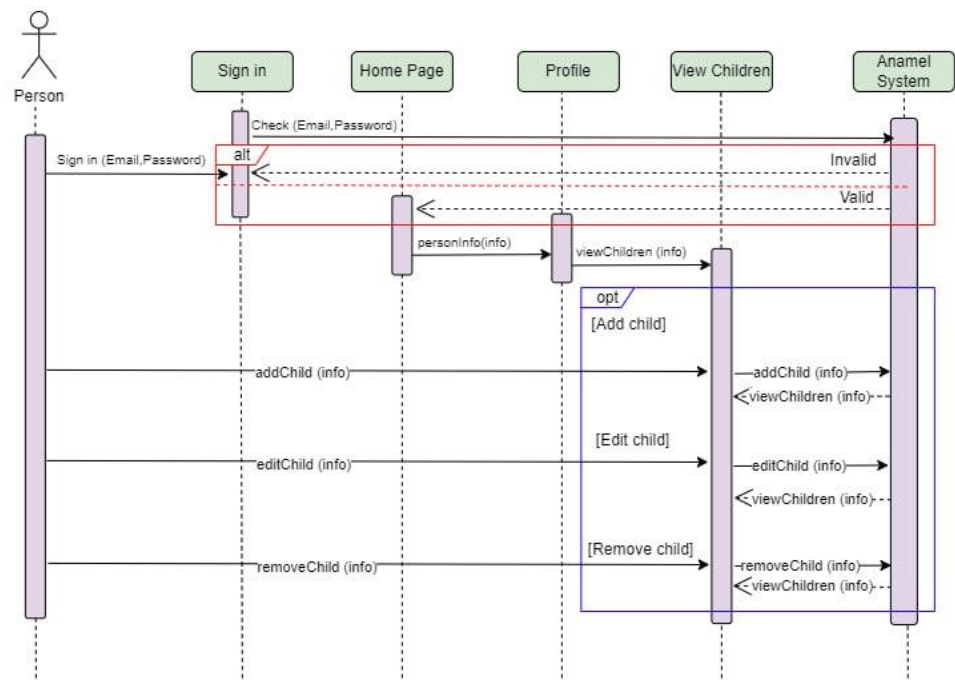


Figure 26: : Person Viewing and Modifying Children Sequence Diagram

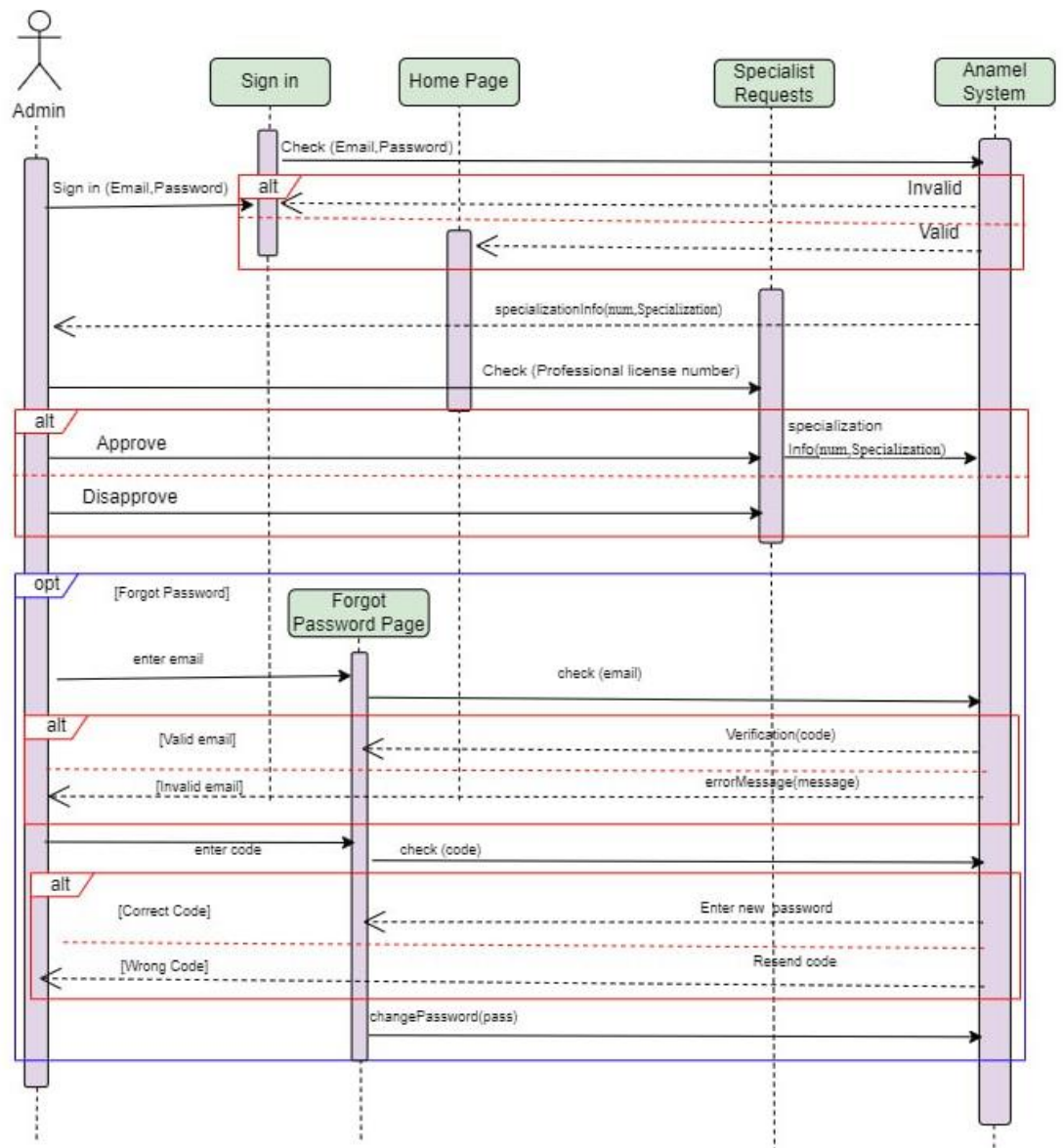


Figure 27: Admin Approving Specialist's Professional License Number Sequence Diagram

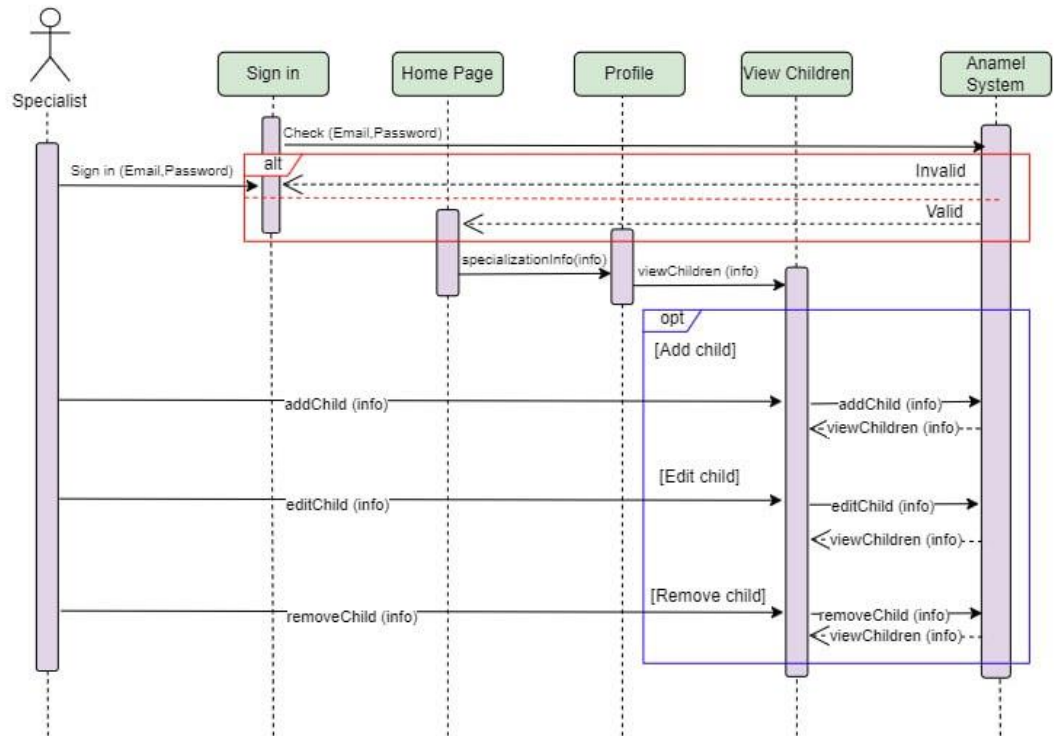


Figure 28: Specialist Viewing and Modifying Children Sequence Diagram

Chapter 2 System Analysis

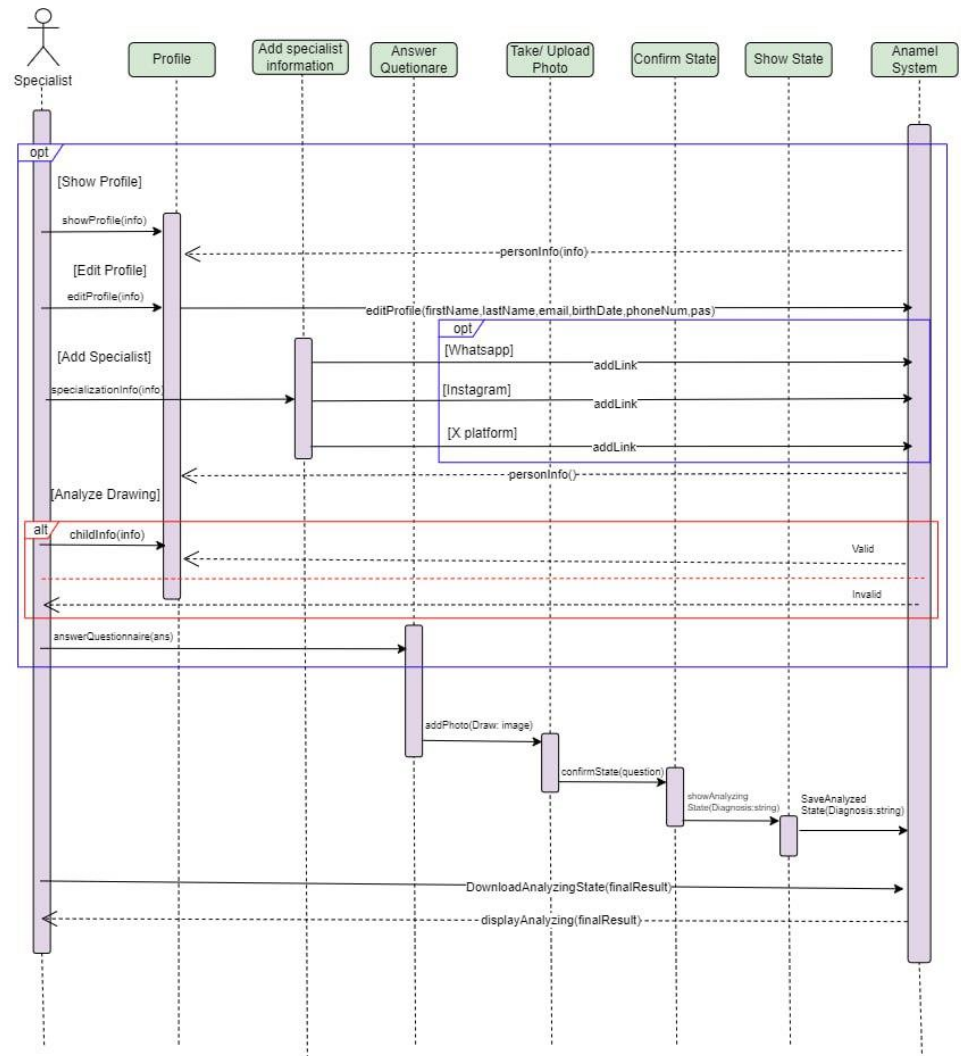


Figure 29: Specialist Editing Profile, Answering Questions, Attaching Drawing, Confirming, and Viewing Results System Sequence Diagram.

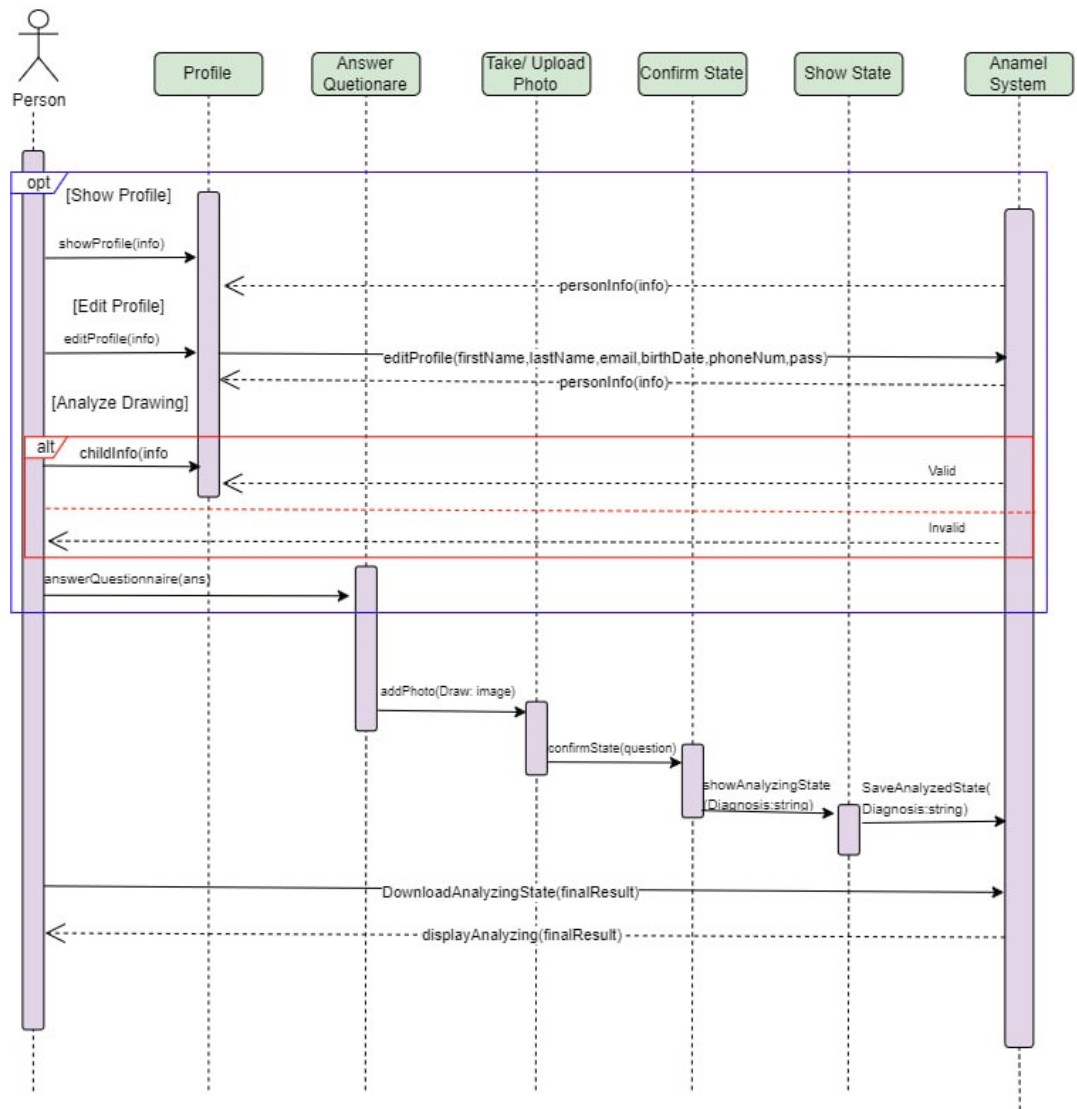


Figure 30: Person Editing Profile, Answering Questions, Attaching Drawing, and Viewing Results System Sequence Diagram.

2.7 Component Diagram

The component diagram demonstrates Anamel components and their dependencies on each other.

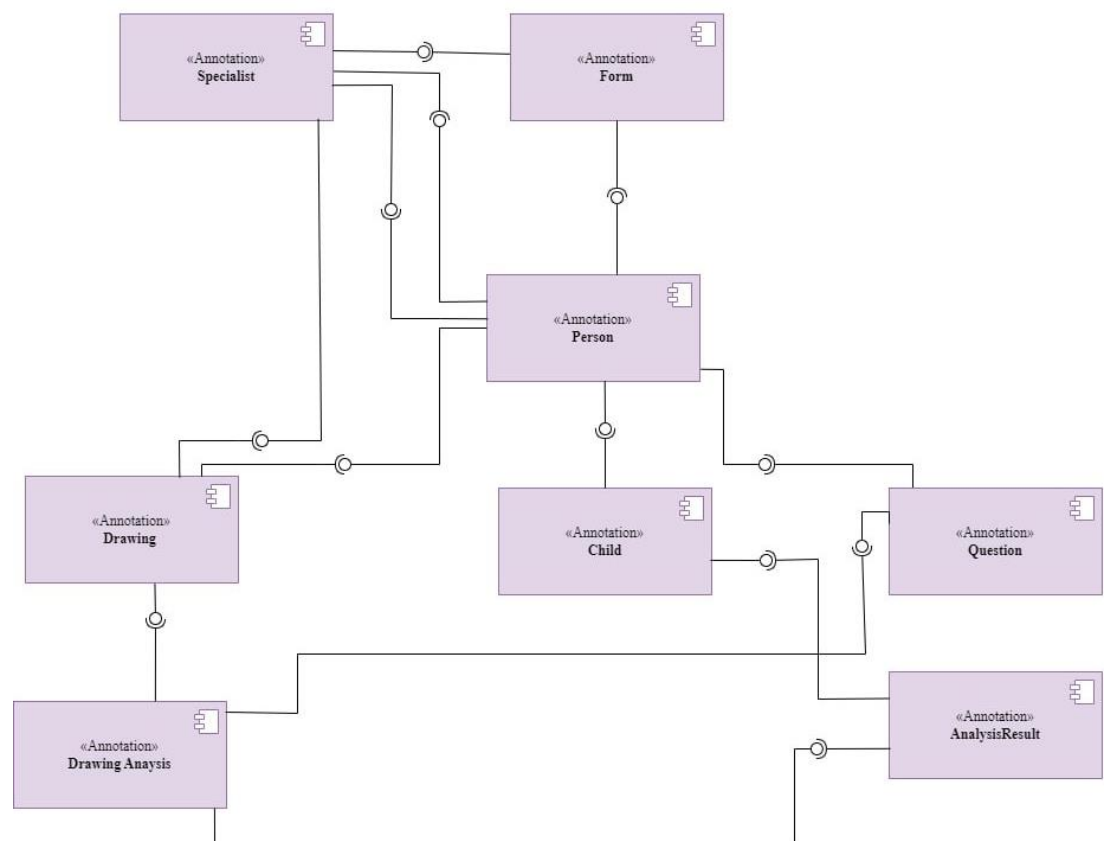


Figure 31: Anamel's Component Diagram.

2.8 Class Diagram

UML Diagram

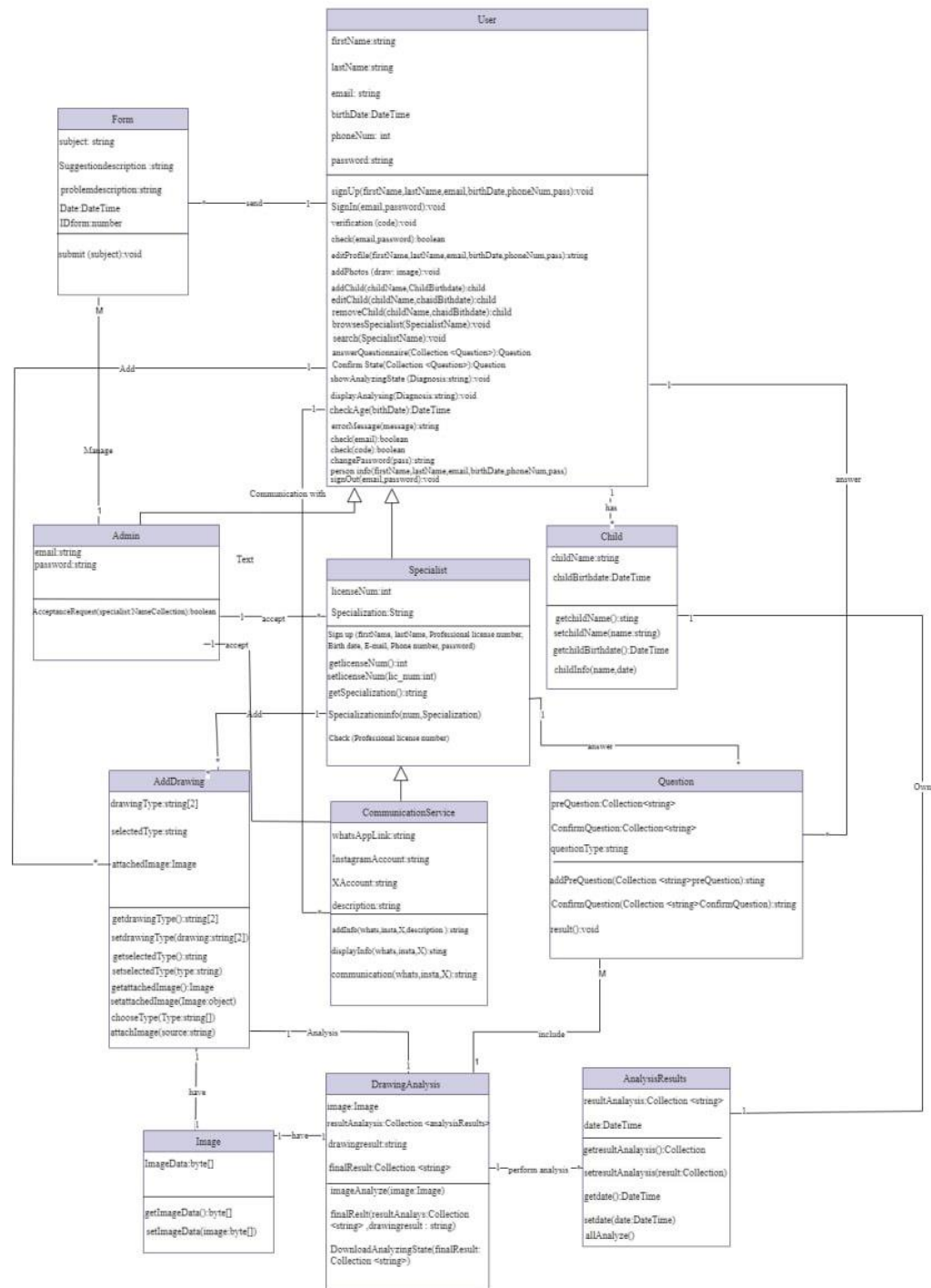


Figure 32: Anamel's Class Diagram.

2.9 Entity-Relationship Model Diagram

Entity-relationship model depicts the representation of a database's structures in the Anamel system, it displays the entity and the connection between them.

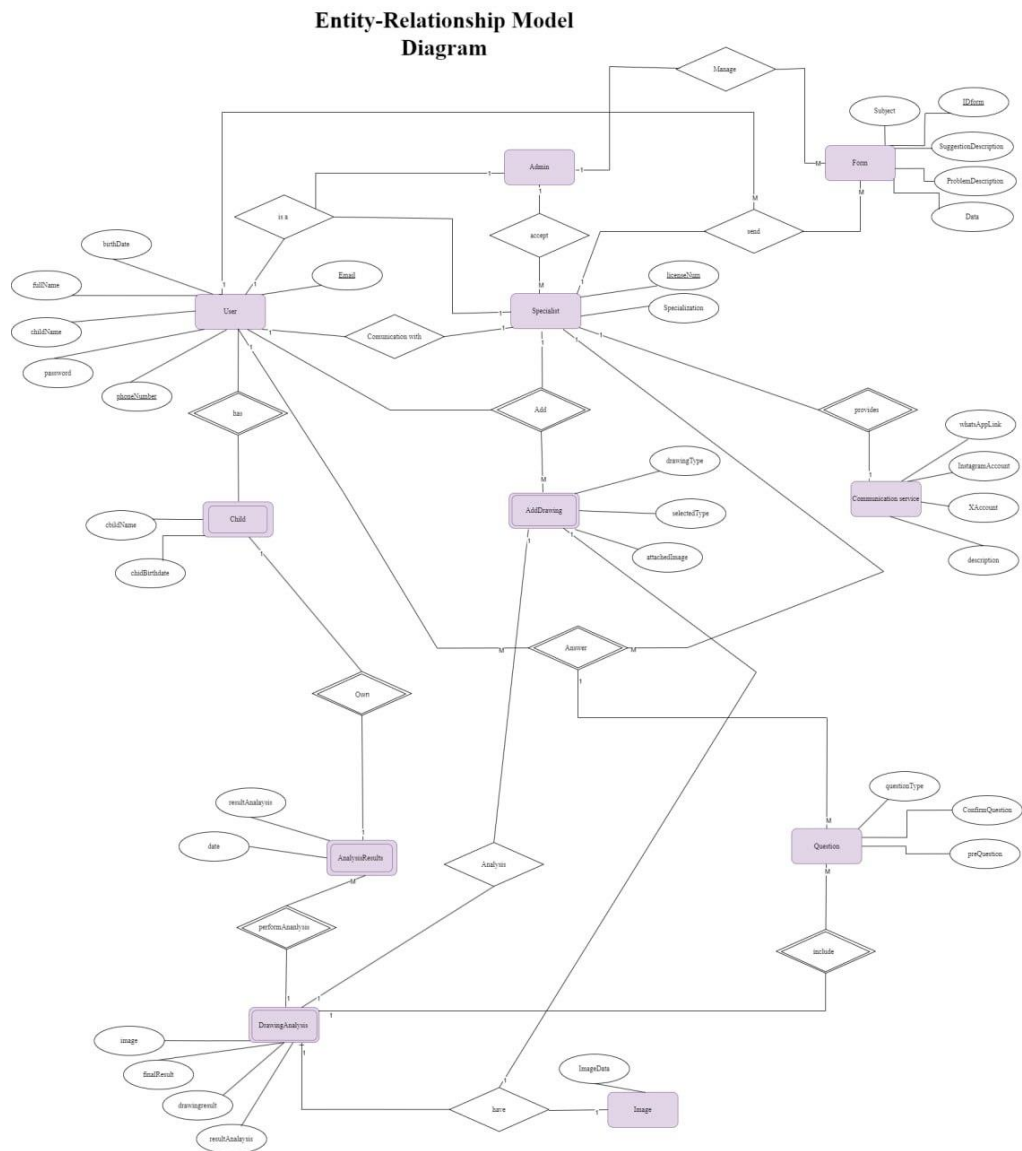


Figure 33: Entity-Relationship Model diagram

2.10 Proposed solutions

According to the interviews we conducted with specialists and the questionnaire we published to parents, we notice that they face many problems for both. Specialists face a longer time for diagnosis and they believe that the application will help them make an accurate diagnosis according to the approved clinical indications. While it is difficult for parents to understand their children's feelings and they see that the application provides them with assistance, it will be a personal assistant for both specialists and parents to understand their children's feelings using artificial intelligence.

2.11 Alternative Solutions

The initial diagnosis is made by psychologist first, by questions or by analysis of drawings before artificial intelligence.

Chapter 3 -Design Considerations

3.1 Design Constraints

3.1.1 Hardware Environment

3.1.2 Software Environment

3.1.3 End user characteristics

3.2 Architectural Strategies

3.2.1 Algorithm to be used.

3.2.2 Reuse of existing software components

3.2.3 Project management strategies

3.2.4 Development Method

3.2.4 Future enhancements/plans

3.1 Design Constraints

3.1.1 Hardware Environment

The system will be operated by four personal computers that work on Windows Operating System version 11, with RAM (Random Access Memory) ranging from 8 to 16 GB.

3.1.2 Software Environment

3.1.2.1 Utilizing Flutter for Application Development

For developing the application and its interfaces. Flutter, a UI toolkit by Google, offers a comprehensive framework for building natively compiled applications for mobile, web, and desktop from a single codebase. Its extensive widget catalog, reactive framework, and stateful hot reload feature significantly enhance the development experience.

3.1.2.2 Flutter Overview

Flutter provides a rich set of pre-designed widgets for creating beautiful, fast, and native user interfaces. With its layered architecture, Flutter allows for customization at every level, ensuring flexibility and scalability throughout the development process. Its reactive framework enables seamless updates to the UI in response to user interactions or data changes.

3.1.2.3 Integrating Firebase for Database Connectivity

In our project, we integrate Firebase for database connectivity, enhancing the application's backend capabilities. Firebase offers a comprehensive suite of services, including real-time database, authentication, cloud storage, and hosting. By utilizing Firebase with Flutter, we establish a robust backend infrastructure, facilitating data storage, retrieval, and synchronization across devices.

3.1.2.4 Firebase Features

Firebase's real-time database feature provides seamless data synchronization across clients in real-time, enabling collaborative and responsive applications. Additionally, Firebase Authentication ensures secure user authentication and authorization, while Firebase Cloud Firestore offers a scalable NoSQL database for storing and syncing app data at a global scale.

3.1.2.5 Implementing YOLO for Image Analysis:

Our project incorporates YOLO (You Only Look Once) for image analysis, utilizing versions 5 and 7 for advanced object detection and recognition. YOLO algorithms excel in real-time object detection tasks, providing high accuracy and efficiency. By integrating YOLO into our application, we empower the capability to analyze images accurately.

3.1.2.6 YOLO (You Only Look Once) Overview:

YOLO is a state-of-the-art, real-time object detection system that processes images in a single pass, making it incredibly fast and efficient. YOLO divides images into grids and predicts bounding boxes and probabilities for each grid cell, enabling simultaneous detection of multiple objects within the image.

3.1.3 End User Characteristics

The end users of the Anamel system are people over 12 years old, in particular parents, teenagers, and specialists. We target users who want to analyze their children's drawings, specialist doctors to diagnose their patient's drawings, or even teenagers who want to analyze their own drawings.

3.2 Architectural Strategies

3.2.1 Algorithm to be used:

For image analysis and recognition of shapes and symbols within images, we opt for a Convolutional Neural Network (CNN) approach. Specifically, we leverage YOLO (You Only Look Once) versions 5 and 7 for superior accuracy and efficiency.

- **CNN for Image Analysis:**

CNNs are a class of deep neural networks widely used in computer vision tasks, including image classification, object detection, and segmentation. CNNs employ convolutional layers to automatically learn hierarchical features from input images, enabling them to recognize complex patterns and structures. This makes CNNs particularly effective for tasks like shape and symbol recognition within images.

- **YOLO (You Only Look Once):**

YOLO is a state-of-the-art object detection algorithm known for its exceptional speed and accuracy. YOLO operates by dividing the input image into a grid and predicting bounding boxes and class probabilities for each grid cell simultaneously. Versions 5 and 7 of YOLO offer advancements in speed and accuracy, making them ideal choices for our image analysis requirements.

3.2.2 Reuse of existing software components

Through the reuse of existing software components such as Flutter, Python training models, and YOLO for image analysis, we aim to develop a robust and feature-rich application. By integrating these components and exploring API-based integration with other software programs, we ensure scalability, flexibility, and interoperability, ultimately delivering a versatile solution that meets the diverse needs of our users and stakeholders.

3.2.3 Project Management Strategies

A Gantt chart was used to visualize the project management strategy, where tasks are distributed over 10 consecutive weeks.

Task	Week									
	1	2	3	4	5	6	7	8	9	10
Abstract/ proposal										
Chapter 1: Introduction										
Chapter 2: System Analysis										
Chapter 3: Design Considerations										
Chapter 4 – System Design										
Review										
Prototype										
Review and audit										

Figure 34: Project Plan

3.2.4 Development Method

- **Requirements and Analysis:**

The first step in our Software Engineering life cycle involves understanding users' needs through personal interviews with psychologists and drawing analysts. This information will be used to draw up comprehensive application requirements.

- **System Design:**

Once the requirements are collected, the focus shifts to planning and designing the system architecture. This phase involves creating diagrams and design documents that define how the system will function.

- **Data Collection and Preparation:**

Following the design phase, attention is turned to gathering and counting data for the drawings. This is a crucial step to ensure that the system has the necessary information for effective implementation.

- **Implementation with Detection and Analysis Algorithms:**

The next phase involves the actual implementation of the system. Detection and analysis algorithms, including the CNN algorithm and other detection algorithms, are applied. Models will be trained on the collected data, aligning with the designed system architecture.

- **Application, Testing, and Maintenance:**

The final phase encompasses the implementation of the application, rigorous testing to ensure functionality and ongoing maintenance. This ensures that the system operates as intended and can adapt to future needs.

3.2.4 Future enhancements/plans

The current Anamel system is used to analyze drawings of children and teenagers by answering specific questions and uploading a picture of the drawing. However, in the future, we plan to train the model to analyze the child's movements and expressions while drawing, and that's by allowing the user to upload a video of the child. Additionally, we also plan to use NLP to find and analyze any letters or text written by the child, such as: family members or friends names. Finally, we will work to expand the target users and include all ages.

Chapter 4 – System Design

4.1 System Architecture and Program Flow

4.2 Detailed System Design

4.2.1 Detailed component description

Note : [Click on the Diagram Name for a Closer View](#)

4.1 System Architecture and Program Flow

This diagram shows the system's interconnection with users, its connection to the Internet, and its connection with various APIs to integrate and integrate the system.

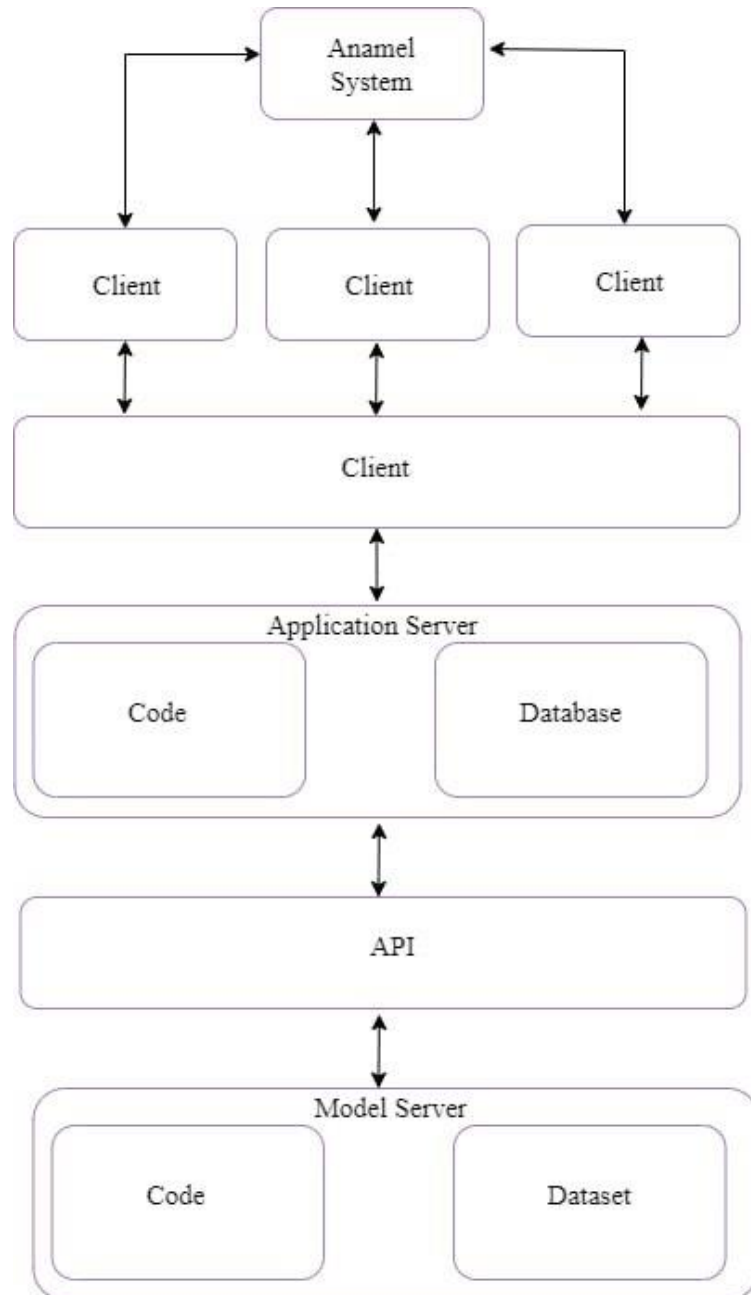


Figure 35: Anamel's System Architecture Diagram

4.1.1 Model – View – Controller Diagram

MVC (Model-View-Controller) is a software architectural style for developing user interfaces in applications. It consists of three main components:

- **Model:** Represents application data and handles data processing and management, as well as defining the application's functionality. In the Anamel system, the model would include database tables.
- **View:** Represents the user interface components. It receives user input and forwards it to the controller for processing. In the Anamel system, the view encompasses all application interfaces.
- **Controller:** Serves as a mediator between the model and the view. It receives user input from the view, processes it, updates the model, and retrieves data from the model to update the view with the latest information.

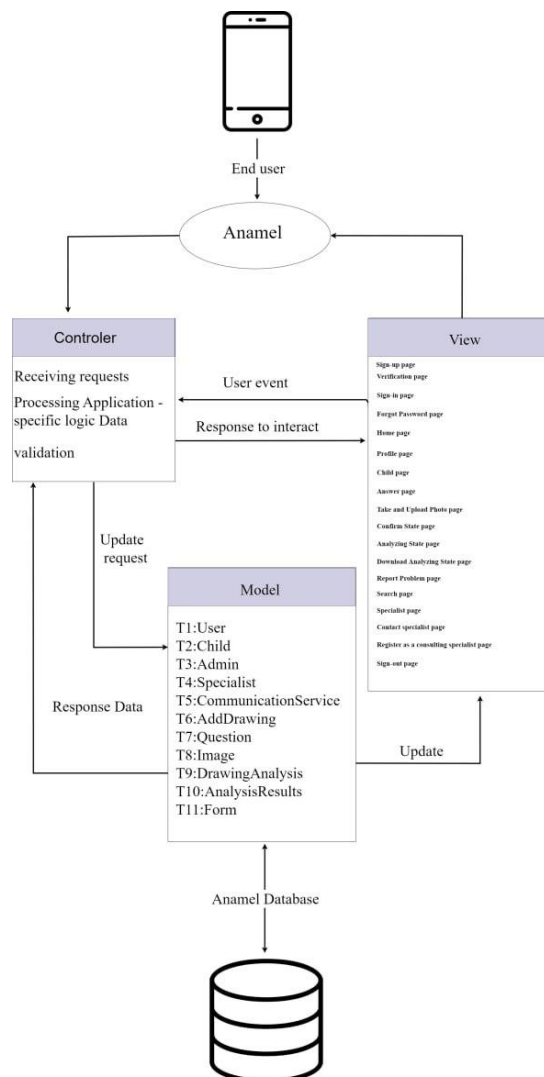


Figure 36: MVC Diagram

4.1.2 Methodology Diagram

The "methodology diagram" is a representation that explains the mechanism of operation of the artificial intelligence model and the approach used to implement it from algorithms in a simplified manner, with an explanation of the training and testing of the dataset according to certain percentages out of 100%.

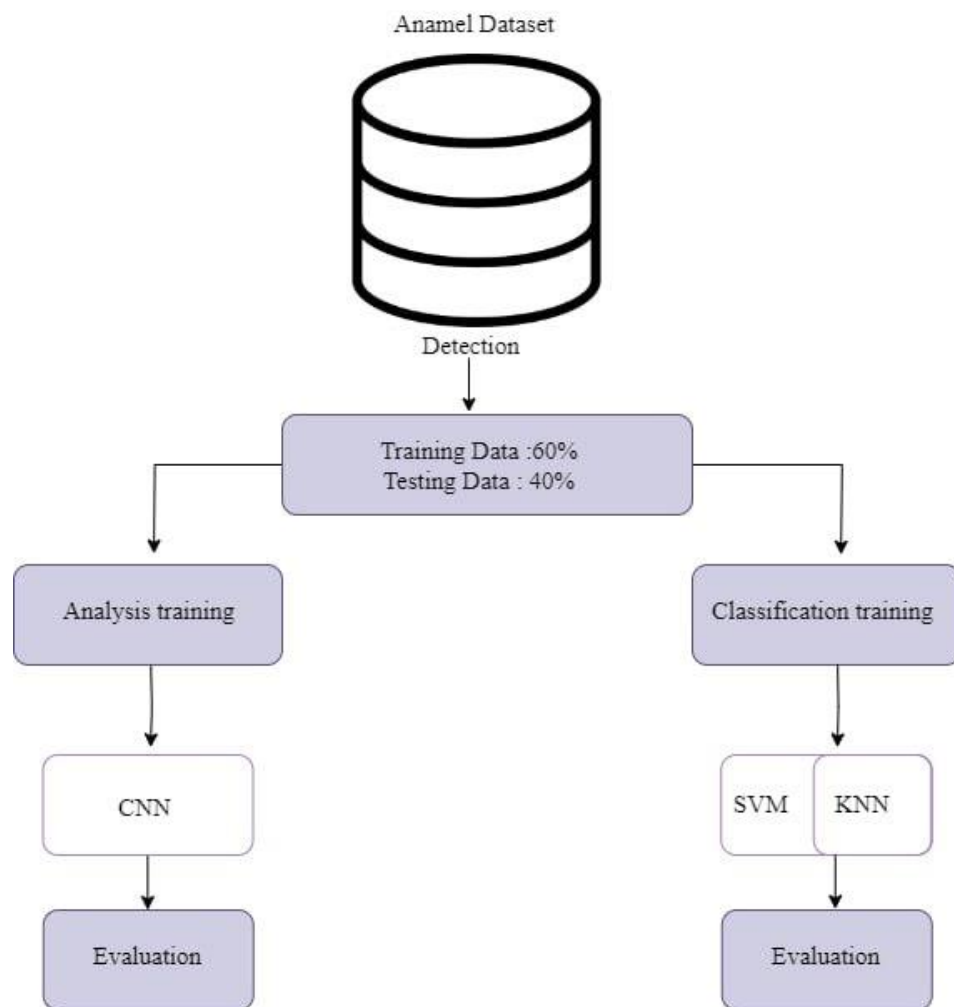


Figure 37: Methodology Diagram

4.2 Detailed System Design

4.2.1 Detailed component description

In this section, we will outline the Prototype and Initial Interface Division, where the prototypes of the application's administrative interface and users will be presented. This division serves as the foundation for the subsequent development stages, providing a structured approach to designing and implementing key components of the system.



Figure 38: Anamel's loading screen and Sign in/Sign up

The image displays two mobile application screens for user authentication, presented side-by-side. Both screens have a teal header bar with white text.

Left Screen: تسجيل الدخول (Sign In)

- Header: تسجيل الدخول
- Form fields:
 - Dropdown: نوع المستخدم (User Type)
 - Text: رقم الرخصة المهنية (Professional License Number) with a license icon
 - Text: كلمة المرور (Password) with a lock icon and a toggle for 'هل نسيت كلمة المرور?' (Forgot password?)
- Button: تسجيل الدخول (Sign In)
- Text: ليس لديك حساب؟ انشئ حساب جديد (Don't have an account? Create a new account)
- Icons: Microsoft, Email, and Google

Right Screen: تسجيل جديد (Sign Up)

- Header: تسجيل جديد
- Form fields:
 - Dropdown: نوع المستخدم (User Type)
 - Text: الاسم الاول (First Name) with a person icon
 - Text: الاسم الاخير (Last Name) with a person icon
 - Text: رقم الرخصة المهنية (Professional License Number) with a license icon
 - Text: رقم الهاتف (Phone Number) with a phone icon
 - Text: كلمة المرور (Password) with a lock icon
 - Text: التأكيد من كلمة المرور (Confirm password) with a lock icon
- Button: تسجيل (Sign Up)
- Text: لديك حساب؟ تسجيل الدخول (Have an account? Sign in)

Figure 39: Specialist's Sign in/Sign up screens.

The image displays two mobile application screens for user authentication, presented side-by-side. Both screens have a teal header bar with white text.

Left Screen: تسجيل جديد (Sign Up)

- Header: تسجيل جديد
- Form fields (all with light pink backgrounds and rounded corners):
 - Dropdown: نوع المستخدم (User Type)
 - Text: الاسم الأول (First Name) with a person icon
 - Text: الاسم الأخير (Last Name) with a person icon
 - Text: تاريخ الميلاد (Date of Birth) with a calendar icon
 - Text: البريد الإلكتروني (Email) with an envelope icon
 - Text: رقم الهاتف (Phone Number) with a phone icon
 - Text: كلمة المرور (Password) with a lock icon and an eye icon for visibility toggle
 - Text: التأكد من كلمة المرور (Confirm Password) with a lock icon and an eye icon for visibility toggle
- Buttons:
 - Teal button: تسجيل (Sign Up)
 - Text: لديك حساب؟ تسجيل الدخول (Have an account? Sign in)

Right Screen: تسجيل الدخول (Sign In)

- Header: تسجيل الدخول
- Form fields (all with light pink backgrounds and rounded corners):
 - Dropdown: نوع المستخدم (User Type)
 - Text: البريد الإلكتروني (Email) with an envelope icon
 - Text: كلمة المرور (Password) with a lock icon and an eye icon for visibility toggle
- Text: هل نسيت كلمة المرور؟ (Forgot your password?) with a toggle switch.
- Buttons:
 - Teal button: تسجيل الدخول (Sign In)
 - Text: ليس لديك حساب؟ انشئ حساب جديد (Don't have an account? Create a new account)
- Footer: Three circular icons for social login: Microsoft, Email, and Google.

Figure 40: Person's Sign in/Sign up screen



Figure 41: Forget password screen.



Figure 42: Anamel's Home Page and Children's page.

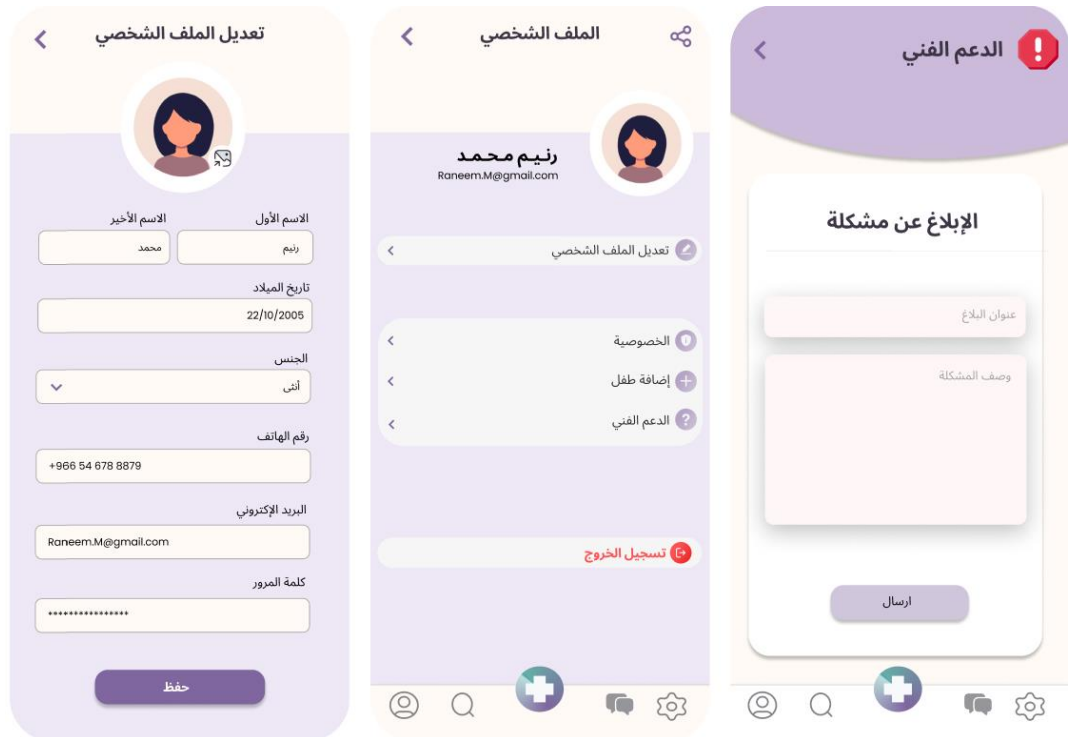


Figure 43: : Anamel's Profile Page.



Figure 44: Anamel's Pre-questions page.



Figure 45: Anamel's Upload/Take photo and Analyze drawing screen.

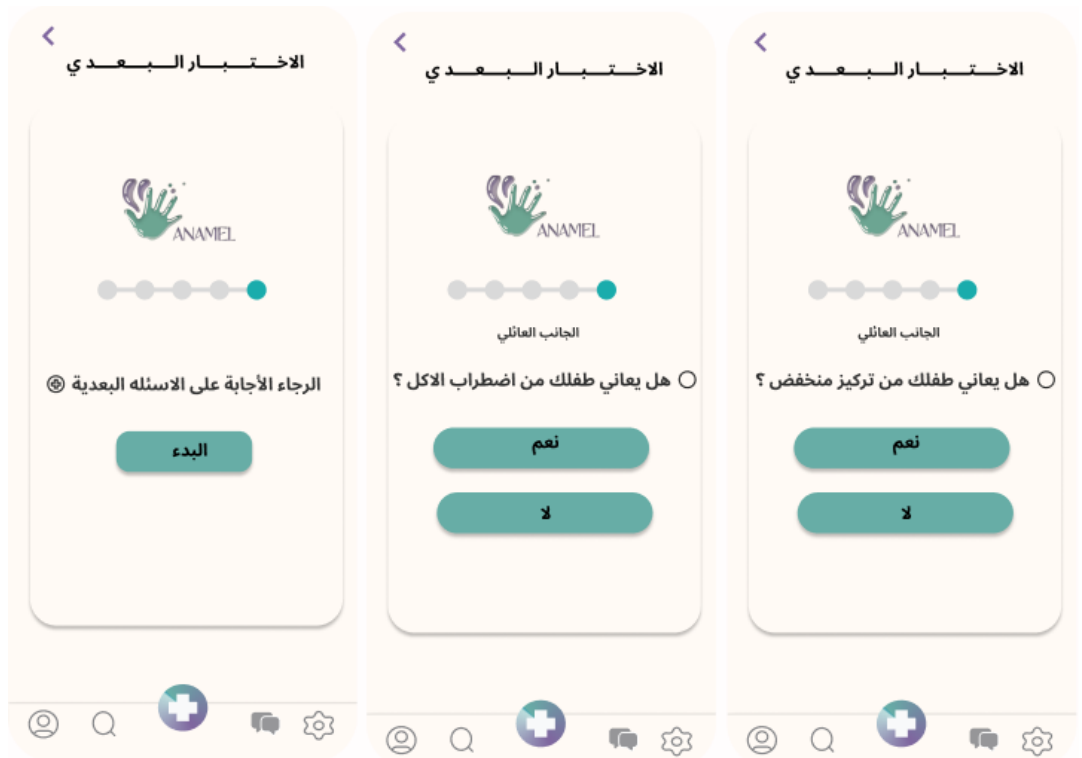


Figure 46: Anamel's Post-questions page.



Figure 47: Anamel's Results and Contact Specialist page.



Figure 48: Specialist Profile and Register as a Consultative Screens

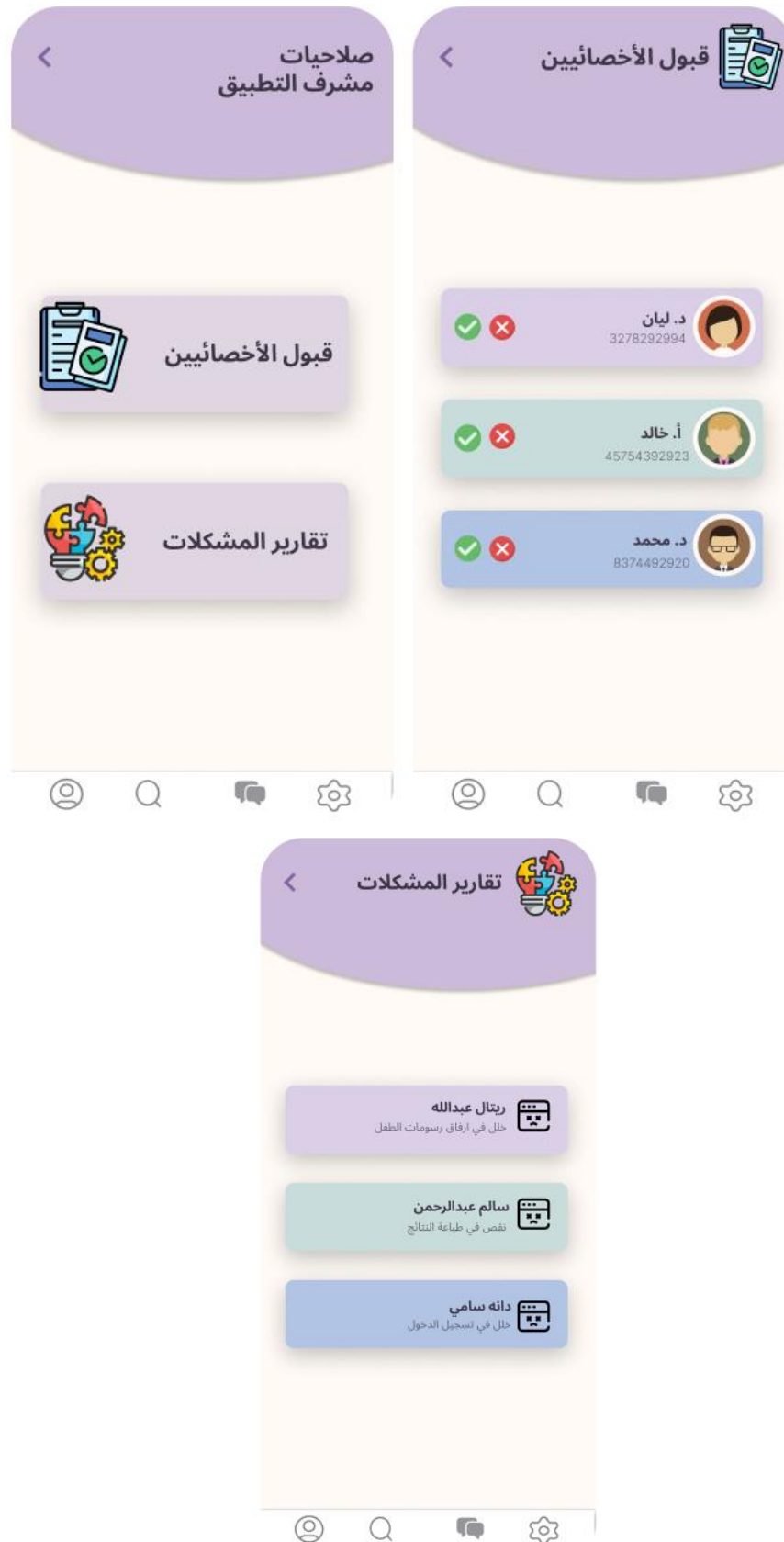


Figure 48: Specialist Profile and Register as a Consultative Screen

Chapter 5 – System Implementation

5.1 Implementation Plan

5.1.1 Implementation Approach and Decisions

5.1.2 Implementation Tools and Environment

5.2 Technical Work

5.2.1 Input/Output Design Implementation

5.2.2 Database Implementation

5.2.3 Algorithm used

5.2.4 Code Module (Key parts)

5.2.5 System Implementation

5.3 Building AI model

5.4 Implementation Issues/Challenges faced

5.1 Implementation Plan

This chapter explains the steps followed by the team to implement the project's idea, from the artificial intelligence model to implementing the whole application. The chapter also provides screenshots of code snips used in the implementation process and a short explanation of each step.

5.2 Introduction

At this stage, the team worked on three steps as follows:

- Create appropriate environment to allow team members to work in parallel.
- Work on the artificial intelligence model in Google Colab.
- Add team members to the project.
- Create an Anamel database in Firebase (Cloud Firestore)
- Create the application in Visual Studio Code and connect to the AI model.

1-Building the application:

- **Frontend**

- Design the interfaces using Figma.
- Develop the designs using Flutter & Dart language.
- Improve the application's interfaces to become more interactive and user-friendly.

- **Backend**

- Connect the database with the application.
- Create project schedules.
- Write CURD operations functions.

2-Build the artificial intelligence model:

- Collect the dataset.
- Preprocess the data.
- Train the model.
- Evaluate the model.
- Analyze the results.
- Connect the model to the application.

5.1.2 Implementation Approach and Decisions

- **Dataset**

After we finished the previous 4 chapters, we started collecting the dataset from children, where we asked them to draw a family drawing and a free drawing. The family drawing has always contained a clear idea about how the child currently felt. Due to that, it was very helpful for our model.

- **Model**

To find a suitable model for our idea we have read many objects detection and classification papers and we choose to instigate in ResNet50 and YOLO with its versions 8 and 9 as they are widely used in objects detection and classification in images.

- **Classification**

To classify the Anamel dataset for the training, and after meeting with specialists we classified the dataset into four class: happy, sad, anxiety, anger and aggression. Those are the most emotions seen in the children's drawings, and the objects that can be detect in these emotions are a lot which will help in the training. The classification of the dataset done by annotating them in Roboflow[16].

- **Training**

For the training we used the pretrained ResNet50 and YOLOv9 objects detection models and trained them on the Anamel dataset. ResNet50 achieved a 0.70 accuracy rate and YOLOv9 achieved 0.72 (details in section 5.3). As the accuracy rate was not that satisfying, we tried to use YOLOv8 for classification the Anamel dataset and the accuracy rate showed a great improvement where it achieved 0.94 accuracy rate (details in section 5.3).

5.1.3 Implementation Tools and Environment

The destinations are designed to fully visualize the progress of the project, then later implement the code and model, using many tools such as:

- **Figma**

Figma is a collaborative web application tool that helps developers design and prototype user interfaces. This performance creates a clear understanding of how to build the application more realistically. Figma also helps ensure that all details of the application are consistent with reality when starting the implementation programmatically.

- **Flutter**

Flutter platform is an open-source software development kit (SDK) created by Google for building native mobile, web, desktop, and applications that run on Android and iOS systems, which makes it easier for developers to put in a lot of effort and work. Flutter is also characterized by speed, ease, and flexibility because it is built in DART programming language.

- **IDE (Visual Studio Code)**

Visual Studio Code is a source code editor by Microsoft that can run on your desktop. It is designed to support a wide range of programming languages and provides all the developer's needs. It is also lightweight but powerful and is available on Windows, macOS, and Linux operating systems.

- **Colab**

Google Colaboratory (Colab) is a free tool from Google for teaching and researching machine learning models. It also provides valuable resources such as GPU and Python libraries. Moreover, Colab is considered a Jupyter Notebook service, so there is no need to set up to use it. It also allows you to store your work in Google Drive.

- **Database**

Firebase is a cloud computing platform by Google, specifically designed for mobile and web applications. It is classified as a NoSQL database, as it stores data in a document. It also provides several services to developers that enable them to easily develop, build, manage, and scale applications with high flexibility.

5.2 Technical Work

5.2.1 Preparing the environment

A folder for the AI model has been uploaded to drive. Then the link to the folder was shared with the team members to work on it in Google Lab, as shown in Figure 49.

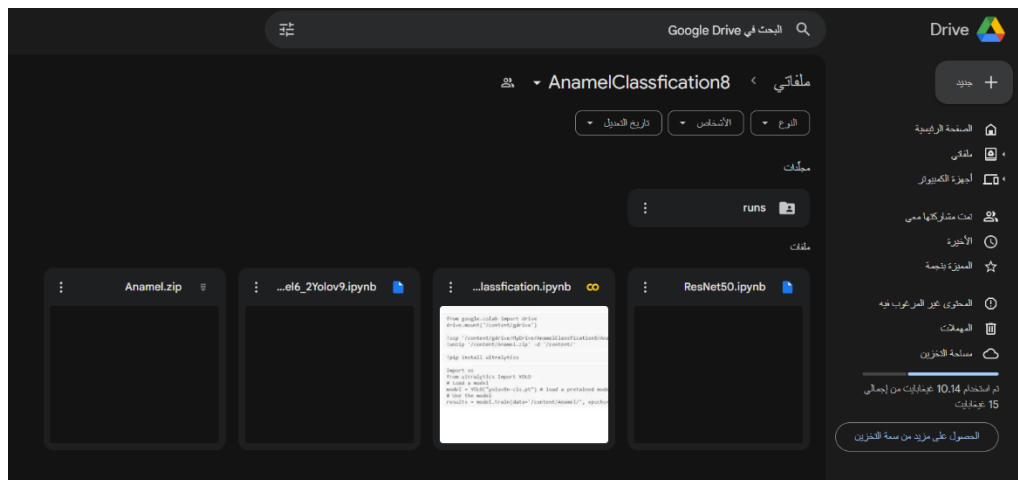


Figure 49: Uploading the AI Model to Drive

After training and testing the model, the project file was created in Visual Studio Code, and the Anamel database was created in Firebase, as shown in Figure 50.

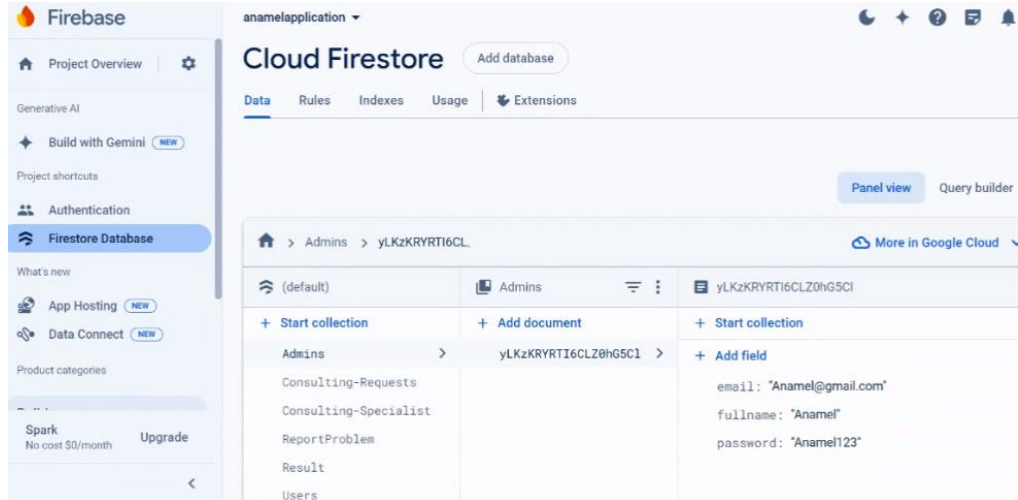


Figure 50: Anamel's application database.

The database contains authentication for the user to create a new account or log in, by storing the data in Firestore, which is the cloud that contains the system's various collections.

5.3 System Implementation

5.3.1 Frontend

Dart was used to write the content of the application. Figure 51 shows the dart files created in this project.

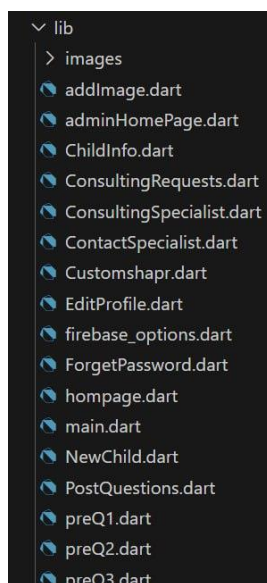


Figure 52: Profile file 1.

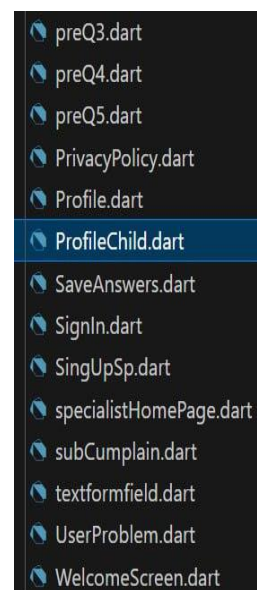


Figure 51: Profile file 2.

The system contains 30 user interfaces, in addition to basic files and database links, which will be presented in the following section.

5.4 Database Implementation

5.4.1 Backend

The backend consists of connection to a database. The Anamel database has been connected to Firebase. With many tables, it contains an automatic authentication when creating an account via email, and a Cloud Fire Store for the rest of the collection. Bg Figure 53 shows the code used for the user's authentication.

```
92  
93  
94 // Access Firestore collection  
95 CollectionReference users = FirebaseFirestore.instance.collection('Users');  
96 // Add a new document with user input data  
97 await users.add(  
98   'fullName': _fullNameController.text,  
99   'userName': _userNameController.text,  
100   'email': _emailController.text,  
101   'password': _passwordController.text,  
102   'userType': _userType,  
103   'age': age,  
104   // Add other fields as needed  
105 );  
106 await _auth.createUserWithEmailAndPassword(  
107   email: _emailController.text,  
108   password: _passwordController.text,  
109 );
```

Figure 53: User Collection & Auth

User data is stored and retrieved successfully to log in and navigates to the home page. According to the type of user, the user can add a new child. Figure 54 shows the code for adding and storing a new child.

```
CollectionReference newChild = FirebaseFirestore.instance.collection('newChild');  
  
Future<void> addNewChild() async {  
  // Calculate age  
  int age = calculateAge(DateTime.parse(_dateController.text));  
  
  // Add child data to Firestore  
  await newChild  
    .add(  
    'nameChild': nameChild.text,  
    'birthDate': _dateController.text,  
    'selectedAge': age.toString(),  
    )  
    .then((value) => print("Child Added"))  
    .catchError((error) => print("Failed to add child: $error"));  
  
  // Navigate to the addChildren screen  
  Navigator.push(  
    context,  
    MaterialPageRoute(builder: (context) => AddChildren()),  
  );  
  
  // Print child data for verification  
  print('Child Name: ${nameChild.text}');  
  print('Birthdate: ${_dateController.text}');  
  print('Age: $age years');
```

Figure 54: Add new child collection

After adding the child, the user can view his data, then add his drawings for analysis. The results appear after the analysis and are stored in the results as shown in Figure 55.

```
// Insert data into the 'Result' collection
await FirebaseFirestore.instance.collection('Result').doc().set({
  'email': userEmail,
  'fullname': fullName,
  'childName': childName,
  'childAge': childAge,
  'result': result,
  // Add other necessary fields
});
```

Figure 55: results collection 1.

If the user is a specialist, can add his information via the registration form as a specialist, and his information is stored in the database as shown in Figure 56 to then be displayed on the contact page with a specialist.

```
class ContactSpecialistPage extends StatelessWidget {
  Widget build(BuildContext context) {
    // Center // AppBar
    body: StreamBuilder<QuerySnapshot>(
      stream: FirebaseFirestore.instance.collection('Consulting-Specialist').snapshots(),
      builder: (context, snapshot) {
        if (snapshot.hasError) {
          return Center(
            child: Text(
              'حدث خطأ: ${snapshot.error}',
              style: TextStyle(color: Colors.red),
            ), // Text
          ); // Center
        }
        if (snapshot.connectionState == ConnectionState.waiting) {
          return Center(
            child: CircularProgressIndicator(),
          ); // Center
        }
        final specialists = snapshot.data!.docs;
        return ListView.builder(
          itemCount: specialists.length,
          itemBuilder: (context, index) {
            final specialist = specialists[index];
            final name = specialist['fullName'];
            final phoneNumber = specialist['phoneNumber'];
            // Building
          },
        );
      },
    );
  }
}
```

Figure 56: Specialist collection 1

The user can also report a problem, and this problem is stored in the database as shown in Figure 57 to be displayed to the admin later.

```
CollectionReference ReportProblem = FirebaseFirestore.instance.collection('ReportProblem');
Future<void> addProblem() {
    return ReportProblem
        .add({
            "Report_Subject":ReportSubject.text,
            "Description": descriptionController.text,
            "email":email.text
        })
        .then((value) => print("User Added"))
        .catchError((error) => print("Failed to add user: $error"));
}
```

Figure 57: Report problem collection 1

5.5 Building AI model

5.5.1 Algorithm Used

After completing the construction of the basic codes for the application and connecting them to the database, the drawing analysis model is linked to the application after trying several models and choosing the best of them, and among these models are:

- **ResNet50**

Residual Network (ResNet50) is a deep learning model used for computer vision applications. ResNet is adaptable and scalable, which makes it important in computer vision. ResNet was used to train our model, hoping to achieve high accuracy and speed in object detection within images.

- **YOLOv9**

The YOLO algorithm is an innovative method that combines speed and accuracy in object detection [17]. Object detection methods are usually three-stage processes: proposed region selection in the image using bounding boxes; feature extraction; and then the trained classifier is used to perform classification [18]. YOLOv9 depends on YOLOv8's strengths and introduces new features for even better object detection and classification. Some of these features are Programmable Gradient Information (PGI): This technique allows for the dynamic adjustment of gradients during training, which enables the model to prioritize learning from specific types of errors, leading to more targeted improvements in accuracy. The second feature that improved the YOLOv9 model is the Generalized Efficient Layer Aggregation Network (GELAN), which refines the feature extraction process by efficiently combining features from various network layers. It contributes to both speed and accuracy by providing a richer feature set for object classification.

- **YOLOv8**

YOLOv8 has a good balance between accurate object detection and fast inference. It achieves this through the anchor-free design where YOLOv8 predicts bounding boxes directly. This makes the model simpler and reduces parameter numbers resulting in faster processing. YOLOv8 also has its own classification model "yolov8-cls". The basic Task of this model is to classify directly and using the features of convolutional neural networks.

5.5.2 Collecting Dataset

The dataset serves as the foundation for training the model. After conducting comprehensive research and several interviews with psychologists to meet the necessary requirements with high accuracy. However, no data set was found that includes all the requirements. Hence, children's drawings were collected from different places in Mecca's community, after conducting interviews with children, forming a questionnaire that was shared with families, and interviewing psychologists to collect drawings. Furthermore, the next step was to use Roboflow to annotate images by dividing them into categories as: happy, sad, anxiety, angry, and aggressive as shown in Figure 58. Next, the dataset was divided into 80% for training, 10% for testing, and 10% for validation, making a total of 400 drawings for training, 50 for testing, and 50 for validation as Figure 59 shows.

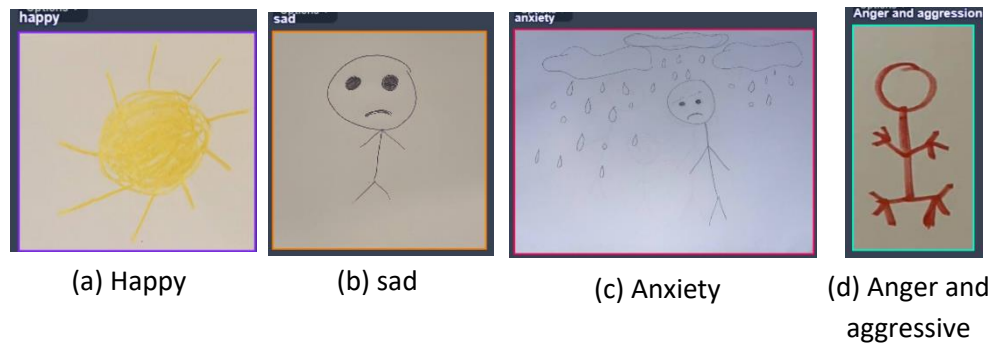


Figure 58: The four categories of the annotate

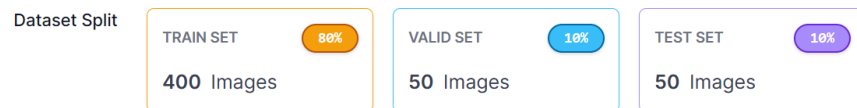


Figure 59: Splitting the Dataset into Train Set, Valid Set and Test Set

5.5.3 Training methodology

This project aims to analyze children's drawings with the highest level of performance while maintaining the minimum size suitable for use in mobile applications. This is done using convolutional neural networks, especially the recent developments in the YOLO versions, which support convolutional neural networks. Resnet50 was used to perform the same task, and this project compared all these results to determine the best performance. YOLO offers different sizes for their models (nano, small, medium, large, and extra-large). In this project, YOLOv8n and YOLOv9c were used, which effectively improves the architecture and achieves comparable accuracy. The evaluation has been extended to include resnet50. This allows us to comprehensively evaluate and compare the effectiveness of different models. These models are trained using the Anamel dataset, basic training settings include batch size, learning rate, momentum, and weight decay. These models were trained using a set of hyperparameters including varying durations from 10 to 150 epochs, a learning rate of 0.01, batch size of 8 for YOLOv9c, 16 for YOLOv8n, and 32 for YOLO. ResNet50 with a training dataset of 500 images, a test set of 50 images, and a validation set of 50 images. Table 24 shows the training Settings.

Model	Hyperparameter	Value
YOLOv8c	Epochs	1 – 50
	Learning rate	0.01
	batch size	16
YOLOv9c	Epochs	1 -150
	Learning rate	0.01
	batch size	8
ResNet50	Epochs	1-50
	Learning rate	0.01
	batch size	32

Table 24: Training settings for the models

5.5.4 Training Environment

For the model training environment, advanced computational power like GPUs was needed. Due to that, Google Colab was chosen to run the Python code on its cloud platform. And the free version of Google Colab gives access to a GPU T4 graphics card with 12GB of VRAM which proved sufficient for a significant portion of our training needs.

5.5.6 Training The Model

Yolov9:

In this project, the Yolov9 model was used, and to begin training, there were several commands to set up an appropriate environment for running the model. First, we used the commands as shown in Figure 60 to create an environment with the required resources to facilitate the training process.

```
[ ] !git clone https://github.com/SkalskiP/yolov9.git
    %cd yolov9
    !pip install -r requirements.txt -q
```

Figure 60: Set up an environment to implement the Yolov8 model

The Roboflow platform provided us with a code to install the Roboflow Python package, and it also configures the connection to the platform through an API key as shown in Figure 61.

```
[ ] !pip install -q roboflow
    from roboflow import Roboflow
    rf = Roboflow(api_key="hmbyKsLJ6deOH8f4NeF1")
    project = rf.workspace("anamel").project("anamel7")
    version = project.version(2)
    dataset = version.download("yolov9")
```

Figure 61: Install the Roboflow Python package and establish a connection with the Roboflow platform.

In addition, the “weight” directory will be created in the main directory {HOME}, after which four pre-trained weight files for the YOLOv9 model will be obtained via specific URLs and stored in the “Weights” directory. Finally, as shown in Figure 62, the “data” directory will be created. To store data related to the YOLOv9 model.

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```
[ ] !mkdir -p {HOME}/weights

[ ] !wget -P {HOME}/weights -q https://github.com/WongKinYiu/yolov9/releases/download/v0.1/yolov9-c.pt
!wget -P {HOME}/weights -q https://github.com/WongKinYiu/yolov9/releases/download/v0.1/yolov9-e.pt
!wget -P {HOME}/weights -q https://github.com/WongKinYiu/yolov9/releases/download/v0.1/gelan-c.pt
!wget -P {HOME}/weights -q https://github.com/WongKinYiu/yolov9/releases/download/v0.1/gelan-e.pt

!ls -la {HOME}/weights

Show hidden output

[ ] !mkdir -p {HOME}/data
```

Figure 62: Code to download pre-trained weight files for YOLOv9 model

The directory will be changed to Yolov9, then the data set is downloaded from the “Anamel7” project, as the Roboflow platform will be accessed through the project’s API key as shown in Figure 63.

```
%cd {HOME}/yolov9
from roboflow import Roboflow
rf = Roboflow(api_key="hmbyKsLJ6deOH8f4NeF1")
project = rf.workspace("anamel").project("anamel7")
version = project.version(2)
dataset = version.download("yolov9")
```

Figure 63: To change Directory to Yolov9.

To train the Yolov9 model for a specific data set, the train.py file must be called and several variables must be specified, such as the batch size, epoch count, image dimensions, and other variables, as the goal is to conduct training and improve the efficiency of the model in execution.

```
[ ] %cd {HOME}/yolov9
!python train.py \
--batch 8 --epochs 150 --img 640 --min-items 0 --close-mosaic 15 \
--data {dataset.location}/data.yaml \
--weights {HOME}/weights/gelan-c.pt \
--cfg models/detect/gelan-c.yaml \
--hyp hyp.scratch-high.yaml
```

Figure 64: Training Yolov9 model.

5.5.7 Result and Discussions of YOLOv9

After conducting several experiments training the models, a comparative analysis was performed to evaluate the models and determine optimal performance based on accuracy and model size. Table 25 shows the YOLOv9c results.

Model	Epoch	mAP0.5
YOLOv9c	10	0.53
	20	0.617
	30	0.745
	50	0.818
	100	0.949

Table 25: YOLOv9c results.

The YOLOv9c model shows a noticeable increase in accuracy at Epoch 100. The model has been re-evaluated to Epoch 150. Table 26 portrays the YOLOv9c results for evaluation metrics comparison.

Model	Epoch	(P%)	(R%)	mAp0.5 (%)
YOLOv9c	10	57%	0.53	53%
	20	75%	0.617	61.7%
	30	85.5%	0.745	74.5%
	50	72.6%	0.818	81.8%
	100	90.7%	88.6%	94.9%
	150	88.9%	91.1%	95.1%

Table 26: YOLOv9c results for evaluation metrics comparison

The table above shows the improvement in accuracy in Epoch 150 compared to the 100. Accordingly, Table 27 presents details of the results for Epoch 150.

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Class	(P%)	(R%)	mAp0.5 (%)	mAp0.5 -0.95 (%)
All	88.9%	91.1%	95.1%	72.6%
Anger -and -aggression	95.6%	97.3%	61.7%	59.7%
Anxiety	91.8%	86.2%	74.5%	75.7%
Happy	90.9%	81%	81.8%	68.2%
Sad	77.2%	100%	94.9%	86.8%

Table 27: YOLOv9 results for Epoch 150.

Confusion Matrix

The results presented in the previous tables show results based on the confusion matrix, and Figures 65 - 69 show both the confusion matrix, recall-confidence curve, precision- recall curve, and precision-confidence curve – F1 confidence curve.

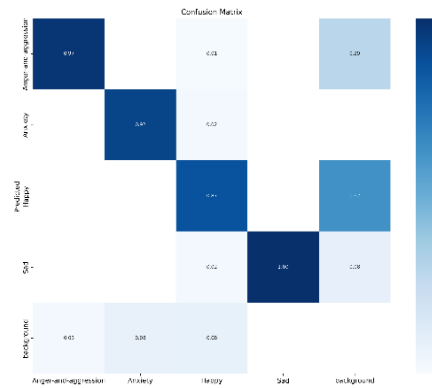


Figure 65: Confusion matrix for YOLOv9c

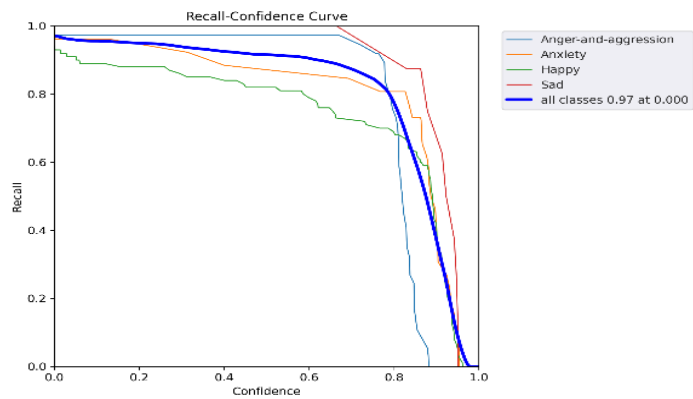


Figure 66: Recall-Confidence Curve for YOLOv9c

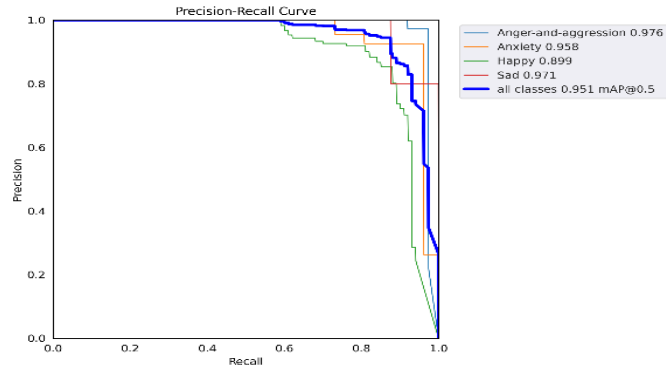


Figure 67: precision- recall curve for YOLOv9c

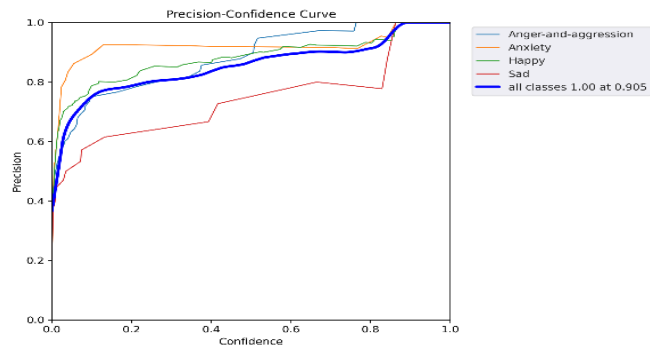


Figure 68: precision- confidence curve for YOLOv9c

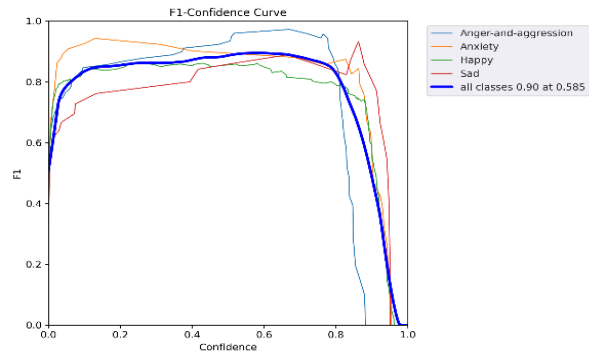


Figure 69: F1 confidence curve for YOLOv9c

- **ResNet50:**

The same steps are performed for the data set, the difference is in the model structure where TransferLearn_env, tensorflow and opencv are used. Figure 70 shows a code snippet.

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```
import tensorflow_hub as hub
resnet50_url = "https://tfhub.dev/google/imagenet/resnet_v1_50/feature_vector/4"
resnet50_model = tf.keras.Sequential([
    hub.KerasLayer(resnet50_url, trainable=False),
    tf.keras.layers.Dense(10, activation='softmax') # Adjust output units according to your dataset classes
])

resnet50_model.compile(optimizer='adam',
                      loss='categorical_crossentropy',
                      metrics=['accuracy'])

import tensorflow as tf

num_classes = 3 # Adjust this according to your dataset
# Load pre-trained ResNet50 model without top classification layer
base_model = tf.keras.applications.ResNet50(weights='imagenet', include_top=False, input_shape=(224, 224, 3))
# Freeze the pre-trained layers
base_model.trainable = False
# Add classification layers on top of ResNet50
model = tf.keras.Sequential([
    base_model,
    tf.keras.layers.GlobalAveragePooling2D(),
    tf.keras.layers.Dense(512, activation='relu'),
    tf.keras.layers.Dropout(0.5),
    tf.keras.layers.Dense(num_classes, activation='softmax')
])
# Compile the model
model.compile(optimizer='adam',
             loss='categorical_crossentropy',
             metrics=['accuracy'])
# Assuming you have properly configured generators
# Train the model
history = model.fit(
    train_generator,
    steps_per_epoch=train_generator.samples // train_generator.batch_size,
    validation_data=validation_generator,
    validation_steps=validation_generator.samples // validation_generator.batch_size,
    epochs= 25 # Adjust epochs as needed
)
```

Figure 70: Code snippet

5.5.8 Result and Discussions of ResNet50

The results presented in the previous tables show the results of the resnet50 model, which shows the highest accuracy at Epoch 24 at 70.3%. Figures 71 and 72 show both Model Accuracy and Model Loss for the ResNet50 model.

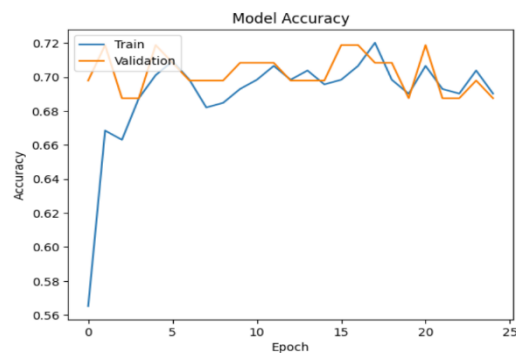


Figure 71: Model Accuracy for ResNet50

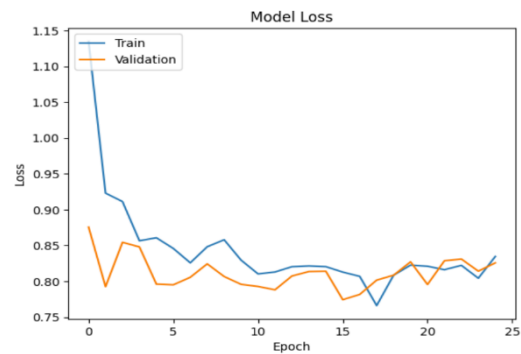


Figure 72: Model Loss for ResNet50

Model	Epoch	accuracy
RESTNET50	10	69.29%
	24	70.3%
	25	69%
	50	68%

Table 28: RestNet50 results

Yolov8:

The figure shows the YOLO8 model code used for classification, which showed good results compared to the rest of the models.

```
!pip install ultralytics

import os
from ultralytics import YOLO
!scp '/content/gdrive/MyDrive/AnamelClassification8/Anamel.zip' '/content/Anamel.zip'

!unzip '/content/Anamel.zip' -d '/content/'
# Load a model
model = YOLO("yolov8n-cls.pt") # load a pretrained model
# Use the model
results = model.train(data='/content/Anamel/', epochs=50) # train the model

Downloading https://github.com/ultralytics/assets/releases/download/v8.2.0/yolov8n-cls.pt to 'yolo'
```

Figure 73: YOLOv8 code snippet

5.5.9 Result and Discussions of YOLOv8

Model	Epoch	top 1_acc
YOLOv8n-cls	10	94%
	15	92%
	20	94%
	25	92%
	50	94%

Table 29: YOLO v8 results

Table 29 shows the accuracy of the YOLOv8n-cls model prepared for classification, and it showed better results than previous models, as the accuracy was 94% for all classes in Top 1 compared to Epoch 10. Figures 74 to 77 show the confusion matrix, accuracy, and loss for both training and validation.

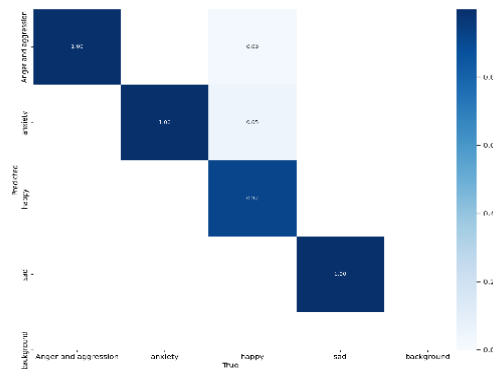


Figure 74: confusion matrix for YOLOv8n-cls

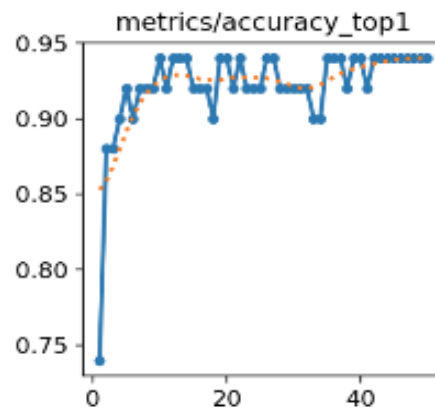


Figure 75: Matrix for accuracy top1

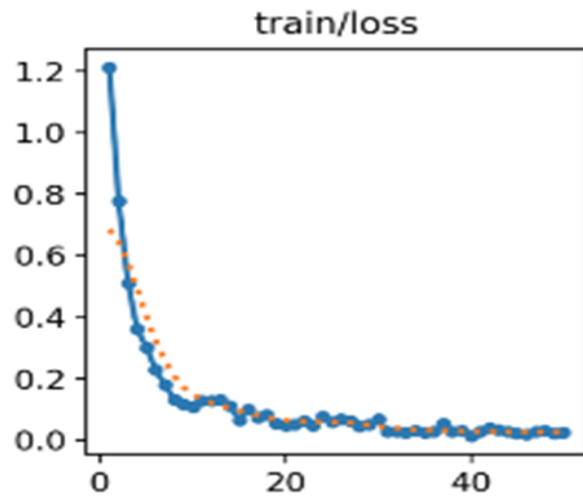


Figure 76: YOLOv8 Loss for train

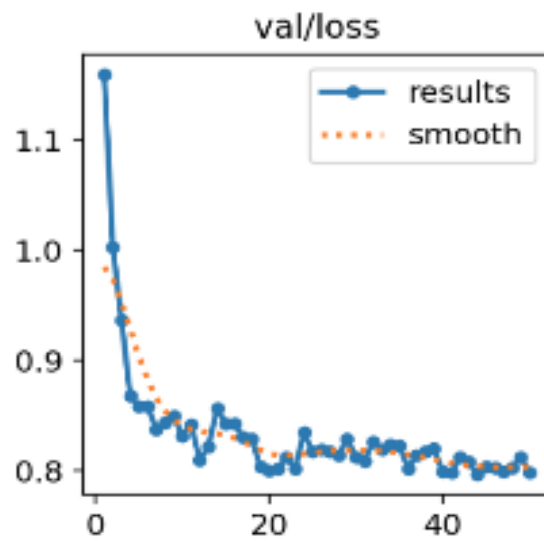


Figure 77: YOLOv8 Loss for validation

5.5.10 Testing The Model

5.5.10.1 Sample of The Testing

After training all the models and varying the quality of detection in terms of object detection or classification by several models with different versions, Figure 78 to 81 shows the results of the test by classification and object detection.

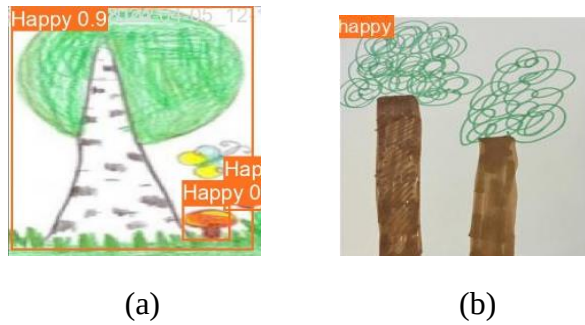


Figure 78: Figure A shows the detection of happy using object detection models, and Figure B shows the same feeling through classification using YOLOv8n-cls

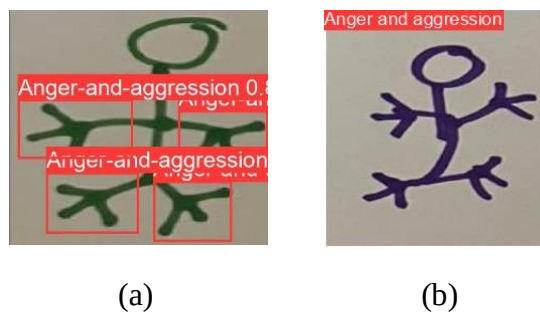


Figure 79: . Figure A shows the detection of Anger and aggression using object detection models, and Figure B shows the same feeling through classification using YOLOv8n-cls.

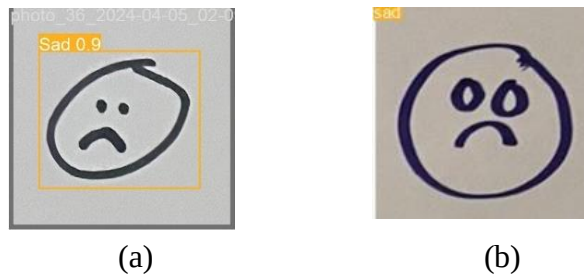


Figure 80: Figure A shows the detection of Sad using object detection models, and Figure B shows the same feeling through classification using YOLOv8n-cls.

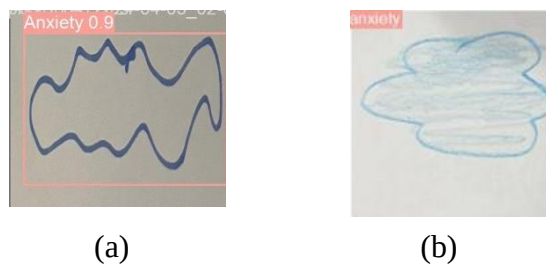


Figure 81: Figure A shows the detection of Anxiety using object detection models, and Figure B shows the same feeling through classification using YOLOv8n-cls.

5.5.10.2 Model Size

After all models were completed and accuracy tested, Table 30 shows the comparison between the models in terms of size.

Model	Model size (MB)
YOLOV8-cls	2.83
YOLOv9c	5.26
ResNet50	94.3

Table 30: Model size

This project focuses on the highest level of performance while maintaining the minimum size suitable for use in mobile applications and based on Table 30, the YOLOV8-cls model for classification shows the minimum size.

5.5.10.3 Compare results.

Table 31 shows a general comparison between the models in this project and the model of the Esraa study [1], as the results in this project showed the superiority of the model for classification in Yolo 8 over all other models.

Study	Model	Type	Accuracy
Esraa	ResNet34	Object Detection	0.76
	ResNet50		0.79
Anamel	ResNet50		0.50
	Yolov9c	Classification	0.951
	Yolov8n-cls		0.94

Table 31: General Comparison

5.5.10.4 Discussion results of the models

The results show that YOLOv8n -cls meets the requirements in terms of overall performance, given the importance of achieving accurate classification with less model size. Table 26 displays the results of Yolov9n, which reached an accuracy of 95.1% at Epoch 150, and an accuracy of 94.9 % at Epoch 100, and the Precision was 88.9% at Epoch 150 for all classes. In Esra's study, the highest accuracy of p was only 74% for the ResNet50 model [1]. Table 28 shows the accuracy of the ResNet50 model which reached 70.3% at Epoch 24. Therefore, the classification made a big difference in terms of performance and accuracy, as its accuracy was 94% when compared to Epoch 10. Figures 78 - 81 show samples of the test with classification and object detection, and the results were ideal in the YOLOv8-cls model, which fully met the requirements in terms of detecting the feeling of the drawing by classification with the smallest size model stands out from the rest of the other models.

Chapter 6: Testing, Results and Discussion

6.1 Testing

6.1.1 System Testing

6.1.2 User Testing

6.2 Results and Discussion

6.2.1 System Testing

6.2.2 User Testing

6.2.2.1 Test Result for parents

6.2.2.2 Test result for specialist

6.1 Testing

6.1.1 System Testing

System testing is the level of testing that validates a complete and integrated software product. It checks how the components interact with each other and with the system as a whole, and conducts a comprehensive test for each input in the system to verify the required output [19]. Anamel was tested using six different scenarios, as depicted in Figures 82, 83, 84 , 85 , 86 and 87.

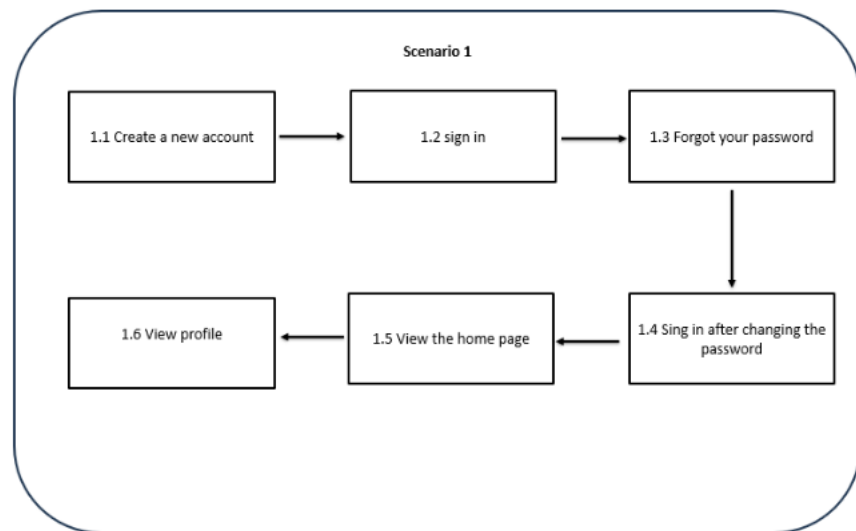


Figure 82: Scenario 1 for user

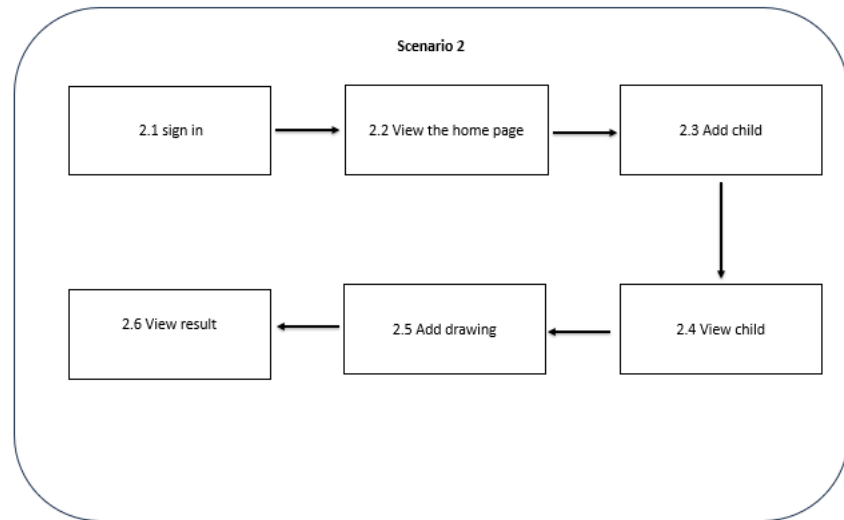


Figure 83: Scenario 2 for user

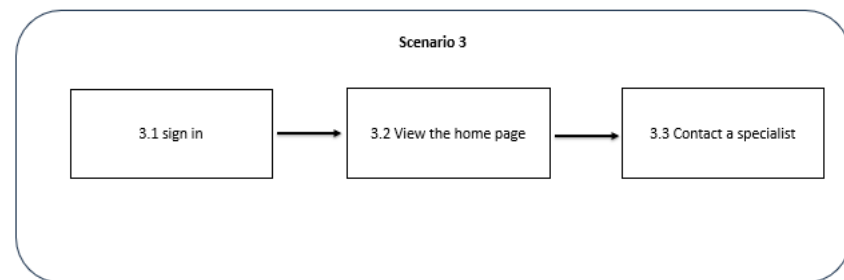


Figure 84: Scenario 3 for user

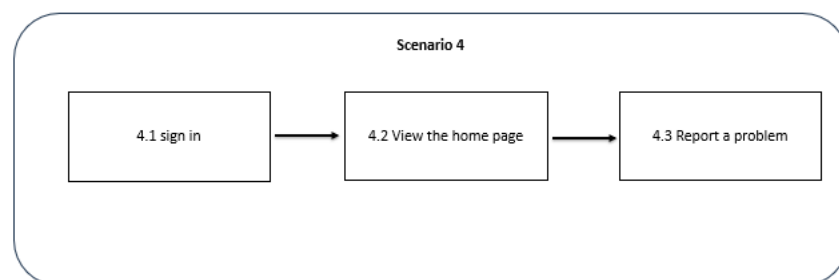


Figure 85: Scenario 4 for user

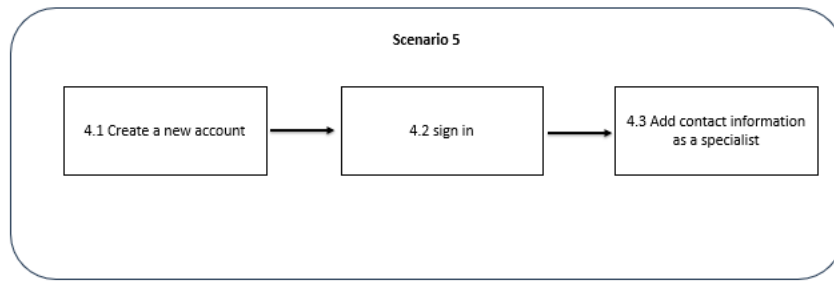


Figure 86: Scenario 5 for specialist

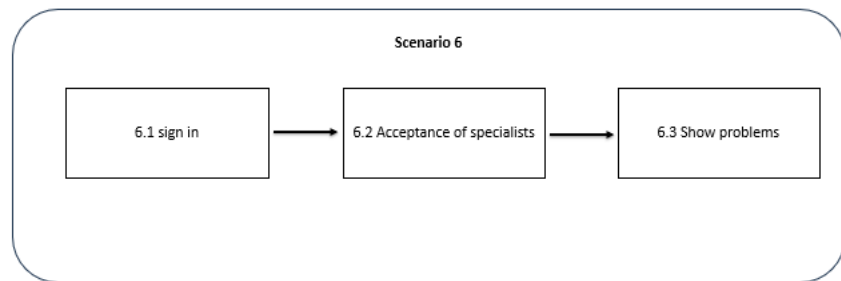


Figure 87: Scenario 6 for the admin

6.1.2 User Testing

A total of 5 users participated in the test, 3 of whom played the role of the parent (user). One specialist and one admin, the Table 32 summarizes the information of the participants.

User Type	Gender	Job	Educational Level	Experience in Using Technology
Parent 1	Female	Teacher	Bachelor	Advance
Parent 2	Female	Teacher	Bachelor	Normal
Parent 3	Male	Accountant	Bachelor	Advance
Specialist	Female	Psychologist	Master	Advance
Admin	Female	Student	Bachelor	Advance

Table 32: participated information

6.2 Results and discussion

In this section the system test results are presented first followed by the results from user testing.

6.2.1 System Testing

As illustrated in Figure 88 , all detected errors were minor and were related to the front-end component. Following the completion of system testing, these errors were resolved, and the system testing was rerun with no errors detected. Next, each participant was given a number of tasks to perform, and they were like that observed during testing. Total time to complete each task for each participant was recorded along with the number of errors for each task.

Parents were asked to perform the following tasks:

- 1- Create a new account
- 2- Sign in and add a child
- 3- Add a drawing, and display the results
- 4- Contact a specialist

The specialist was asked to perform the previous tasks in addition to the following tasks:

- 1- Adding information for communication by users

The admin was asked to perform the following tasks:

- 1- Sign in
- 2- Acceptance of specialists
- 3- Presentation of problems

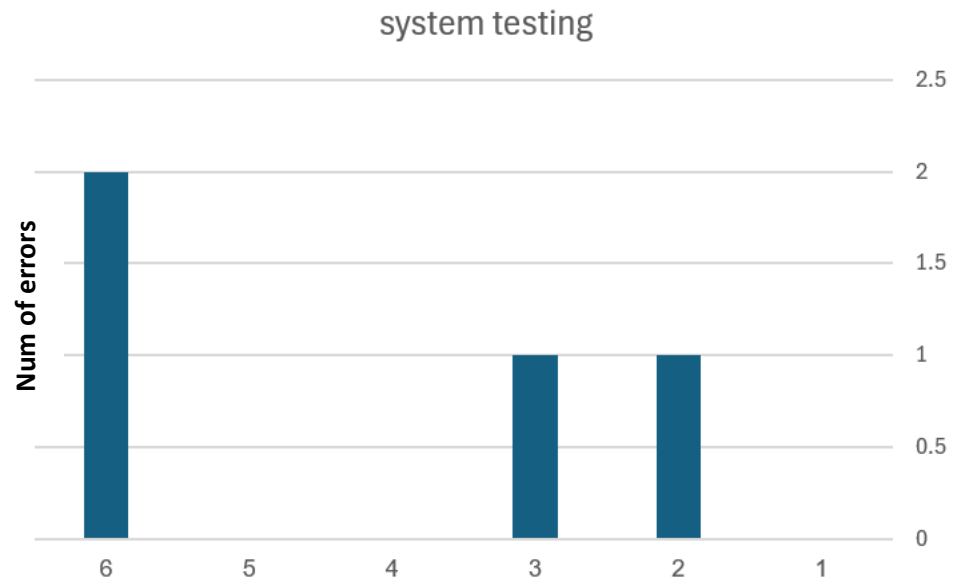


Figure 88: Scenarios

6.2.2 User Testing

Test results for users (parents) are represented first, followed by the results of specialists.

6.2.2.1 Test results for the parents

Task 1: Create an account

In this task, participants were asked to create a new account.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Parent 1	Completed	0	0:58 min	1:003 min
Parent 2	Completed	0	1:29 min	
Parent 3	Completed	0	1:14 min	

Table 33: Create a new account task

It is worth noting that the task took this time as shown in Table 33 because Participants had to wait for a message to confirm that the email was not already registered and then proceed to sign in.

Task 2: sign in and add child

In this task, parents were asked to enter their sign in information and go to the home page to add a new child and view the information of this child.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Parent 1	It was finished with help, after making several mistakes in the fields	1	3:02 min	2:21 min
Parent 2	Completed	0	1:29 min	
Parent 3	Completed	0	2:34 min	

Table 34: Sign in and add child task

Task 3: Add drawing

In this task, parents were asked to enter a drawing of their child for analysis.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Parent 1	Completed	0	0:48 min	1:23 min
Parent 2	The photos cannot be accessed on the device.	2	2:04 min	
Parent 3	Completed	0	0:19 min	

Table 35: Add drawing task

Task 4: Communicate with specialists

In this task, parents were asked to access a page for contacting specialists.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Parent 1	Completed	0	0:10 min	0:1 min
Parent 2	Completed	0	0:06 min	
Parent 3	Completed	0	0:14 min	

Table 36: Contact specialist task

6.2.2.3 Test results for the Specialists

Task 6: Add specialist information

In this task, then ask the specialist to sign in and then add his information to the means of communication by other users.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Specialists	Completed	0	0:10 min	-

Table 37: Add specialist information task

6.2.2.4 Test results for the admin

Task 7: Accept the specialist and present the problems

In this task, then ask the specialist to sign in and then add his information to the means of communication by other users.

User	Effectiveness		Efficiency	
	Task Success	No of error	Completion Time	Average Time
Admin	The specialist's admission is not saved correctly, problems are not displayed to users	2	0:23 min	-

Table 38: Accept the specialist and present the problems task

6.2.2.5 integration testing of the entire system

As part of our comprehensive system testing, we conducted thorough integration testing to ensure that all components of the system functioned correctly together. This included integrating our AI model for analyzing children's drawings into the mobile application. Our AI model, designed to analyze and interpret children's drawings, was successfully integrated into the mobile application using the `flutter_tflite` package. The integration process involved several key steps:

1. Model Conversion and Optimization:

- The AI model was initially developed and trained in a Python environment. To deploy it within the mobile application, we converted the model to TensorFlow Lite format, optimizing it for mobile use to ensure efficient performance and lower resource consumption.

2. Integration with Flutter:

- Using the `flutter_tflite` package, we integrated the TensorFlow Lite model into our Flutter application. This package provides a convenient interface for running TensorFlow Lite models on both Android and iOS platforms.
- We implemented the necessary logic to load the model, preprocess the input (children's drawings), run the inference, and interpret the results within the Flutter application.

3. Testing and Validation:

- Comprehensive testing was conducted to verify that the AI model performed accurately and efficiently when integrated into the mobile app.
- We tested the system with a variety of children's drawings to ensure that the model's predictions were correct and that the application's user interface responded appropriately to the model's output.

Interface Integration

The integration of the AI model was further enhanced by the `addImage` and `postQuestions` interfaces, which utilized several libraries such as `flutter/material.dart` for material design components, `flutter_tflite/flutter_tflite.dart` for TensorFlow Lite integration, and `image_picker/image_picker.dart` for uploading images.

1. `addImage` Interface:

- This interface allows users to upload or capture images, which are then processed by a pre-trained TensorFlow Lite model to analyze and display the results.
- The app first initializes the model using the `_tfLteInit()` method and performs image inference through `_runModelOnImage()`.
- Users interact with the app through the `pickImageGallery()` and `pickImageCamera()` methods by using the `ImagePicker` library to select images.
- The UI components include an `AppBar` for the title and a `SingleChildScrollView` to make the interface scrollable.
- After the model analyzes the drawing, users can navigate to the `postQuestions` interface by clicking 'Yes' in the popup window

2. postQuestions Interface:

- This interface is designed to handle user interactions after the initial image analysis by gathering user answers to the post-analysis questions for each result.
- Users can answer additional questions to provide more context and detail, enhancing the accuracy and confidence in the model's results.
- The final results, along with user responses, are then stored in the database.

The integration of these components, coupled with thorough testing and validation, ensured a seamless user experience and reliable functionality across the entire system. The successful integration and testing of the AI model and interfaces demonstrated the robustness and reliability of our application, providing valuable functionality for analyzing and interpreting children's artwork.

After every test and error in the program, it is modified directly. However, it is worth noting that the participants' opinions were also positive. They liked the project idea, the interfaces, and the way it was implemented. They also gave us advice on how to improve the project in the future and the specialist expressed his desire to register as an actual specialist for the launch of Anamal.

Chapter 7: Conclusion and future work

7.1 Conclusion

7.2 Implementation issues/ Challenges Faced

7.3 Future work

7.1 Conclusion

At the end of this report, the Anamel application has achieved the main goal, which is to help parents and psychologists to make the process of analyzing children's drawings to discover their psychological and mental health easier and faster, by creating an artificial intelligence model to analyze children's drawings with high accuracy and a size suitable for building a phone application. The entire system was completed after the efforts of the team members under the guidance of the supervisor. It took about 26 weeks of hard work, including disseminating the questionnaire to parents, interviewing psychologists, collecting drawings of children in Saudi Arabia, and analyzing the entire system using different graphs. Moving to the Designing stage of the prototype and system interfaces. Next, the dataset was built from scratch. There are five main psychological emotions: happiness, sadness, anxiety, anger, and aggression. This project has been implemented Drawing on everything we learned at the university over the past years And rely on self-learning. The system was implemented and the model trained Using three CNN-supported models. They are Yolo9, Yolo8, and Resnet50. After that, the most accurate model with the lowest size was chosen to be published on the system, which is the Yolo8 model, for classification with an accuracy of 94%. After linking the model to the system, many tests were conducted, and errors were corrected after they were discovered. The system was tested on five participants, including three parents, a specialist, and an addict. The results showed that the system needed simple improvement for several functions, and after repairing them, they were reused and demonstrated high performance and quick response. The interfaces were clear and easy to use, and the Anamel application system succeeded in fully achieving the goal.

7.2 Implementation Issues/ Challenges faced

During the implementation of our project, we encountered several significant challenges that impacted our progress and required careful management and resolution. These challenges are detailed below:

- **Limited GPU Usage in Google Colab**

One major issue we faced was the limited availability and usage restrictions of GPUs in Google Colab. While Google Colab provides a convenient and cost-effective platform for developing and testing AI models, it has several limitations, including:

- **Limited GPU Runtime:** The availability of GPU resources in Google Colab is limited and not guaranteed. Users often face interruptions due to session timeouts or resource unavailability.
- **Resource Constraints:** The computational power provided by Google Colab's free tier is often insufficient for training large, complex AI models, resulting in prolonged training times and inefficient experimentation.
- **Session Limits:** The maximum runtime per session and the total usage limits per day can hinder continuous model training and testing, requiring frequent restarts and session management.

- **Version Compatibility Issues in Flutter for Mobile Application**

Another significant challenge was the frequent release of new versions in Flutter, which resulted in compatibility issues with our mobile application. Specific issues included:

- **Breaking Changes:** New Flutter releases often introduce breaking changes that are not compatible with previous versions. This forced us to regularly update our codebase to comply with the latest API changes and features.
- **Dependency Conflicts:** Updating Flutter versions frequently led to conflicts with third-party packages and dependencies. Some packages were not immediately updated to be compatible with the latest Flutter version, causing build errors and functionality issues.
- **Rework and Testing:** Each update required substantial rework, including refactoring code, resolving conflicts, and retesting the application to ensure it remained functional and stable. This repeated effort consumed significant time and resources, diverting attention from developing new features and enhancements.

7.3 Future work

As future work, we plan to improve Anamel by adding a number of features. The most important of which are:

- Adding more data set and psychological disorders that can be diagnosed.
- Adding more psychological tests that can be conducted and then measured.
- Add a drawing board in the application without the need to attach drawings.
- Analyze the phrases and words in the drawing.
- Filming a video of the child while drawing to analyze physical movements and diagnose mental health.

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